FACULTY OF COMPUTING

SEMESTER 1

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SECP3723-01 SYSTEM DEVELOPMENT TECHNOLOGY

SECTION 01

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1.0 Client Background

Lembaga Kemajuan Pertanian Kemubu (KADA) is a Malaysian organization dedicated to advancing the agricultural sector and supporting farming communities. Besides focusing on farming, KADA also runs a cooperative that offers financial services to its members. These services include managing memberships, providing different types of loans, offering savings accounts, and distributing dividends. The goal is to help members with their financial needs and support their personal and work-related growth.

The cooperative provides different loan options, such as Pinjaman Al-Bai, Pinjaman Al-Innah, and Pinjaman Qardhul Hasan, which is an interest-free loan. These loans help members with expenses like road tax, insurance, and other financial needs. Members who have savings accounts in the cooperative also receive dividends. External auditors check the calculations to ensure fairness and accuracy.

Even though KADA offers many financial services, its cooperative still depends on manual processes. Tasks like registering members, applying for loans, and keeping financial records are done on paper. This can lead to mistakes and slow processing. However, KADA is working on improving its system to make services faster and more reliable, helping both the farming community and its members' financial stability.

2.0 Problem Background

Koperasi KADA currently relies on a manual system for membership and loan application processing. This approach presents several challenges that impact on operational efficiency, data accuracy and security. The key issues are detailed below:

1. **Lack of an automated system for membership and loan application**
   Currently, there is no digital system for handling membership registrations and loan applications. This means applicants must submit paper-based forms, which require manual processing by staff. As a result, the Koperasi can only process a limited number of applications at a time which causes delays and restricts the

number of new members and loan applicants. This limitation slow down the overall growth and efficiency of the Koperasi

2. **Manual notification process for membership and loan approval**

   Employees must inform applicants about their membership and loan application status through phone calls. This manual process is time-consuming, especially when there are a large number of applicants. Additionally, it increases the risk of miscommunication as employees may miss some applicants or provide information inconsistently. This leads to frustration for both applicants and staff.

3. **Risk of data inaccuracies due to manual record-keeping**

   After an application is approved, employees manually record the applicant's information into logs or spreadsheets. However, this process is prone to errors such as misspellings, incorrect numerical entries or issues caused by handwriting. These small mistakes can cause major issues in record accuracy, leading to confusion, incorrect loan calculations or delays in processing requests.

4. **Manual loan calculation**

   Staff manually calculate monthly repayment amounts for each loan. This process is slow and can result in errors, especially when handling large volumes of data. Mistakes in loan calculations could lead to financial problems, affecting both Koperasi and its members.

5. **Risk of data loss due to paper-based records**

   The Koperasi stores all the membership and loan applications in physical paper form. This method is of a high risk for theft, natural disasters and human negligence. Losing such records can cause severe disruptions in operations and lead to disputes with members due to missing or incomplete information.

3.0 Proposed Solution

We proposed a web-based system that streamlines the entire process to improve the accuracy in managing all applications. This solution replaces manual paperwork with digital automation to ensure faster processing, better data accuracy and enhanced security. Below are the key improvements:

1. **Implementation of an Automated Membership and Loan Application**

   To eliminate the inefficiencies of paper-based applications, we introduce a web-based system that allows applicants to register for membership and apply for loans online. This system will enable users to fill out forms digitally and upload required documents to reduce the dependency on manual paperwork. By integrating input validation, the system can ensure that applications are complete before submission to prevent errors and delays. Additionally, a digital system allows staff to process multiple applications simultaneously to fasten the reviews and approvals process.

2. **Automation of Membership and Loan Applications Notifications**

   Instead of relying on manual phone calls to inform applicants about their status, Koperasi KADA can implement an automated notification system using email. This system will instantly send confirmation messages when an application is received, keeping applicants informed at every stage of the approval process. By implementing email notifications, applicants will receive real-time updates.

3. **Enhancement of Data Accuracy Through Digital Record-Keeping**

   Manual data entry increases the risk of errors such as miscalculations and inconsistencies in record. To address this, we implement a centralized database that securely stores all membership and loan records. With this implementation, staff can effectively update, retrieve and manage applicant data without relying on paper logs or spreadsheets. Additionally, a searchable database will allow staff to quickly access applicant details to improve workflow efficiency.

4. **Automation of Loan Calculation Process**

   To reduce human errors in financial computations, we integrate an automated loan calculator into its system. This feature will instantly compute monthly repayment amounts based on factors such as the loan amount, interest rate and repayment period.

5. **Secure Digital Storage to Prevent Data Loss**

   To prevent the high risk of data loss, we do the transition to a digital storage where all membership and loan records are maintained on a MySQL Database, phpMyAdmin server. Digital storage or databases provide access control to ensure that only authorized personnel can view or modify sensitive information.

4.0 Development Approach

The Software Development Life Cycle (SDLC) is a structured approach used to develop the system for KADA Cooperative. This development process consists of five main phases which are planning, analysis, design, implementation and testing. Each phase ensures that the system meets the stakeholders' requirements and functions effectively.

The first phase in SDLC is planning. In this phase, we gathered initial requirements from stakeholders, during the Industry Talk event. The representatives from KADA Cooperative, Mr. Ahmad Rohalian and Ms. Norzamani Ismail, explained their current system and the changes they needed. Based on information gathered, we planned the system development by outlining the project's scope, goals, required resources, and potential risks. This phase ensures that all stakeholders have a clear understanding of the project's direction and feasibility before moving forward.

After the initial planning phase, we conducted a detailed analysis to understand user needs and system requirements. We used the Joint Application Development (JAD) approach collaboratively gathering requirements and discussing possible solutions during the Industry Talk session. Additionally, follow-up interviews in WhatsApp groups allowed us to ask specific questions regarding KADA Cooperative's operations and

expectations. These discussions provided crucial insights into loan types, system usage, dividend calculations, membership appeal processes, and the application approval workflow. We also identified the need for an improved system to aid accounting and transaction tracking, including generating monthly and yearly invoices. This phase ensured that we had all the necessary information to design a system that meets KADA Cooperative's needs.

In this phase, we converted the gathered requirements into a structured system model. We designed system architecture, data models, and user interfaces. Wireframe designs were created to visualize workflows and database structures, ensuring scalability and efficiency. The finalized design served as a guide for implementation.

After finalizing the designs, we started developing and documenting the system. The system was developed in modules, with each team member handling specific functionalities. Initial testing was done locally before deploying the system to the hosting server (jimathosting). Minor bugs encountered during deployment were fixed immediately. A user manual was also created to assist users in navigating through the system.

Comprehensive testing was conducted on January 23, 2025, to ensure all modules functioned correctly. KADA Cooperative representatives reviewed the system and provided feedback. Based on their input, we identified new requirements and made necessary modifications. The user manual was updated accordingly. Follow-up discussions helped refine the system further, ensuring it met KADA Cooperative's expectations. This phase ensured the system was stable and ready for deployment.

Following the SDLC approach allowed us to systematically develop a system that met KADA Cooperatives needs. Engaging stakeholders throughout the process helped us create a system that improves membership and loan application processing and financial record management. The structured approach ensured the successful completion of the project.

5.0 Development Software, Language, Technology, API and Tools

Development software used to write and edit the code includes text editors like Visual Studio Code (VS Code) and Sublime Text. These editors help by providing features such as automatic text coloring, error checking, and code suggestions, which make coding easier and more efficient.

For database management, phpMyAdmin is used along with XAMPP, a software package that creates a local testing environment. XAMPP includes Apache, a web server that processes requests, MySQL for handling database operations, and PHP for writing server-side code. This setup allows the system to be tested on a computer before making it available online.

The system is developed using several programming languages, each serving a different purpose. HTML is used to create the basic structure of web pages, while CSS controls the design, including colors, fonts, and layout. JavaScript is used to make the website interactive by adding features like pop-up messages, form validation, and animations. Bootstrap is also used to help design a responsive website, ensuring that the layout adjusts properly on different screen sizes, including mobile devices. On the server side, PHP is responsible for processing user requests, managing system logic, and handling communication with the database. SQL is used to store, retrieve, update, and delete data efficiently in the MySQL database. These languages work together to create a system that is both functional and user-friendly.

Several tools are used throughout the development process to make it more organized and efficient. Draw.io is used to create diagrams, such as use case diagrams and Entity-Relationship Diagrams (ERDs), which help in planning how different parts of the system connect and interact. Hosting services like Jimat Hosting provide a platform for deploying the system online so that users can access it from any location.

6.0 Module Explanation

There are seven modules for koperasi KADA's automated system:

1. Membership Application Module

   This module allows applicants to apply for membership by submitting their personal details, family information, and bank details, along with the required fees. Once the application is submitted it will be reviewed by the admin before being forwarded to ALK for final approval. This process ensures that all membership applications are properly assessed before acceptance.

2. Terminate Membership Module

   This module involves a Member, who can request to terminate their membership from the system. To initiate the termination process,the member submits a termination request through their account at the "Borang Berhenti" page. Before confirming the request, the system prompts the member to review the consequences such as loss of access to services. Once the request is submitted, the system processes the termination and notifies the Member of the status via email. However, the member can also track the status of their termination request through the "Status Berhenti" page.

3. Review Module

   This module enables admin to review membership application forms, loan application forms, and membership termination forms. The admin will go through the list of applicants and assess their submissions. If an application is complete, the admin will click the "Lengkap" button, sending the application to ALK for approval. If the application is incomplete, the admin will click the "Tidak Lengkap" button, requiring the applicant to resubmit the form.

4. Loan Application Module

   This module involves an applicant, who has an active membership to apply for a loan by filling out an application form with necessary details. Before

submission, the Applicant can review and modify their loan application to ensure accuracy. In terms of accuracy, the system will also check whether the applicant filled out all the required fields. Once submitted, the form is forwarded to admin and Ahli Lembaga Koperasi (ALK) for review and approval. Furthermore, the applicant can track their loan status through the “Status Pembiayaan” page.

5. Approve Application Module

This module involves alk, who will approve or reject membership and loan applications based on company’s policy and board members meeting. Before alk perform decision-making, all of the applications must be reviewed by admin first and status “Lengkap” or “Tidak Lengkap” will be given. Only complete applications will be forwarded to alk. Alk then perform decision-making by clicking button “Diterima” or “Ditolak” and the decision made will be passed to admin for further processing.

6. Statement Report Production Module

This module is used to create financial reports for cooperative members. It prepares financial statements that show important details about members’ financial records. It also includes the cooperative report, which contains statistics on applications, approvals, and other related information. Membership status, loan status updates and financial report will also be sent via email in this module. This helps members and stakeholders see financial and cooperative activities clearly.

7. Financial Tracking Module

This module keeps track of all transactions in the cooperative. It ensures that financial operations are accurately tracked and organized. This module also allows applicants to track their payment regarding their savings and loan. With a proper transaction history, this module helps in creating reports, checking finances, and making better financial decisions.

7.0 System Credentials (Hosting)

Hosting URL: http://kadakoperasi.ct.ws/

7.1. Applicant

Email= farranurzahin@graduate.utm.my
Password= 12345678

7.2. Admin

Email= aliabu@gmail.com
Password= 12345678

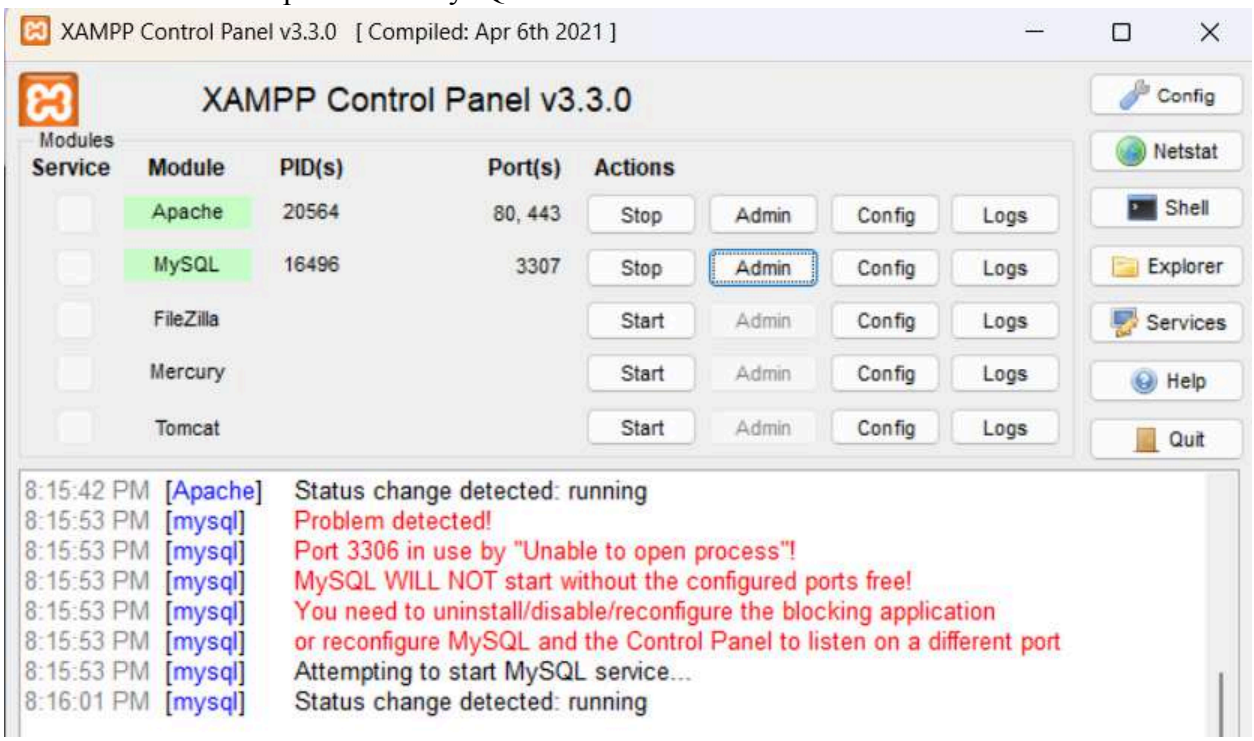
7.3. Ahli Lembaga Koperasi (ALK)

Email= aminahkassim@gmail.com
Password= 12345678

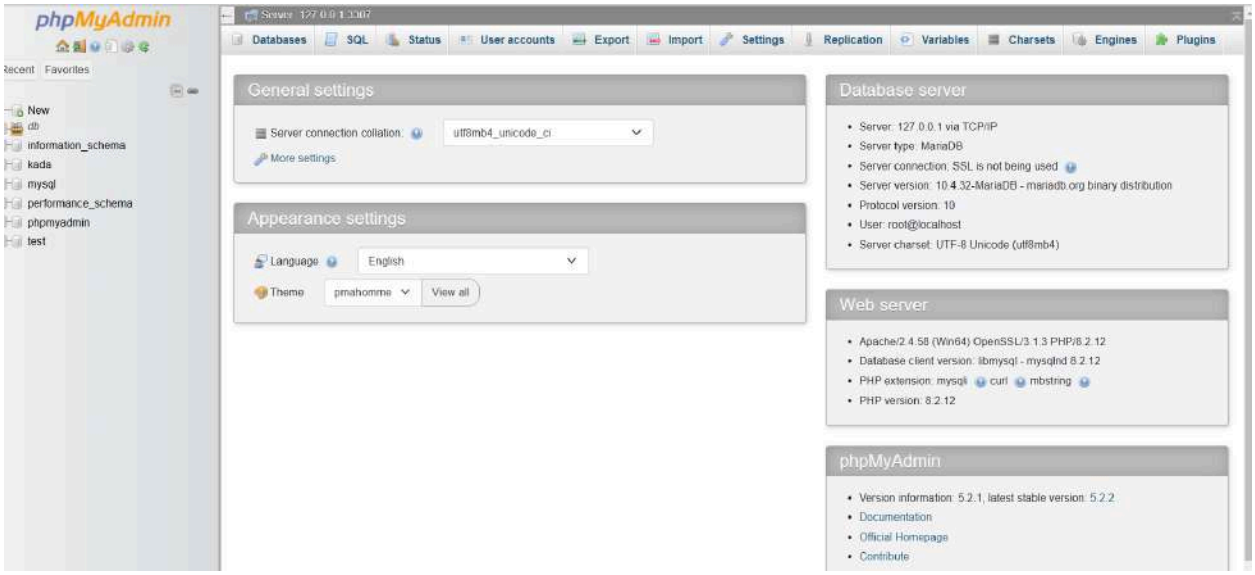
8.0 Localhost Setup

1. Download and run XAMPP Control Panel - https://www.apachefriends.org/

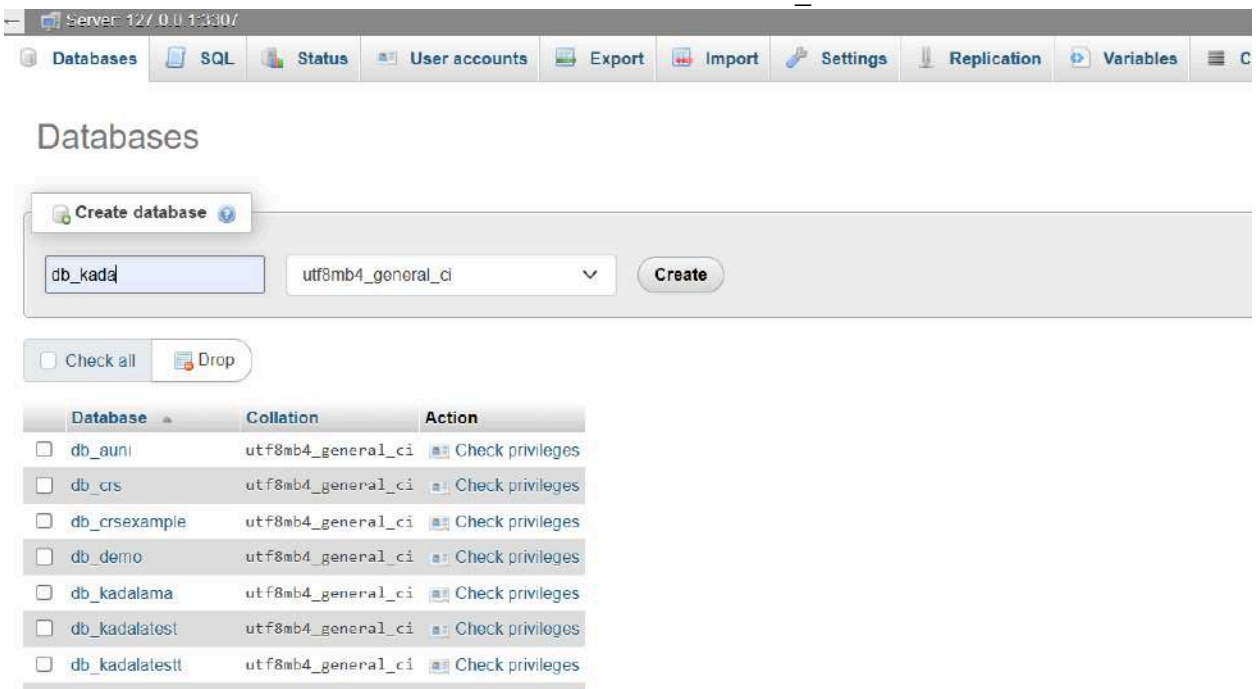
2. Start Apache and MySQL



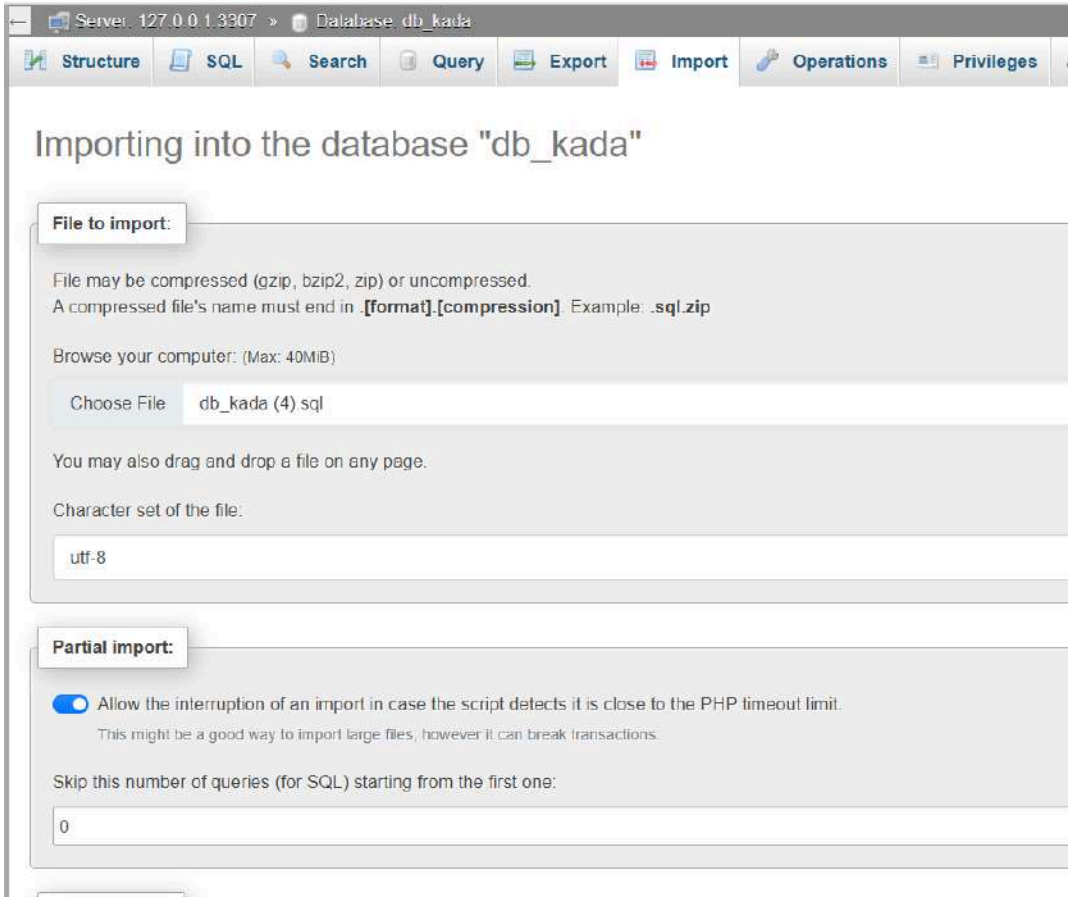
3. Click Admin on MySQL row to navigate to phpMyAdmin



4. Click New and create new database named “db_kada”



5. Click Import and choose database file



9.0 System Interface for All Users

The interface in this document will be organized based on modules:

1. Membership Module

Figure 9.1: Login Page

Figure 9.2: Applicant's Menu

Figure 9.3: Membership Form

Figure 9.4: Family Information

Figure 9.5: Bank Information

Figure 9.6: Saving Information

Figure 9.7: Applicant's Agreement

Figure 9.8: Membership Status

2. Terminate Membership Module

Figure 9.9 : Terminate Application

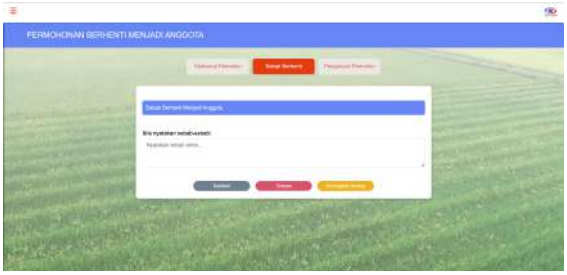


Figure 9.10 : Reason of Termination

Figure 9.11 : Agreement

Figure 9.12 : Terminate Application Status

3. Review Module

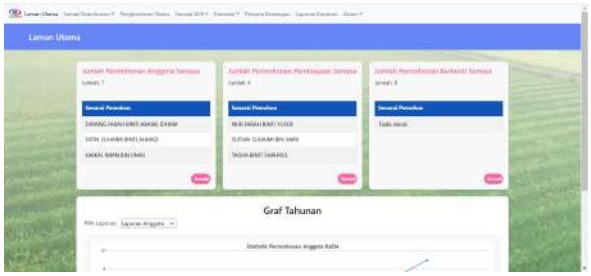


Figure 9.13 : Admin Menu

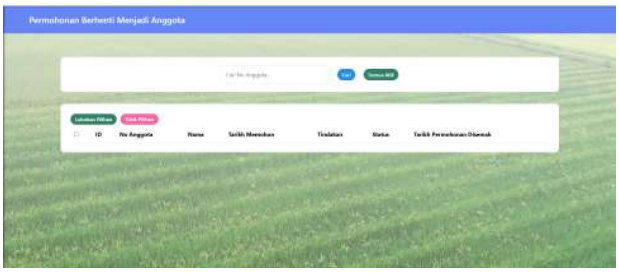


Figure 9.14 : Membership Termination Application Name List

Figure 9.15 : Membership Applicant's Name List

Figure 9.16: Loan Application Name List

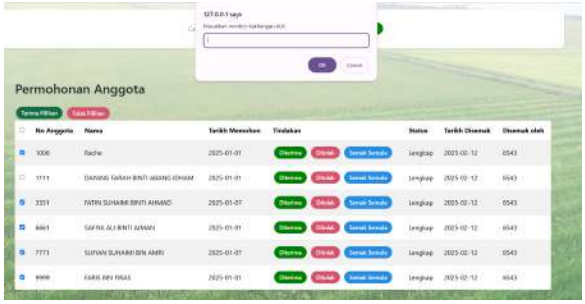


Figure 9.17 : Batch Processing

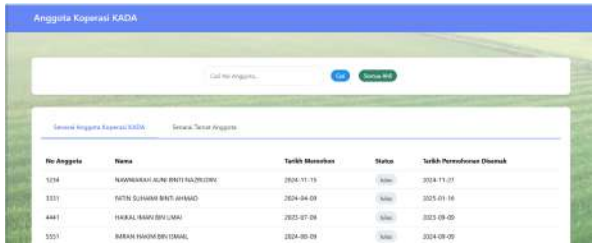


Figure 9.18 : Reviewed applicant's list

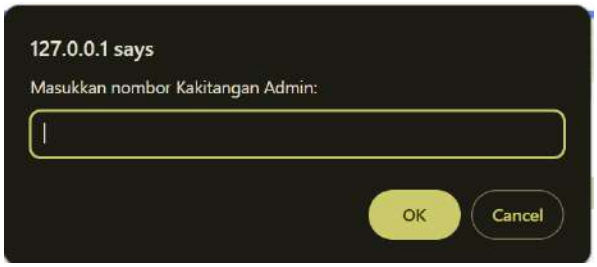


Figure 9.19 : Membership Applicant's Name List

Figure 9.20: Loan Application Name List

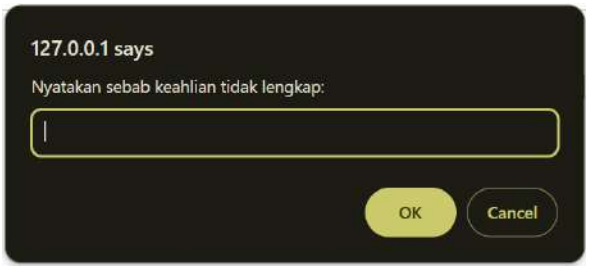


Figure 9.21 : Membership Applicant's Name List

Figure 9.22: Loan Application Name List

4. Loan Application Module

Figure 9.23: Loan Calculator

Figure 9.24: Loan Application Form

Figure 9.25: Applicant's Information

Figure 9.26: Guarantor Information

Figure 9.27: Agreement

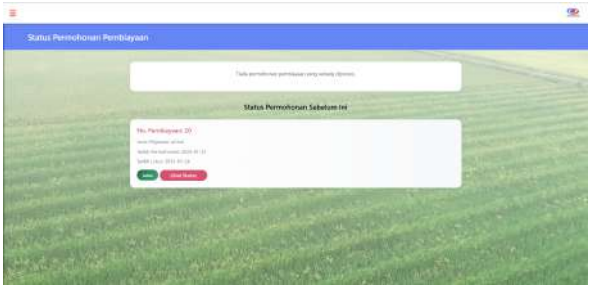


Figure 9.28: Loan Status

5. Approve Application Module

Figure 9.29: ALK' Main Menu

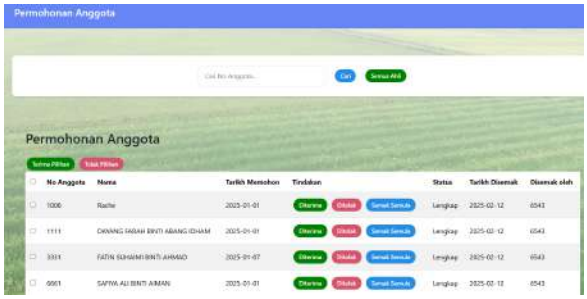


Figure 9.30: Pending Membership List

Figure 9.31: Applicant's Info

Figure 9.32: Button in Applicant's Info

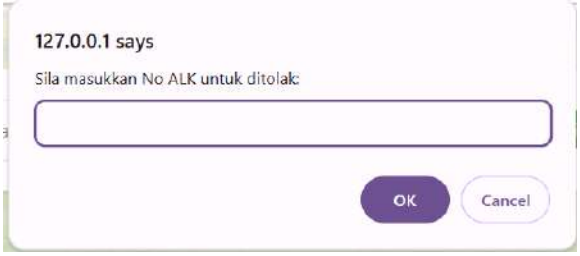


Figure 9.33: Prompts ALK Staff No

Figure 9.34: Prompt Reason

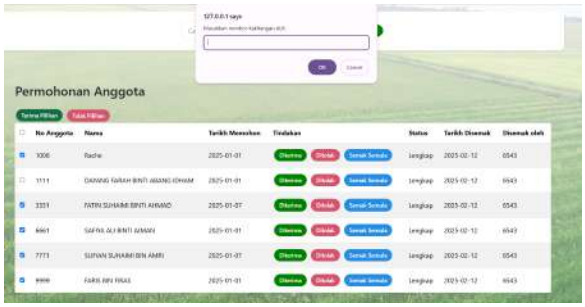


Figure 9.35: Batch Processing

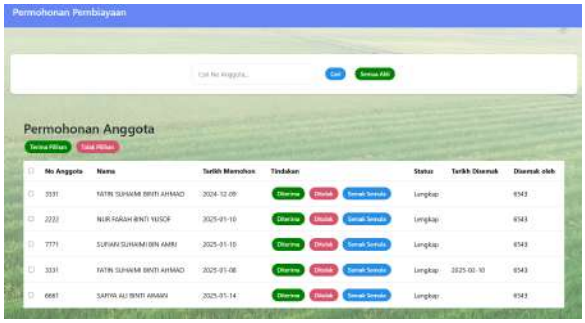


Figure 9.36: Pending Loan List

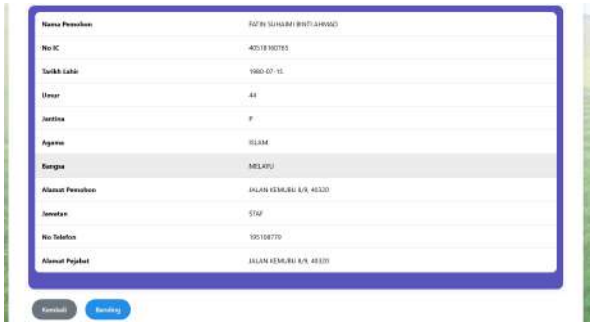


Figure 9.37: Applicant's Info

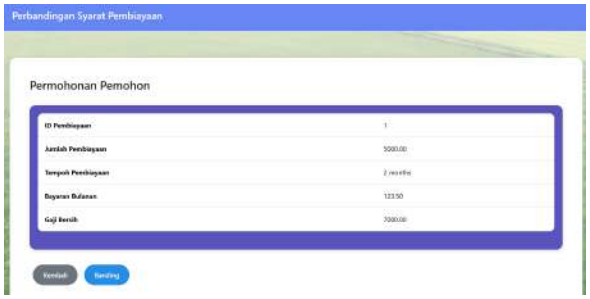


Figure 9.38: Summary Page

Figure 9.39: Results of Comparison

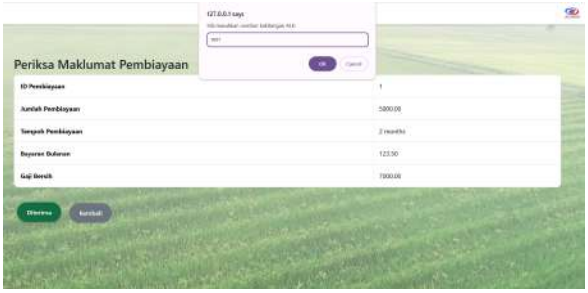


Figure 9.40: Approving

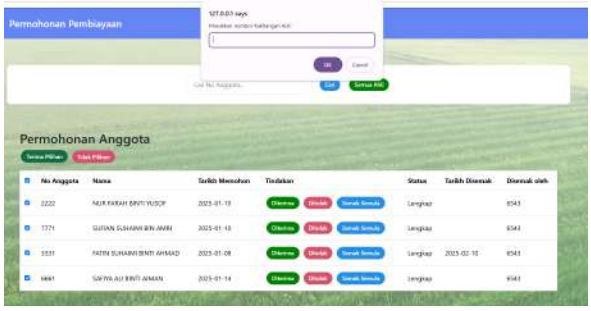


Figure 9.41: Batch Processing

Figure 9.42: Pending Terminate Membership List

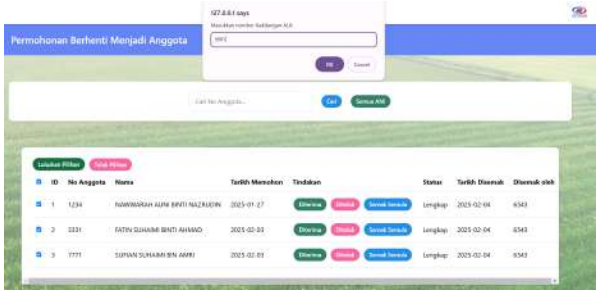


Figure 9.43: Batch Processing

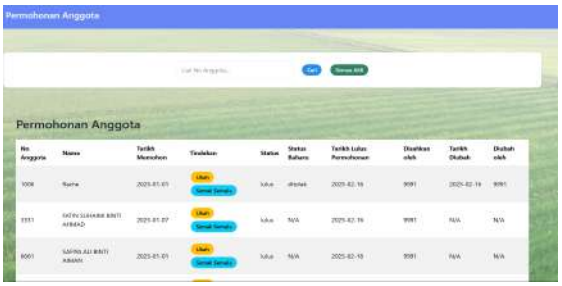


Figure 9.44: List of Past Membership Application

Figure 9.45:Confirmation to Change Status

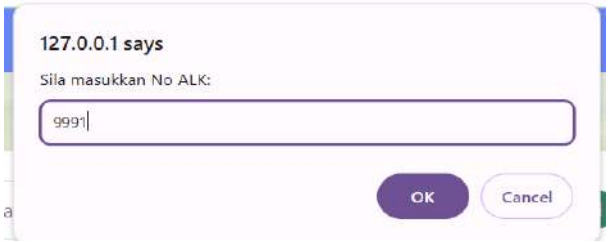


Figure 9.46: Prompt ALK Staff No

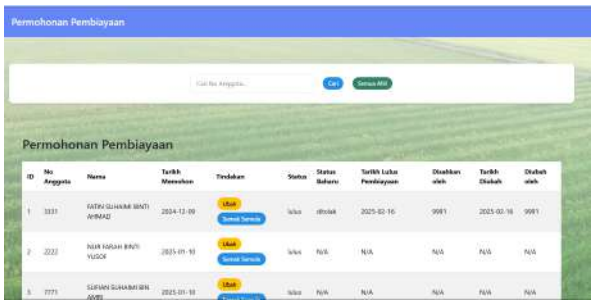


Figure 9.47: List of Past Loan Applications

Figure 9.48 : After Modification

6. Statement Report Production Module

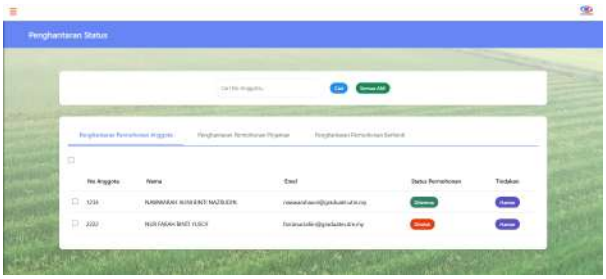


Figure 9.49: Send Status for Membership Application

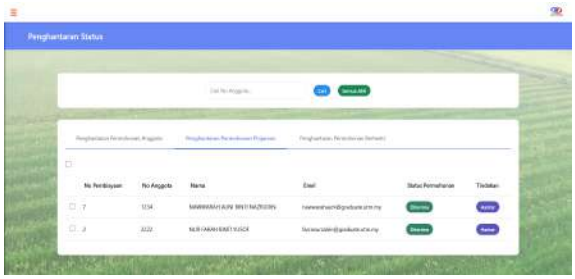


Figure 9.50: Send Status for Loan Application

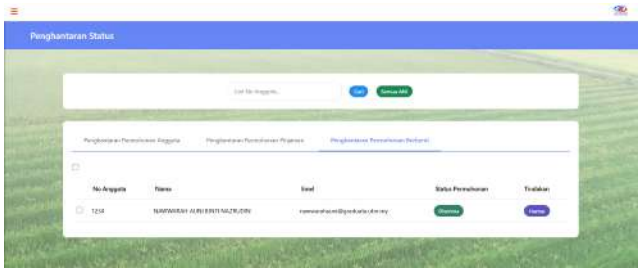


Figure 9.51: Send Status for Terminate Application

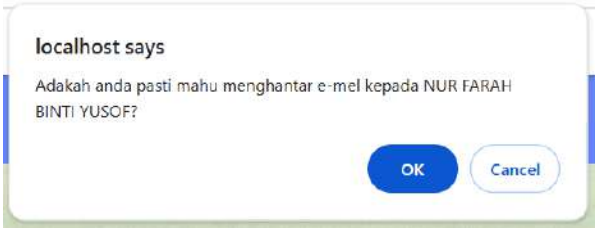


Figure 9.52: Confirmation Message

Figure 9.53: Successful Email Notifcation

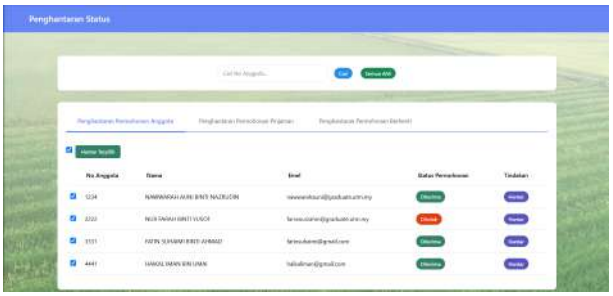


Figure 9.54: Batch Processing

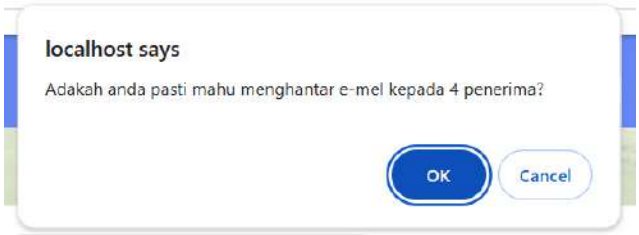


Figure 9.55: Batch Confirmation Message

Figure 9.56: Batch Successful Notification

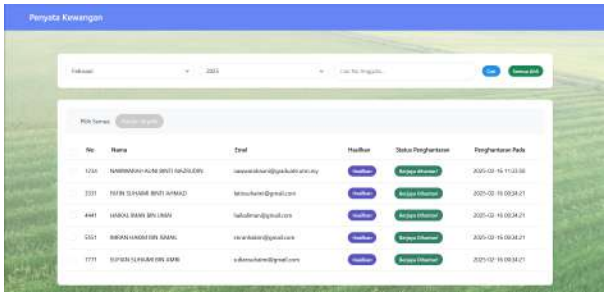


Figure 9.57: Generate Statement Page

Figure 9.58: Generated Statement

Figure 9.59: Sending Generated Statement

Figure 9.60: Successful Email Notification

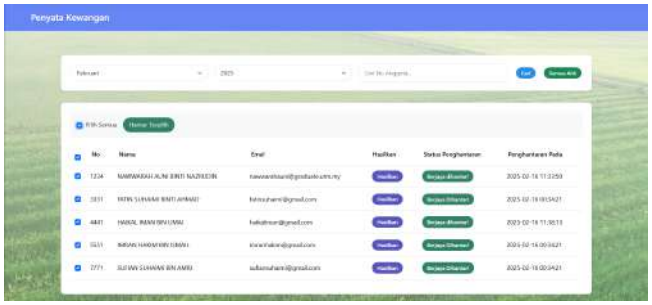


Figure 9.61: Batch Processing Financial Statement

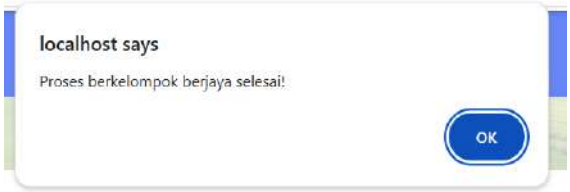


Figure 9.62: Batch Successful Notification

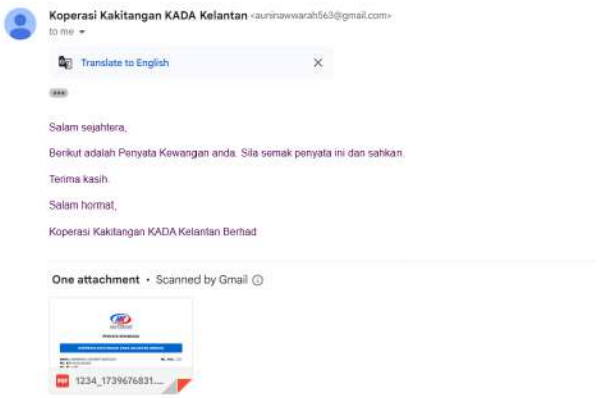


Figure 9.63: Email Notification Received

Figure 9.64: Generated Statement Received

7. Financial Tracking Module

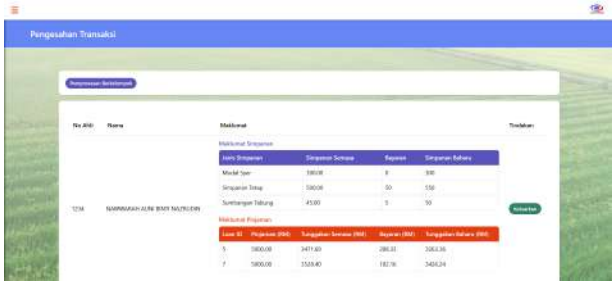


Figure 9.65: Monthly Transaction

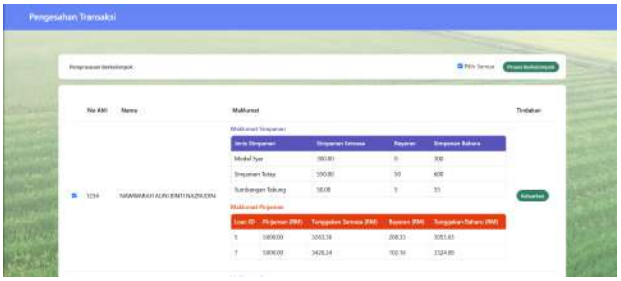


Figure 9.66: Batch Processing

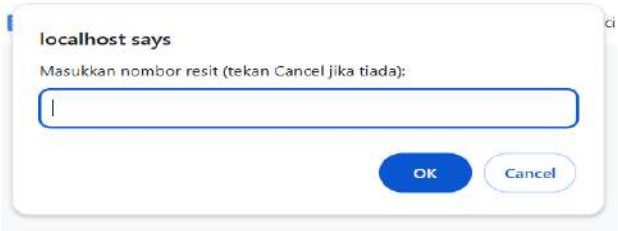


Figure 9.67: Receipt Form

Figure 9.68: Successful Transaction Notification

Figure 9.69: Irregular Saving Transaction Form

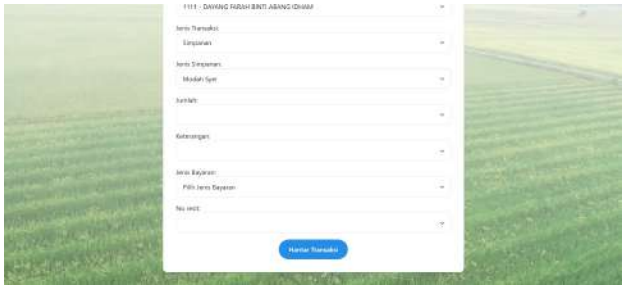


Figure 9.70: Irregular Saving Transaction Form

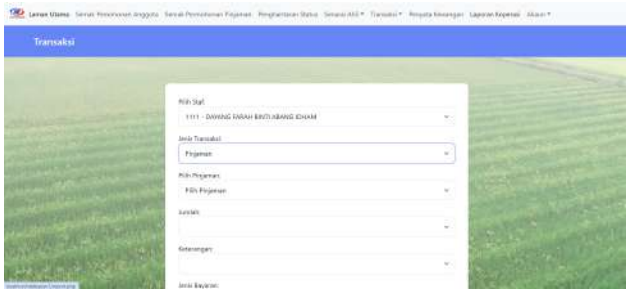


Figure 9.71: Irregular Loan Transaction Form

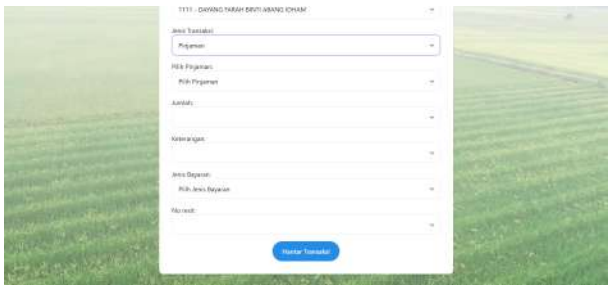


Figure 9.72: Irregular Loan Transaction Form

Figure 9.71: Saving Transaction Record

Figure 9.72: Loan Transaction Record

10.0 Alpha and Beta Testing Reports

Alpha testing is an early phase of software testing conducted internally to identify and resolve critical bugs, usability issues and system inconsistencies before the software is released for external testing. In the context of this system, alpha testing covers three main modules which are **Membership Application**, **Loan Application** and **Terminate Membership Application**. The purpose of this testing phase is to ensure that all functionalities within these modules operate correct;y and meet business and system requirements.

Object 1 : Membership Application

Object Class	Attributes	Equivalence Partition and Input Range	
		Equivalence Classes	Boundary Value
users	email	Valid : Correct Email format: “safiyamasnoor@gmail.com” “farranurzahin@gmail.com” “nawwarahauni@gmail.com”	Minimum: a@gmail.com
		Invalid: Incorrect format: “dayang.com” “auni.gmail.com”	Maximum: Length = 50
	firstName	Valid: “Dayang Farah Farzana” “Nawwarah Auni”	Minimum : Length = 1 “D”
		Invalid: Contain numbers: “123Farra” “safiyasyuhadah2004”	Maximum : Length = 255
	lastName	Valid:	Minimum :

		“Abang Idham” “Nazrudin”	Length = 1 “A”
		Invalid: Contain numbers: “123Farra” “safiyasyuhadah2004”	Maximum : Length = 255
applicant	staffNo	Valid : “1234” “2234”	Minimum : “0001”
		Invalid : “0” “12” “09870”	Maximum : “9999”
	photo	Valid : File Type =	Minimum : 1 photo
		Invalid : File type other than valid file type: “Xxx.php” “xxx.cpp”	Maximum: 1 photo
	applicantName	Valid: “Dayang Farah Farzana binti Abang Idham” “Nawwarah Auni binti Nazrudin”	Minimum : Length = 1 “A”
		Invalid: Contain numbers: “123Farra” “safiyasyuhadah2004”	Maximum : Length = 255
	applicantIC	Valid :	Minimum :

		Correct IC format : “040130010789”	Length = 12
		Invalid : Incorrect IC format : “090807020301937”	Maximum : Length = 12
	applicantPF	Valid : “1234” “2234”	Minimum : “0001”
		Invalid : “0” “12” “09870”	Maximum : “9999”
	applicantGender	Valid : Choose a bullet point ● Lelaki ● Perempuan	Minimum : Choose 1
		Invalid : Not choosing any choices	Maximum : Choose 1
	applicantReligion	Valid : Select choices 1. Islam 2. Kristian 3. Buddha 4. Hindu 5. Lain-lain	Minimum : Choose 1
		Invalid : Not choosing any choices	Maximum : Choose 1
	applicantDOB	Valid : Correct date format: mm/dd/yyy “01/31/2004”	Minimum : Choose one date

		Invalid : Incorrect date format : “56/43/90”	Maximum : Choose one date
	applicantAge	Valid : Positive number “43” “30”	Minimum : “1”
		Invalid : Negative numbers or zero “-1” “0”	Maximum : “99999”
	maritalStatus	Valid : Choose from interface “Bujang” “Berkahwin” “Bercerai”	Minimum : Choose one
		Invalid : Not choosing from the choices	Maximum : Choose one
	applicantRace	Valid : Choose from interface “Melayu” “Cina” “India” “Lain-lain”	Minimum : Choose one
		Invalid : Not choosing from the choices	Maximum : Choose one
	applicantPostcode	Valid : “93050” “81310”	Minimum : “00001”
		Invalid :	Maximum :

		“909090” “898426847”	“99999”
	applicantState	Valid : Choose from interface “Johor” “Kedah” … “Wilayah Persekutuan Putrajaya”	Minimum : Choose one
		Invalid : Not choosing from the choices	Maximum : Choose one
	applicantPosition	Valid : “Staf” “Pengurus”	Minimum : Length = 1
		Invalid : Contain numbers : “123Staf”	Maximum : Length = 50
	applicantGrade	Valid : “GD34”	Minimum : One grade
		Invalid : “12345”	Maximum : One grade
	officePostcode	Valid : “93050” “81310”	Minimum : “00001”
		Invalid : “909090” “898426847”	Maximum : “99999”
	officeState	Valid :	Minimum :

		Choose from interface “Johor” “Kedah” … “Wilayah Persekutuan Putrajaya”	Choose one
		Invalid : Not choosing from the choices	Maximum : Choose one
	officeFax	Valid : “082660654”	Minimum : “000000000”
		Invalid : “090-9099099099”	Maximum : “999999999”
	applicantPhoneNumber	Valid : “01166065432”	Minimum : “0000000000”
		Invalid : “090-9099099099”	Maximum : “999999999999”
	applicantPhoneHome	Valid : “082660654”	Minimum : “000000000”
		Invalid : “090-9099099099”	Maximum : “999999999”
	applicantSalary	Valid : “10000”	Minimum : “1”
		Invalid : “-1” “0”	Maximum : “9999999999.99”
	salaryStatement	Valid :	Minimum :

		File Type : pdf “penyata.pdf”	One file
		Invalid : “Xxx.php” “xxx.cpp”	Maximum : One file
familyinfo	familyIC	Valid : Correct IC format : “040130010789”	Minimum : Length = 12
		Invalid : Incorrect IC format : “090807020301937”	Maximum : Length = 12
	relationship	Valid : “Anak-Beranak” “Adik-Beradik”	Minimum : One relationship
		Invalid : “12345”	Maximum : One relationship
	familyName	Valid: “Dayang Farah Farzana binti Abang Idham” “Nawwarah Auni binti Nazrudin”	Minimum : Length = 1 “A”
		Invalid: Contain numbers: “123Farra” “safiyasyuhadah2004”	Maximum : Length = 255
bankinfo	accountBankNumber	Valid : “16601600167548” “7640578987”	Minimum : “1”
		Invalid :	Maximum :

		“011-090-877”	“9999999999999999999999999”
	bankAccountName	Valid : Choose bank from interface “RHB Bank” “Hong Leong Bank”	Minimum : One bank
		Invalid : Not choosing any choices	Maximum : One bank
	bankStatement	Valid : File Type :	Minimum : One file
		Invalid : “Xxx.php” “xxx.cpp”	Maximum : One file
savingtype	feeMasuk	Valid : “100” “55”	Minimum : “1”
		Invalid : “0” “-1”	Maximum : “9999999999.99”
	modahSyer	Valid : “300” “350”	Minimum : “300”
		Invalid : “0” “-1”	Maximum : “9999999999.99”
	modalYuran	Valid : “100” “55”	Minimum : “1”

		Invalid : “0” “-1”	Maximum : “9999999999.99”
	wangDepositAnggota	Valid : “100” “55”	Minimum : “1”
		Invalid : “0” “-1”	Maximum : “9999999999.99”
	sumbanganTabung	Valid : “100” “55”	Minimum : “1”
		Invalid : “0” “-1”	Maximum : “9999999999.99”
	simpananTetap	Valid : “100” “55”	Minimum : “1”
		Invalid : “0” “-1”	Maximum : “9999999999.99”
	lainLain	Valid : “100” “55”	Minimum : “1”
		Invalid : “0” “-1”	Maximum : “9999999999.99”

Object 2 : Loan Application

Object Class	Attributes	Equivalence Partition and Input Range	
		Equivalence Classes	Boundary Value
loan	loanAmount	Valid : “20000” “10000”	Minimum : “1”
		Invalid “-1” “0”	Maximum : “9999999999.99”
	loanDuration	Valid : “10” “20”	Minimum : “1”
		Invalid “-1” “0”	Maximum : “99”
	monthlyPayment	Valid : “500” “1000”	Minimum : “1”
		Invalid “-1” “0”	Maximum : “9999999999.99”
	bankName	Valid : Choose bank from interface “RHB Bank” “Hong Leong Bank”	Minimum : One bank
		Invalid : Not choosing any choices	Maximum : One bank
	accountBankNumber	Valid :	Minimum :

		“16601600167548” “7640578987”	“1”
		Invalid : “011-090-877”	Maximum : “9999999999999999999999999”
	applicantNetSalary	Valid : “10000” “8000”	Minimum : “1”
		Invalid “-1” “0”	Maximum : “9999999999.99”
	applicantBasicSalary	Valid : “10000” “8000”	Minimum : “1”
		Invalid “-1” “0”	Maximum : “9999999999.99”
	pengesahanMajikan	Valid : File Type : pdf “pengesahan.pdf”	Minimum : One file
		Invalid : “Xxx.php” “xxx.cpp”	Maximum : One file
	loanType	Vallid : Choose from interface “1” “2”	Minimum : Choose one type of loan
		Invalid “-1”	Maximum : Choose one type of

		“0”	loan
guarantorinfo	guarantorIC	Valid : Correct IC format : “040130010789”	Minimum : Length = 12
		Invalid : Incorrect IC format : “090807020301937”	Maximum : Length = 12
	guarantorName	Valid: “Dayang Farah Farzana binti Abang Idham” “Nawwarah Auni binti Nazrudin”	Minimum : Length = 1 “A”
		Invalid: Contain numbers: “123Farra” “safiyasyuhadah2004”	Maximum : Length = 255
	guarantorID	Valid : “1234” “2234”	Minimum : “0001”
		Invalid : “0” “12” “09870”	Maximum : “9999”
	guarantorPF	Valid : “1234” “2234”	Minimum : “0001”
		Invalid : “0” “12” “09870”	Maximum : “9999”

	guarantorPhone Numer	Valid : “01166065432”	Minimum : “0000000000”
		Invalid : “090-9099099099”	Maximum : “999999999999”

Object 3 : Membership termination Application

Object Class	Attributes	Equivalence Partition and Input Range	
		Equivalence Classes	Boundary Value
membershipend	reason	Valid : “Gaji tidak sah”	Minimum : “A”
		Invalid : “1234”	Maximum : Length = 255

In conclusion, alpha testing plays a crucial role in ensuring reliability, functionality and usability of the Membership Application, Loan Application and Terminate Membership Application modules. By identifying and resolving defects early in the development cycle, alpha testing helps improve system stability and ensures that all features meet the required business and technical specifications.Successful completion of alpha testing provides confidence that the system is ready for the next phase, beta testing where real users can further evaluate it performance and usability in real world,

Beta Testing

Beta testing is the final phase of testing before a software system is officially released. Unlike alpha testing, which is conducted internally by developers and testers, beta testing involves real users outside the development team. The primary goal is to evaluate the system’s functionality, usability and performance in a real-world environment with actual

user interactions. This phase helps identify bugs, user experience issues and compatibility problems that may not have been detected during internal testing.

Unfortunately, our group has not yet conducted beta testing due to difficulties in contacting the client. Since beta testing requires real user involvement to evaluate the system's functionality and usability in a real-world environment, direct communication with the client is essential for arranging test sessions and gathering feedback.

During our recent meeting with the client, we were only able to present the current progress of the system rather than proceed with beta testing. After the meeting, there are new things to be consider and changes we made on our system ;

1. Add Terminate Membership Module
   This module aims to allow member to end membership by applying terminate membership application with valid reason
2. Add Financial Tracking Module
3. Minimum Modah Syer is RM300 to be eligible to apply for loan
4. Maximum loan for Al-Bai is RM20000
5. Minimum Modal Yuran is RM35

11.0 UAT Reports

Test Case ID	Test Scenario	Test Steps	Expected Results	Test Data	Actual Result	Pass /Fail	Comments
TC001	Register User	1. Fill in required information such as first name, last name, e-mail, password and position. 2. Click button “Hantar” to register	Information can be entered and the applicant will be redirected to the login page.	First Name : AMINAH Last Name : HADI Email : aminah@gmail.com Password : 12345678 Position : Pemohon	As Expected	Pass	All expected results matched the actual results.
TC002	Login As Applicant	1. Enter email address and password	Credentials can be entered	Email : aminah@gmail.com Password : 12345678	As Expected	Pass	All expected results matched the actual results.
		2. Click button “Log Masuk” to login as applicant	The main page for applicants will be displayed after successful login.		As Expected	Pass	All expected results matched the actual results.
TC003	Apply for Membership	1. Click the “Memohon” under the Borang Keanggotaan section.	The applicant will be redirected to the applicant's information page.		As Expected	Pass	All expected results matched the actual results.
		2. Enter applicant’s information	All information required completed	applicantName : AMINAH applicantIC : 547656 applicantDOB : 2025-01-01 applicantAge : 1 maritalStatus :	As Expected	Pass	All expected results matched the actual results.

				Bujang applicantGender : PEREMPUAN applicantReligion : Islam applicantRace : Melayu applicantStreet : No. 11A Jalan Kantansari 3 applicantCity : Rawang applicantPostcode : 48010 applicantState : Selangor staffNo : 10001 applicantPF : 10001 applicantPosition : MANAGER applicantGrade : S21 officeStreet : No. 11A Jalan Kantansari 3 officeCity : Rawang officePostcode : 48010 officeState : Selangor officeFax: 0360223345 applicantPhoneNumber : 0175969209 applicantPhoneHome : - applicantSalary : 1234.00			
		Click “SIMPAN”	Applicants will be redirected to the family information page.		As Expected	Pass	All expected results matched the actual

							results.
		Enter family information	All information required completed	relationship : SUAMI familyName : ABID familyIC : 8977655	As Expected	Pass	All expected results matched the actual results.
		Click “SIMPAN”	Applicants will be redirected to the bank information page.		As Expected	Pass	All expected results matched the actual results.
		Enter bank information	All information required completed	bankAccountName : OCBC Bank Malaysia accountBankNumber : 786554564	As Expected	Pass	All expected results matched the actual results.
		Click “SIMPAN”	Applicants will be redirected to the saving information page.		As Expected	Pass	All expected results matched the actual results.
		Enter saving information	All information required completed	feeMasuk : 1 modahSyer : 1 modalYuran : 11.00 wangDepositAnggota : 0.00 sumbanganTabung : 0.00 simpananTetap : 0.00 lainLain : 1.00	As Expected	Pass	All expected results matched the actual results.
		Click “SIMPAN”	Applicants will be redirected to		As Expected	Pass	All expected results

			the applicant's statement page				matched the actual results.
		Tick the box if the applicant’s agree with the statement page.	Applicants need to tick off the box.		As Expected	Pass	All expected results matched the actual results.
		3. Press the “Hantar” button to register.	The application status page will be displayed after successfully filling out the membership form.		As Expected	Pass	All expected results matched the actual results.
TC004	Apply for Loan	1. Click the “Memohon” under the Borang Pembiayaan section.	The applicant will be redirected to the loan application page.		As Expected	Pass	All expected results matched the actual results.
		Enter loan information required	All information required completed	loanType= Al-Bai loanAmount= 50000 loanDuration=100 bankName=RHB accountBankNumber=1501105123466544 netSalary=3150 basicSalary=3500 gName1=BUNGA BINTI RASHID gIC1=9008176790 gPF1=9076 gID1=6790 gPhoneNumber1=01907667098656	As Expected	Pass	All expected results matched the actual results.

				gName2=CAMELIA HONG gIC2=911111112234 gPF2=2234 gID2=4322 gPhoneNumber2=01223443221			
		Click “Hantar”	The application status page will be displayed after successfully filling out the loan form.		As Expected	Pass	All expected results matched the actual results.
TC005	Calculate Monthly Estimate Repayment	1. Click the “Kiraan” under the Kalkulator Bayaran Balik section.	The loan calculator page should be open		As Expected	Pass	All expected results matched the actual results.
		Enter Loan Amount& Loan Duration (Years)	Loan monthly repayment calculated and displayed	loanAmount= 1500 loanDuration=-3	As Expected	Pass	All expected results matched the actual results.
		Click Submit	Monthly repayment is calculated		As Expected	Pass	All expected results matched the actual results.
TC006	Apply for Terminate Membership	1. Click the “Memohon” under the Borang Berhenti section.	The terminate application page should be open		As Expected	Pass	All expected results matched the actual

							results.
		Enter terminate information required	All information required completed	Reason: Pencen	As Expected	Pass	All expected results matched the actual results.
		Click “Hantar”	The application status page will be displayed after successfully filling out the terminate membership form.		As Expected	Pass	All expected results matched the actual results.
TC007	View Status of Membership Application	1. Click on Status Permohonan Anggota	The Status Permohonan Anggota page should be displayed		As Expected	Pass	All expected results matched the actual results.
TC008	View Status of Loan Application	1. Click on Status Permohonan Pembiayaan	The Status Permohonan Pembiayaan page should be displayed		As Expected	Pass	All expected results matched the actual results.
TC009	View Status of Terminate Membership Application	1. Click on Status Permohonan Berhenti	The Status Permohonan Berhenti page should be displayed		As Expected	Pass	All expected results matched the actual results.
TC010	View Transaction Record	1. Click on Rekod Pembayaran	The Rekod Pembayaran page should be displayed		As Expected	Pass	All expected results matched

							the actual results.
		Clicks on the loan section	The applicant can see the latest amount of financing that has been paid, loan balance and latest transactions/salary deductions		As Expected	Pass	All expected results matched the actual results.
TC011	Login as Admin	Enter User email & Password	Credential can be entered and Admin dashboard page is displayed	email: aliaabu@gmail.com password: 12345678	As Expected	Pass	All expected results matched the actual results.
TC012	Review Membership List Application	Click Senarai Permohonan Anggota	The page display the list of membership applications		As Expected	Pass	All expected results matched the actual results.
		Click on one of the applicant's name	The page will display all the information entered by the applicant.		As Expected	Pass	All expected results matched the actual results.
		Choose one applicant or more than one	If choose more than one, the checkbox are highlighted		As Expected	Pass	All expected results matched the actual results.
		Click "Lengkap"	Confirmation messages of	adminStaffNo = 6543	As Expected	Pass	All expected

			status Lengkap updated successfully will be displayed				results matched the actual results.
		Click “Tidak Lengkap”	Confirmation messages of status Tidak Lengkap updated successfully will be displayed and an email explaining why the form is incomplete will be sent to the applicant	adminStaffNo = 6543 reason = Slip Gaji tidak sah	As Expected	Pass	All expected results matched the actual results.
TC013	Review Loan List Application	Click Senarai Permohonan Pembiayaan	The page display the list of loan applications		As Expected	Pass	All expected results matched the actual results.
		Click on one of the applicant’s name	The page will display all the information entered by the applicant.		As Expected	Pass	All expected results matched the actual results.
		Choose one applicant or more than one	If choose more than one, the checkbox are highlighted		As Expected	Pass	All expected results matched the actual results.
		Click “Lengkap”	Confirmation messages of status	adminStaffNo = 6543	As Expected	Pass	All expected results

			Lengkap updated successfully will be displayed				matched the actual results.
		Click “Tidak Lengkap”	Confirmation messages of status Tidak Lengkap updated successfully will be displayed and an email explaining why the form is incomplete will be sent to the applicant	adminStaffNo = 6543 reason = Slip Gaji tidak sah	As Expected	Pass	All expected results matched the actual results.
TC014	Review Terminate Membership Application	Click Senarai Permohonan Berhenti	The page display the list of membership termination applications		As Expected	Pass	All expected results matched the actual results.
		Click on one of the applicant’s name	The page will display all the information entered by the applicant.		As Expected	Pass	All expected results matched the actual results.
		Choose one applicant or more than one	If choose more than one, the checkbox are highlighted		As Expected	Pass	All expected results matched the actual results.
		Click “Lengkap”	Confirmation messages of status	adminStaffNo = 6543	As Expected	Pass	All expected results

			Lengkap updated successfully will be displayed				matched the actual results.
		Click “Tidak Lengkap”	Confirmation messages of status Tidak Lengkap updated successfully will be displayed and an email explaining why the form is incomplete will be sent to the applicant	adminStaffNo = 6543 reason = Alasan tidak munasabah	As Expected	Pass	All expected results matched the actual results.
TC015	Send status Membership Application	Click “Hantar” button at desired member or choose “pilih semua” button for batch processing	Confirmation message of successful or failure of email notification.		As Expected	Pass	All expected results matched the actual results.
TC016	Send status Loan Application	Click “Hantar” button at desired member or choose “pilih semua” button for batch processing	Confirmation message of successful or failure of email notification.		As Expected	Pass	All expected results matched the actual results.
TC017	Send status Terminate Membership Application	Click “Hantar” button at desired member or choose “pilih semua” button for batch processing	Confirmation message of successful or failure of email notification.		As Expected	Pass	All expected results matched the actual results.
TC018	Manage Monthly Transaction	Click Transaksi Bulanan on navigation bar	List of koperasi members savings and loans		As Expected	Pass	All expected results matched the

			transactions before and after deductions displayed				actual results.
		Click Keluarkan button at any applicants	Notification of successful transaction		As Expected	Pass	All expected results matched the actual results.
TC019	Manage Irregular Transaction	Click Transaksi Tidak Tetap on navigation bar	Transaction form displayed		As Expected	Pass	All expected results matched the actual results.
		Enter Transaction Form	Completed transaction form displayed	Pilih Staf: 3331 - FATIN SUHAIMI BINTI AHMAD Jenis Transaksi: Simpanan Jenis Simpanan: Simbangan Tabung Jumlah: 30.00 Keterangan: Bayran Lebih Jenis Bayaran: Tunai	As Expected	Pass	All expected results matched the actual results.
		Click Hantar Transaksi	Notification of successful transaction		As Expected	Pass	All expected results matched the actual results.
		Click OK	Notification of successful		As Expected	Pass	All expected

			transaction disappears				results matched the actual results.
TC020	Generate Financial Statement	Click Penyata Kewangan on navigation bar	List of koperasi members displayed		As Expected	Pass	All expected results matched the actual results.
		Click the Hasilkan button in the same row as the desired members	System processes and prepares the statement		As Expected	Pass	All expected results matched the actual results.
		View the generated statement	Financial statement of selected members displayed		As Expected	Pass	All expected results matched the actual results.
		Verify the statement data accuracy	Statement contains correct applicant financial information		As Expected	Pass	All expected results matched the actual results.
		Click Laporan Koperasi in navigation bar	Corporation report displayed		As Expected	Pass	All expected results matched the actual results.
		View the corporation report for monthly and yearly	Report contains correct statistic of		As Expected	Pass	All expected results matched

			membership and loan applications				the actual results.
TC021	Send Financial Statement	Click Penyata Kewangan on navigation bar	List of koperasi members displayed		As Expected	Pass	All expected results matched the actual results.
		Locate prepared statements	Prepared statements is visible		As Expected	Pass	All expected results matched the actual results.
		Click send button	Notification of email berjaya dihantar!		As Expected	Pass	All expected results matched the actual results.
		Click Ok	Status penghantaran and penghantaran pada column is updated		As Expected	Pass	All expected results matched the actual results.
		Click penghantaran Status on navigation bar	List of membership and loan applicants displayed		As Expected	Pass	All expected results matched the actual results.
		Click send button for desired applicants	Notification of successful email notification		As Expected	Pass	All expected results matched the actual

							results.
TC022	Login as ALK	Enter User email & Password	Credential can be entered and ALK dashboard page is displayed	email: aminahkassim@gmail.com password: 12345678	As Expected	Pass	All expected results matched the actual results.
TC023	Approval of Membership Application	Click Senarai Permohonan Anggota	The page display the list of membership applications		As Expected	Pass	All expected results matched the actual results.
		Choose one applicant or more than one	If choose more than one, the checkbox are highlighted		As Expected	Pass	All expected results matched the actual results.
		Click Diterima	Javascript display “Status berjaya dikemaskini!” and the application are removed from the list	alkStaffNo = 9991	As Expected	Pass	All expected results matched the actual results.
		Click Ditolak	Javascript display “Status berjaya dikemaskini!” and the application is removed from the list. An email explaining why the application is rejected will	alkStaffNo = 9991 reason = Slip Gaji tidak sah	As Expected	Pass	All expected results matched the actual results.

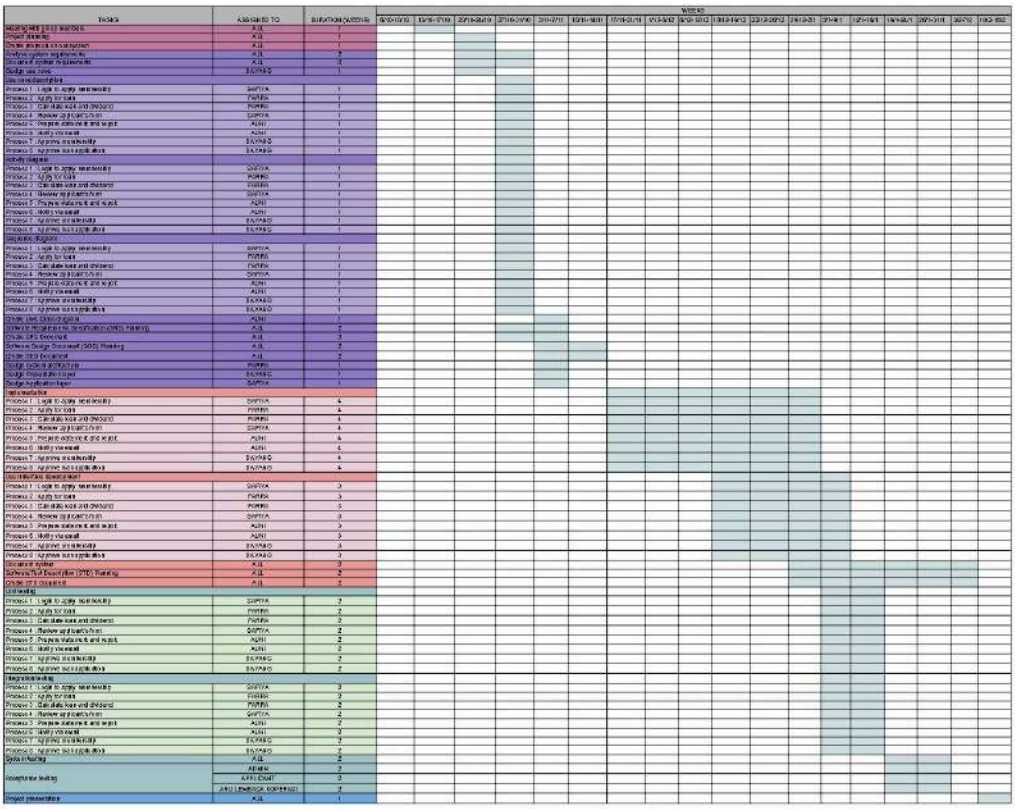
			be sent to the applicant				
		Click Semak Semula	ALK redirected to applicant information page		As Expected	Pass	All expected results matched the actual results.
TC024	Approval of Loan Applicatio n	Choose Senarai Permohonan Pembiayaan	The page display the list of loan applcations		As Expected	Pass	All expected results matched the actual results.
		Click Ditolak	Javascript display “Status berjaya dikemaskini!” and the application is removed from the list. An email explaining why the application is rejected will be sent to the applicant	alkStaffNo = 9991 reason = Slip Gaji tidak sah	As Expected	Pass	All expected results matched the actual results.
		Click Diterima	Javascript display “Status berjaya dikemaskini!” and the application are removed from the list	alkStaffNo = 9991	As Expected	Pass	All expected results matched the actual results.
		Click Semak Semula	The page display applicant		As Expected	Pass	All expected results

			information				matched the actual results.
		Go through the applicant details by clicking every section	The page display applicant information		As Expected	Pass	All expected results matched the actual results.
TC025	Compare Loan Application with Company Policy	Click Banding button	Comparing applicant's information with loan term		As Expected	Pass	All expected results matched the actual results.
TC026	Approval of Terminate Membership	Click Senarai Permohonan Berhenti	The page display the list of membership termination applications		As Expected	Pass	All expected results matched the actual results.
		Choose one applicant or more than one	If choose more than one, the checkbox are highlighted		As Expected	Pass	All expected results matched the actual results.
		Click Diterima	Javascript display "Status berjaya dikemaskini!" and the application are removed from the list	alkStaffNo = 9991	As Expected	Pass	All expected results matched the actual results.
		Click Ditolak	Javascript display "Status	alkStaffNo = 9991 reason = Belum	As Expected	Pass	All expected results

			berjaya dikemaskini!” and the application is removed from the list. An email explaining why the application is rejected will be sent to the applicant	boleh nyahaktif			matched the actual results.
		Click Semak Semula	ALK redirected to applicant information page		As Expected	Pass	All expected results matched the actual results.
TC027	Membership status Modification	Click Sejarah Permohonan Anggota	The page display the list of past applications for membership		As Expected	Pass	All expected results matched the actual results.
		Choose one applicant and click Ubah	Javascript display “Status berjaya dikemaskini!” and status will change	alkStaffNo = 9991 Reason = Salah isi	As Expected	Pass	All expected results matched the actual results.
TC028	Loan status Modification	Click Sejarah Permohonan Pembiayaan	The page display the list of past applications for loan		As Expected	Pass	All expected results matched the actual results.
		Choose one applicant and click Ubah	Javascript display “Status berjaya	alkStaffNo = 9991 Reason = Salah isi	As Expected	Pass	All expected results matched

			dikemaskini!” and status will change				the actual results.

12.0 Gantt Chart



Link of Gantt Chart : [GANTT CHART PROJECT WBL]

13.0 SDD

1. Architectural Rationale

An architectural style refers to a set of design principles and patterns that define the structure and organization of a software system. It provides a standardized way to design system components, their relationships, and communication methods. For Koperasi KADA's automated system, Layered Architecture is used.

1.1 Architecture Style and Rationale

Choosing the right system design is important for making sure a software works well and is easy to manage. The Koperasi Kada Automated System needs a reliable and organized structure to handle membership, loan applications, and statement generation. To achieve this, the system follows a Layered Architecture, which divides it into three main parts which is View Layer, Domain Layer, and Data Access Layer.

The View Layer is the part that users interact with. It displays information and allows users to enter data, such as applying for loans or checking membership details. The Domain Layer is responsible for processing data and applying the system's rules, like approving a loan based on certain conditions. The Data Access Layer connects to the database, making sure that information is stored and retrieved correctly. Separating these layers makes the system more organized and easier to update, since changes in one layer do not affect the others.

Using a Layered Architecture makes the system easier to maintain, test, and expand. For example, if the design of the View Layer needs improvement, it can be updated without affecting the business logic or database. This also helps with security, since the Data Access Layer ensures proper handling of sensitive information. Additionally, this structure allows developers to work on different parts of the system at the same time without causing problems. By following this approach, the Koperasi Kada Automated System can efficiently replace the manual system with a faster and more reliable solution. Figure 1.1 illustrated the Layered Architecture Diagram for Koperasi KADA Automated System.

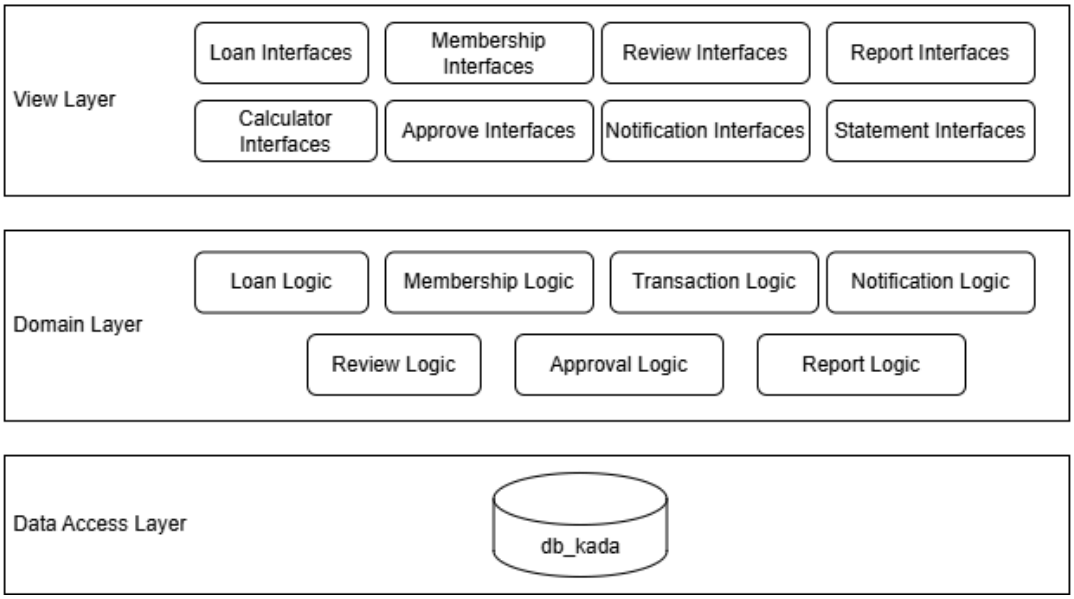


Figure 1.1: Layered Architecture Diagram for Koperasi KADA Automated System

2. Architectural Views

The 4+1 Architectural View Model is an architectural framework that organizes a system into five different views: Logical, Development, Process, Physical and Scenarios (Use Case View). The Logical View represents the system's functionality such as reviewing and approving applications and object-oriented design, defining key classes and relationships. The Development View focuses on the organization of KADA Automated System software modules, which are Membership Application Module, Terminate Membership Module, Review Module, Loan Application Module, Approve Application Module, Statement Report Production Module and Financial Tracking Module, including components and overall code structure.The Process View handles dynamic aspects of the system such as the synchronization mechanisms. The Physical View represents how software components are deployed across hardware nodes, addressing performance and scalability. Lastly, the Scenarios serve as a bridge among all the views, describing the system's functionality from the user's perspective. Figure 2.1 illustrates the 4+1 Architectural View Model.

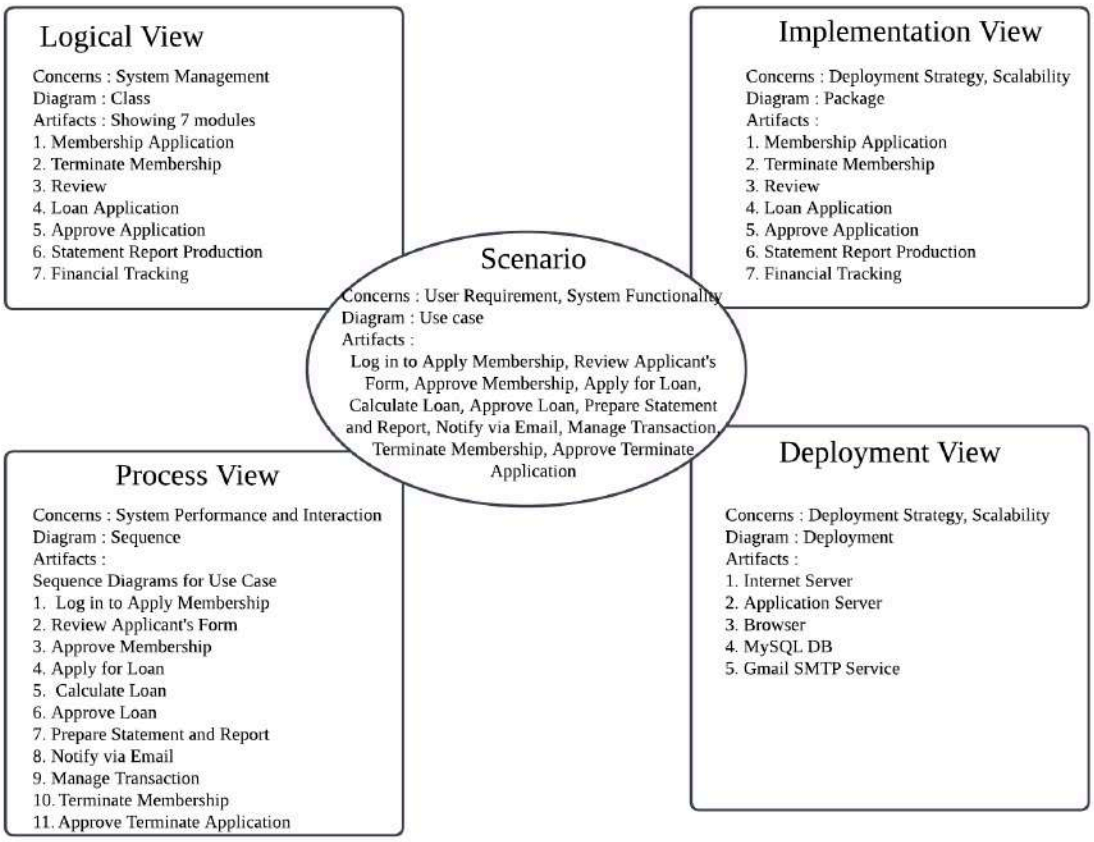


Figure 2.1: 4+1 Architectural View Model

Table 2.1 shows the diagram used for each view in KADA Automated System. Use Case is used in Scenario, Class Diagram is used in Logical View, Sequence Diagram is used in Process View, Package Diagram is used in Implementation View and Deployment Diagram is used in Deployment View. Further descriptions are described in the table below.

Table 2.1: Type of Diagram used

Type of View	Type of Diagram	Description
Scenario	Use Case	shows the interaction between Applicant, Admin, ALK and the system
Logical	Class	represent the structure of the system, focusing on business logic of membership and loan and data relationships

Process	Sequence	shows the flow of operations between different components of KADA Automated System during specific processes
Implementation	Package	represents the organization of system components and how they relate to the storage and management of data
Deployment	Deployment	shows the physical distribution of system components that includes data servers and storage systems of KADA Automated System

2.1 Scenario (Use Case View) - Use Case

This view acts as a connecting thread across the other three views, maintaining consistency and completeness. It shows the functional requirements and connects the views through real-world scenarios or use cases. These use cases serve as a link between the system's conceptual design and technical execution, connecting various components including data models, processes and user interfaces. FInally, this viewpoint contributes to a Broad view by ensuring that all system components work together effortlessly to achieve the desired results while meeting the specific requirements. Figure 2.2 illustrates a use case diagram with its module for Koperasi KADA's Automated System.

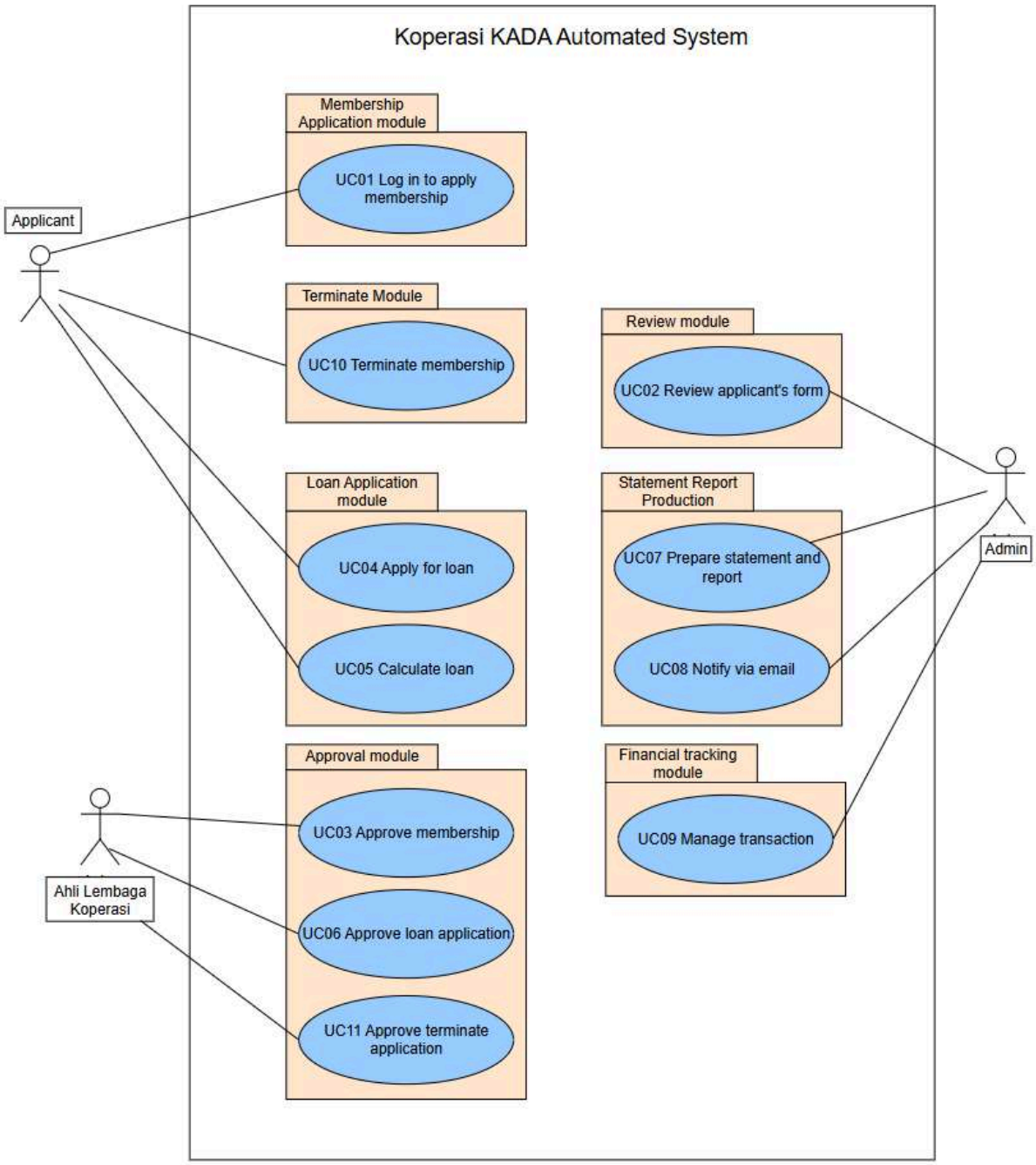


Figure 2.2: Use Case Diagram for Koperasi KADA Automated System

2.2 Implementation View - Package Diagram

Figure 2.3 is the architectural view which highlights the system's software modules that break down into high-level subsystems and their functions which arrange within its development setting. The system includes four main subsystems which are Membership Application Module, Loan Application Module, Terminate Module, Review Module, Approval Module and Statement

Report Production Module.

The Membership Application Module handles user authentication and processes membership applications. The Loan Application Module allows members to apply for loans, calculates monthly repayments loan amount and stores approved loans in a Loan List. Moreover, the Terminate Module allows users to terminate their membership of Koperasi KADA. The Review Module is in charge of going through applications from the Membership List and Loan List, making sure they are properly assessed before being accepted or rejected. The Approval Module receives the reviewed applications after which it makes final choices by accepting or rejecting them and updating the corresponding lists. Finally, data from other subsystems is used by the Statement Report Production Module to create reports and send email notifications.

These subsystems work together seamlessly to provide the intended functionality. To guarantee that only authorised members are able to apply for loans, access to the Loan Application Module and Terminate Module requires the Membership List generated by the Membership Application Module. The Review Module processes the applications for the Membership and Loan lists ensuring a thorough validation before forwarding to the Approval Module. By combining data from every subsystem, the Statement Report Production Module creates thorough reports and alerts relevant users.

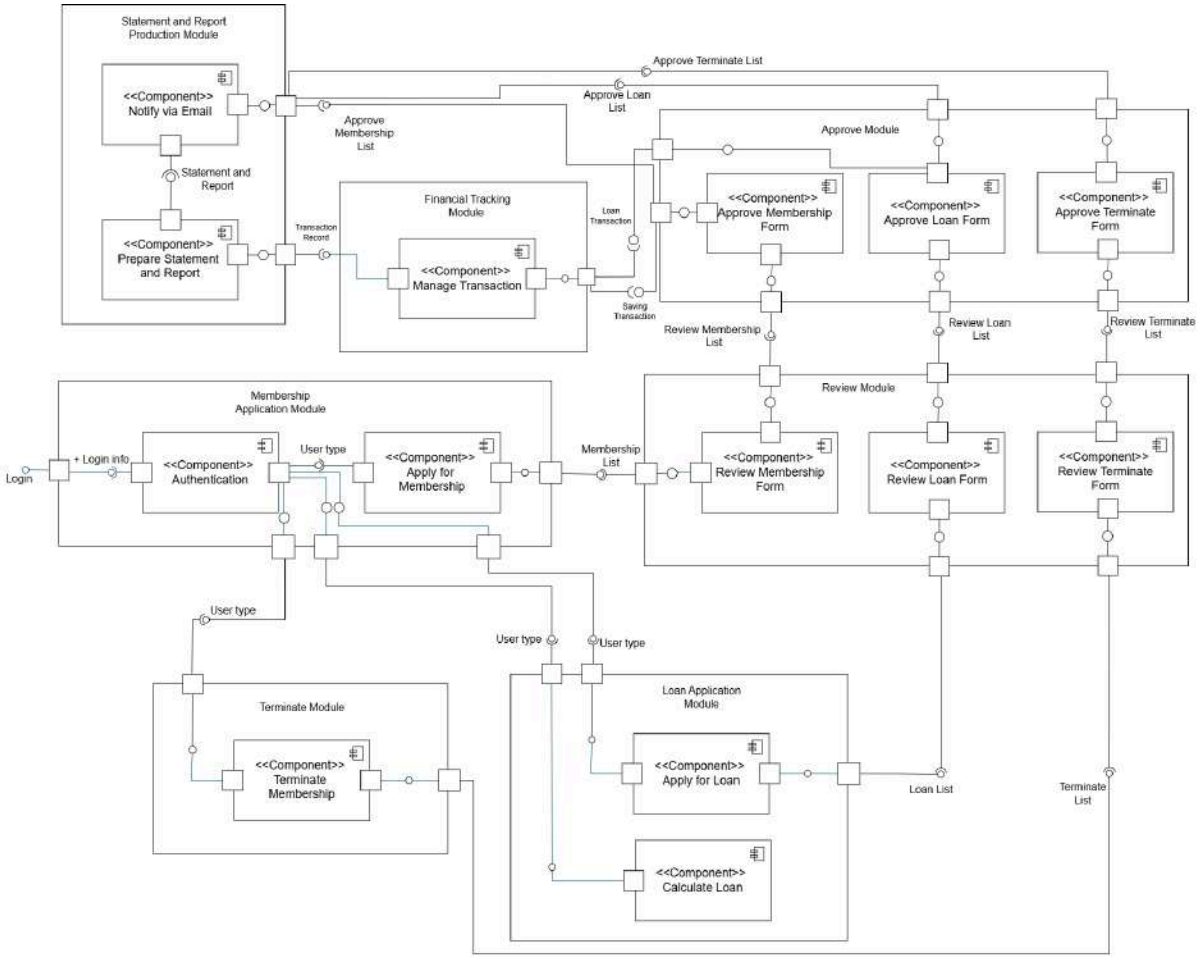


Figure 2.3: Architecture Diagram of Koperasi KADA Automated System

Figure 2.4 shows the subsystems in Koperasi KADA Automated System in Package Diagram form. The User Interface Framework handles user interaction and connects to the other subsystems like the Membership Module, Loan Application Module, Terminate Module, Review Module and Approval Module to process and manage data. The Statement Report Production Module generates reports, which are then sent back to the User Interface Framework for users to access.

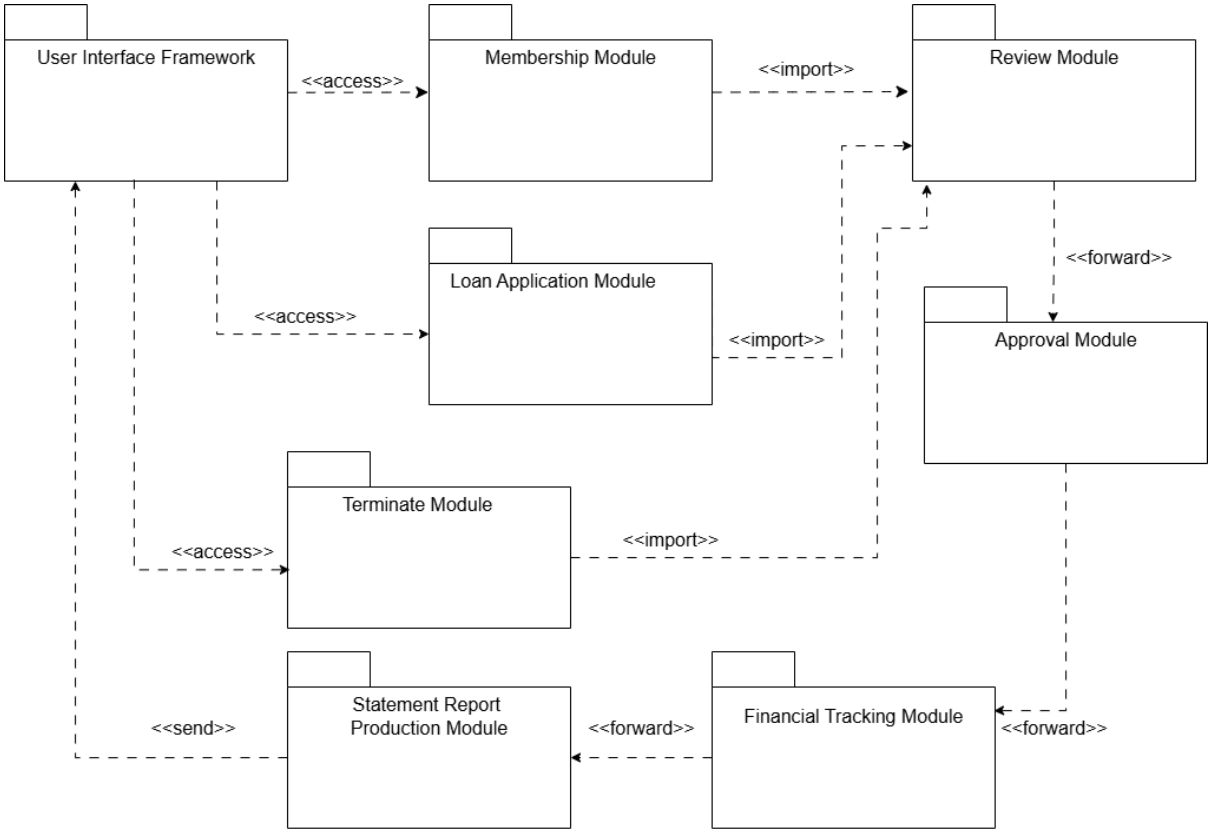


Figure 2.4: Package Diagram for Koperasi KADA Automated System

In Figure 2.5 is the Package Diagram for the Membership Application Module. The View Layer consists of user interfaces such as AuthenticationInterfaces and UserInterface and MembershipStatus Interface, which enables users to interact with the system by logging in, applying for membership and viewing the status of membership application. The Domain Layer includes components such as Applicant, Membership and FamilyInfo, which handle core business logic like processing applicant details and managing membership applications. Lastly, the Data Access Layer manages database interactions through components like MembershipData and familyInfoData, ensuring efficient retrieval and management of membership application data.

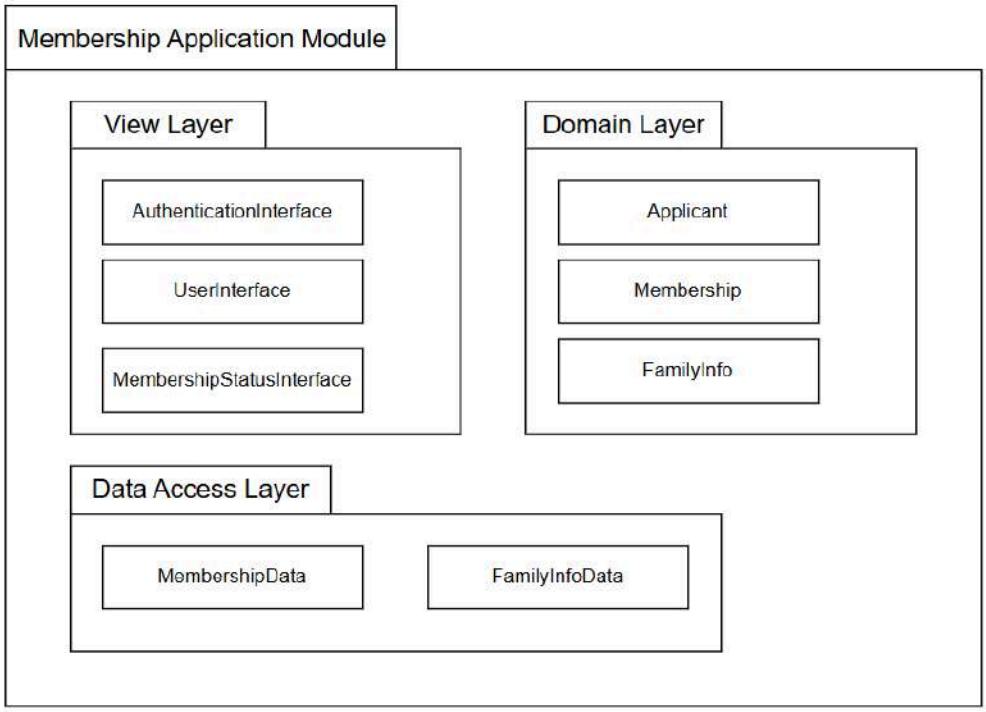


Figure 2.5: P001 Package Diagram for Membership Subsystem

In Figure 2.6 is the Package Diagram for the Review Module. The View Layer features interfaces such as AdministratorInterface, MembershipStatusInterface, Loan StatusInterface and TerminateStatusInterface, enabling administrators to interact with the system to review membership application, loan application, terminate membership application and update the status for all the applications. The Domain Layer consists of components such as Loan, GuarantorInfo, Membership, FamilyInfo, and ApplicationManagement which handle core business logic for loan, membership, and membership termination application reviewing process. Lastly, the Data Access Layer manages database interactions through components like Loan Data, GuarantorInfoData, MembershipData, familyInfoData, and TerminateMembershipData ensuring efficient retrieval and management of application data.

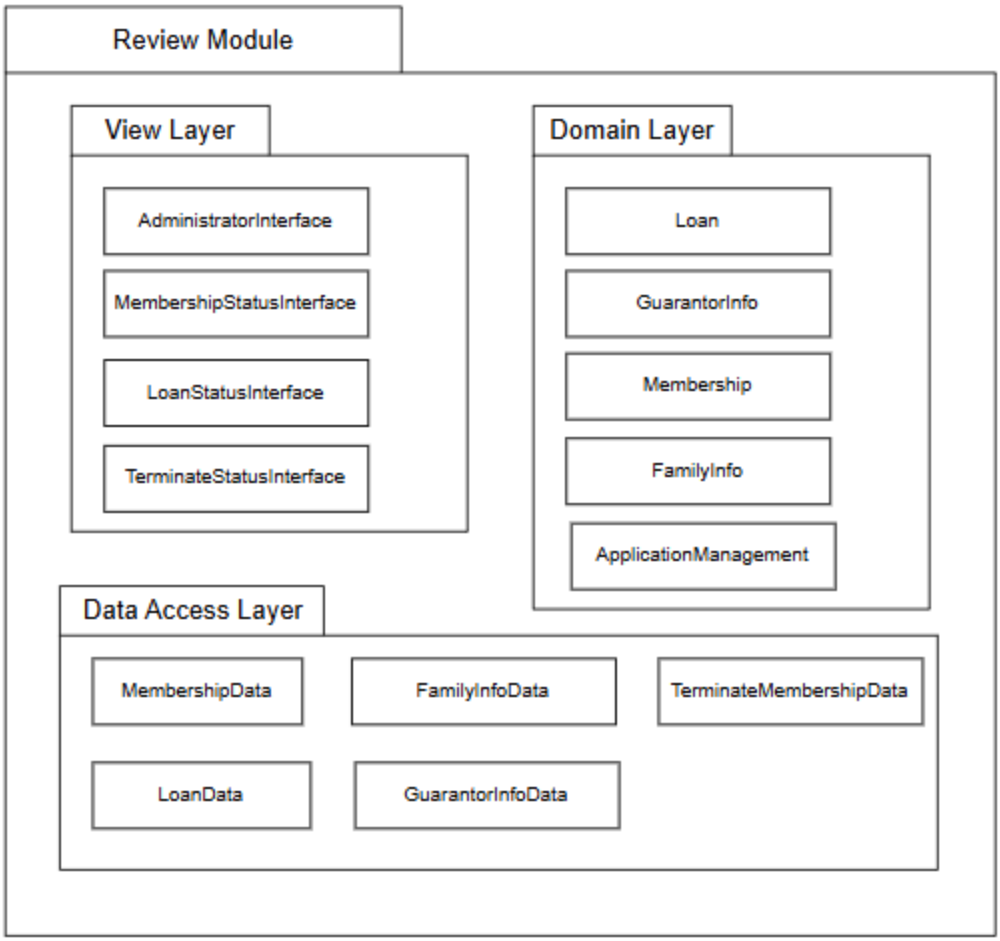


Figure 2.6: P002 Package Diagram for Review Subsystem

In Figure 2.7 is the Package Diagram for Loan Application Module. The view layer has user interfaces such as LoanApplicationInterfaces, LoanStatusInterface and LoanCalculatorInterface, which enables users to interact with the system by applying loans, checking loan status and calculating monthly loan repayment amount. Finally, the Data Access Layer coordinates database interactions within components such as LoanData, LoanStatusData and ApplicantData. This layer assures data storage and retrieval for loan and applicant information.

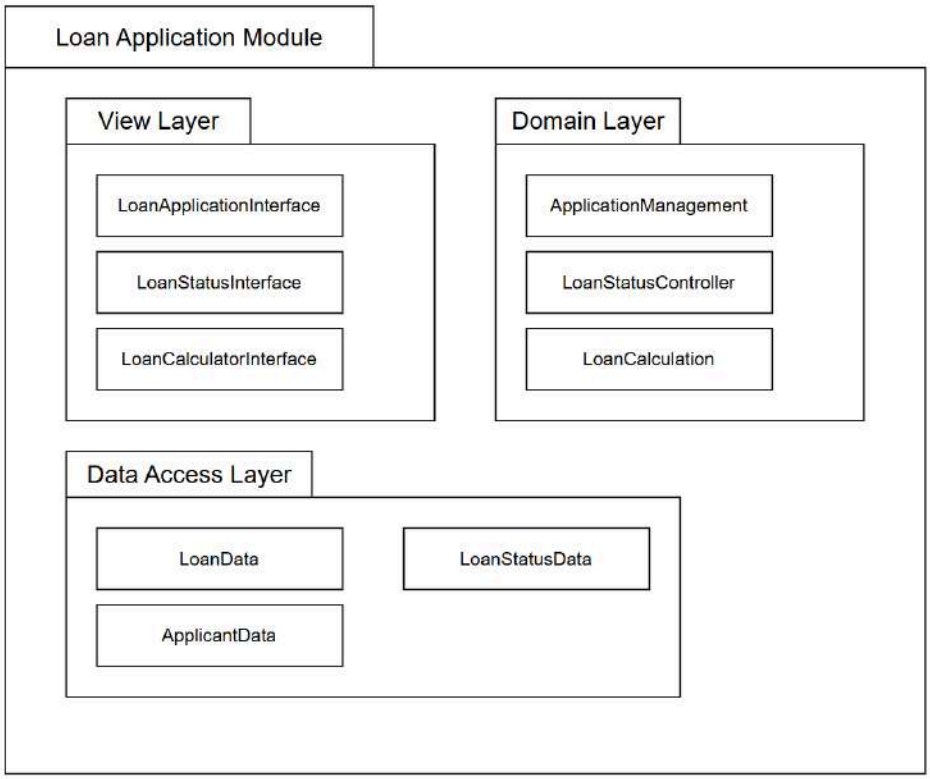


Figure 2.7: P003 Package Diagram for Loan Application Subsystem

Figure 2.8 shows the Package Diagram for Approval subsystem. The view layer represents the view component that interacts with the user. The layer includes ListMembershipUI to display the pending list of membership approval, ListLoanUI to display the pending list of loan approval, MembershipApprovalUI to interact in performing membership approval, LoanApprovalUI to interact in performing loan approval, historyMembershipUI to display past membership applications and historyLoanUI to display past loan applications. Then, the Domain Layer contains the business logic for data processing which includes MembershipApprovalController, LoanApprovalController, LoanComparisonProcessor and ModificationController. All of these are involved in approving or rejecting membership and loan applications. Lastly, the Data Access Layer is used to handle data operations. This layer includes MembershipDataRecord, LoanDataRecord and TerminateMembershipData.

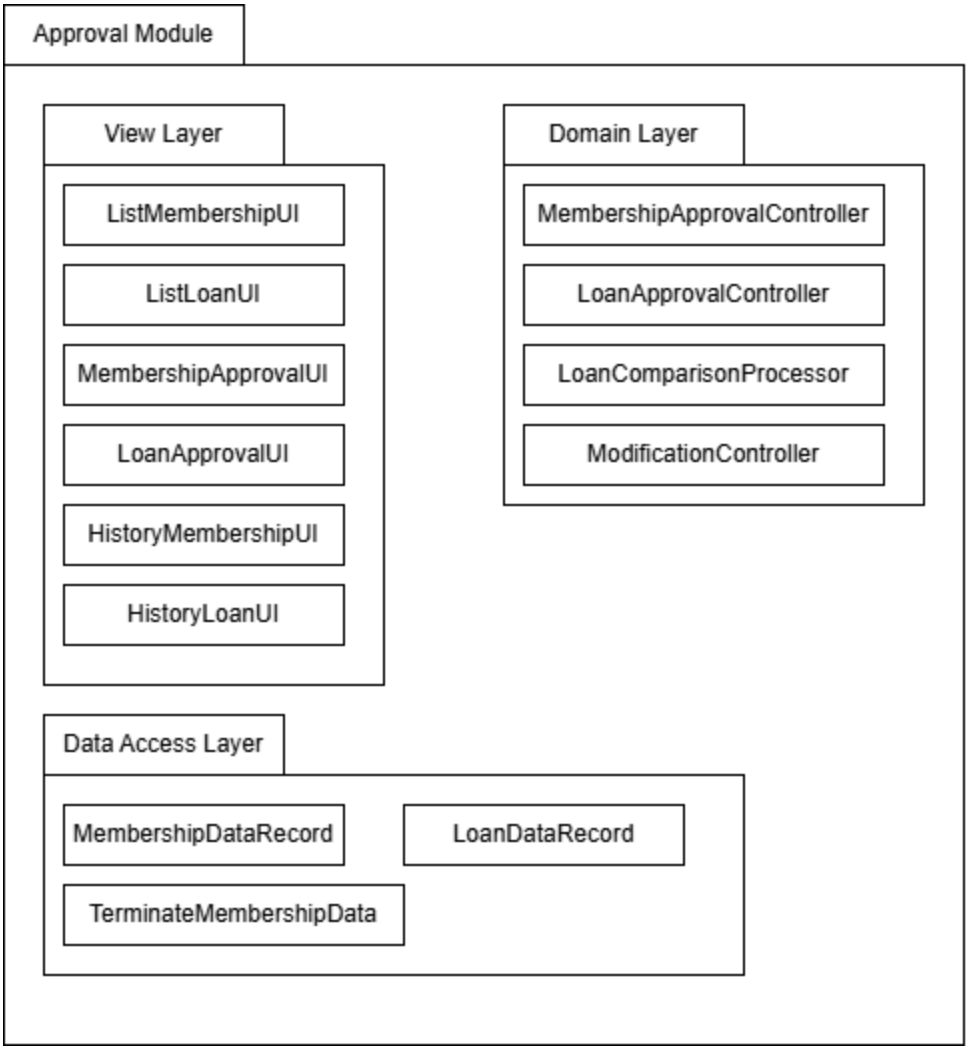


Figure 2.8: P004 Package Diagram for Approval Subsystem

Figure 2.9 illustrates the Package Diagram for the Statement Report Production subsystem. The view layer represents the components that interact with the user, including GenerateReportInterface, which provides a button for the admin to generate the statement report, NotificationInterface, which includes a button for sending email notifications, and TransactionListInterface, which displays the list of transactions related to loans and memberships. The Domain Layer contains the business logic responsible for data processing, including GenerateStatementController, which manages the process of generating the report, NotificationController, which handles the creation and sending of email notifications, and ExportController, which manages the exporting of the report into various formats like PDF or Excel. Finally, the Data Access Layer manages data operations and includes

LoanTransactionData, LoanType, SavingTransactionData, and MembershipData, which handle the retrieval and storage of financial transaction information.

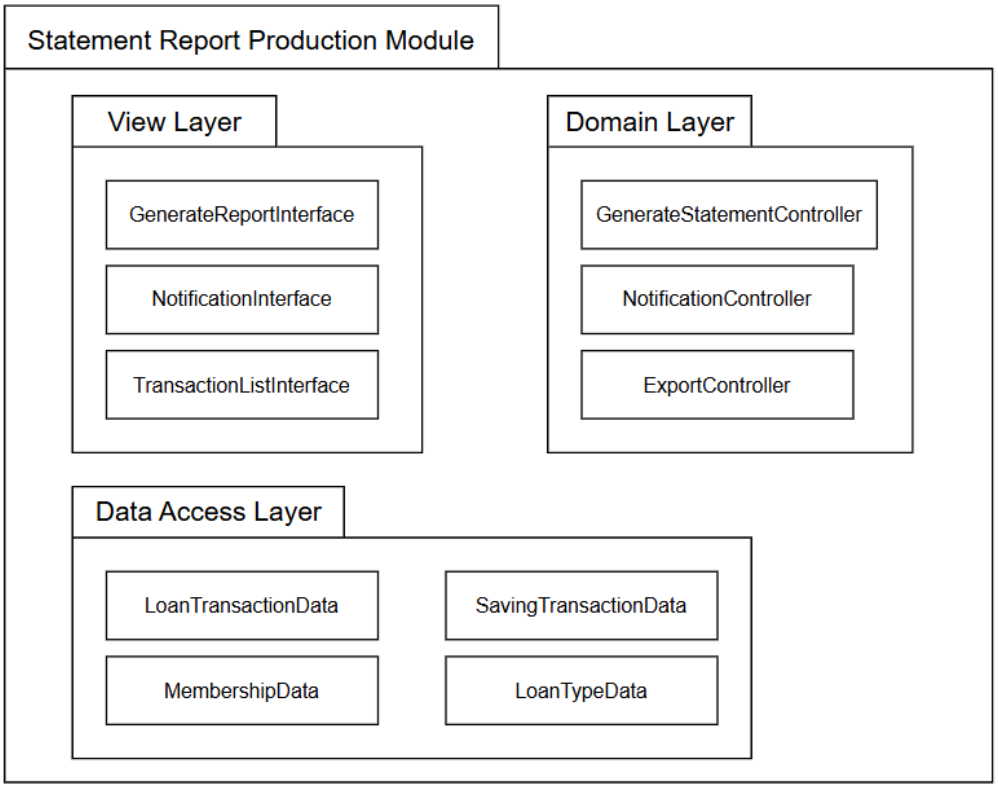


Figure 2.9: P005 Package Diagram for Statement Report Production Subsystem

Figure 2.10 is the Terminate Module packet diagram that organizes the use case into three layers. The View Layer contains interfaces for submitting termination requests and viewing their status. The Domain Layer handles the business logic including termination management and status updates. The Data Access Layer manages data related to members, termination requests and their statuses to ensure seamless interaction across the module.

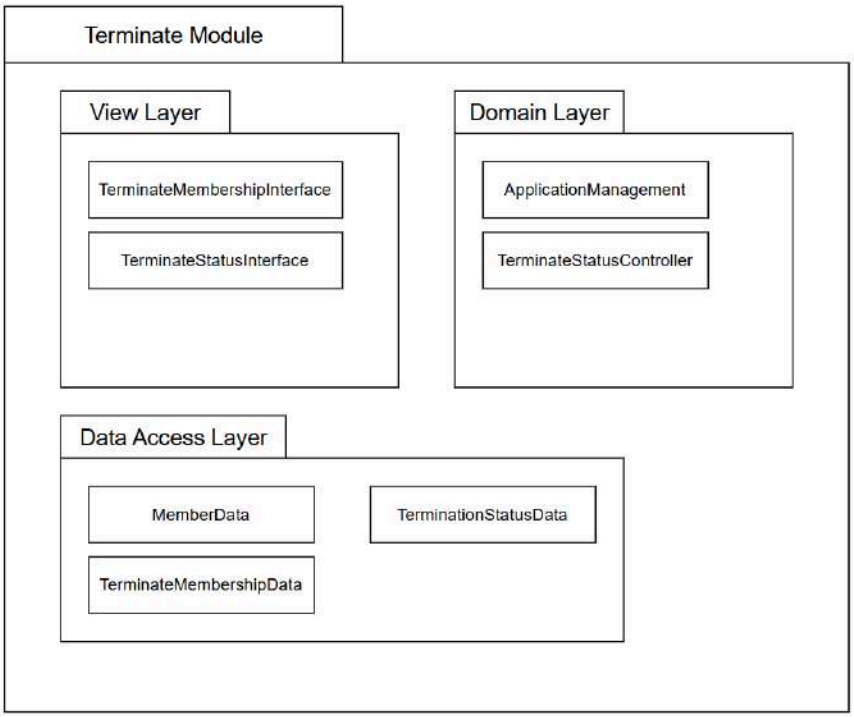


Figure 2.10: P006 Package Diagram for Terminate Module Subsystem

Figure 2.11 presents the Financial Tracking Module packet diagram, which structures the use case into three layers. The View Layer includes interfaces for monthly transactions and non-fixed transactions. The Domain Layer manages business logic, handling the processes for both types of transactions. The Data Access Layer is responsible for managing data related to savings and loan repayments to ensure accurate report statements.

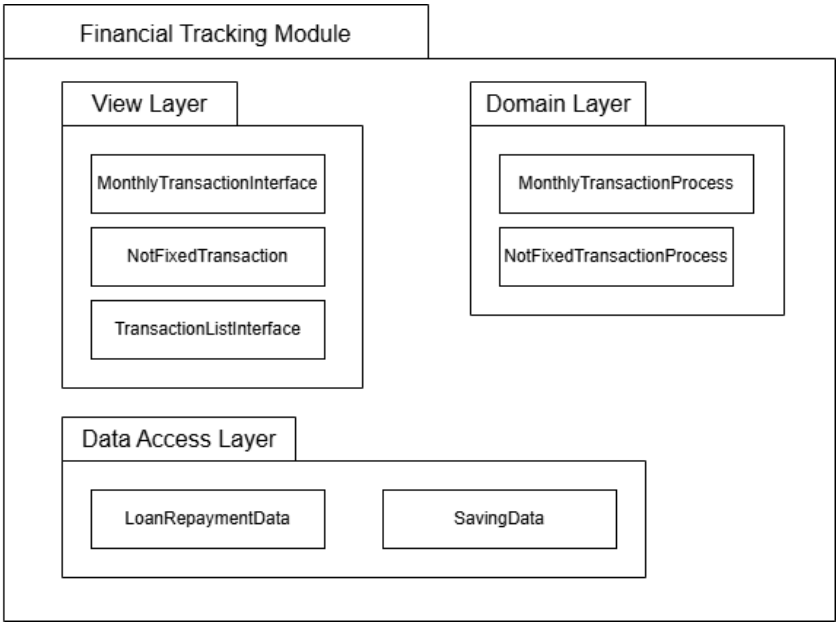


Figure 2.11: P007 Package Diagram for Financial Tracking Module Subsystem

2.3 Logical View - Class Diagram

The logical view provides a perspective on the system's structure from the developer or designer's point of view. It highlights the major abstractions in the system, which in an object-oriented design approach would normally be modeled as objects or object classes. This view is represented through class diagrams, which shows the structure of the system by showing the relationships among classes, along with their attributes and methods.

Figure 2.12 class diagram shows the class diagram for Login to Apply Membership . This Authentication class has a one-to-one relationship with the User class, ensuring each user has unique login credentials. The Applicant class has an association relationship with the User class. The Applicant class has an aggregation relationship with the Membership class, meaning membership details can exist independently of the Applicant. Then, the Membership class has a composition relationship with the familyInfo class, meaning the family information is an essential part of the membership and cannot exist separately.

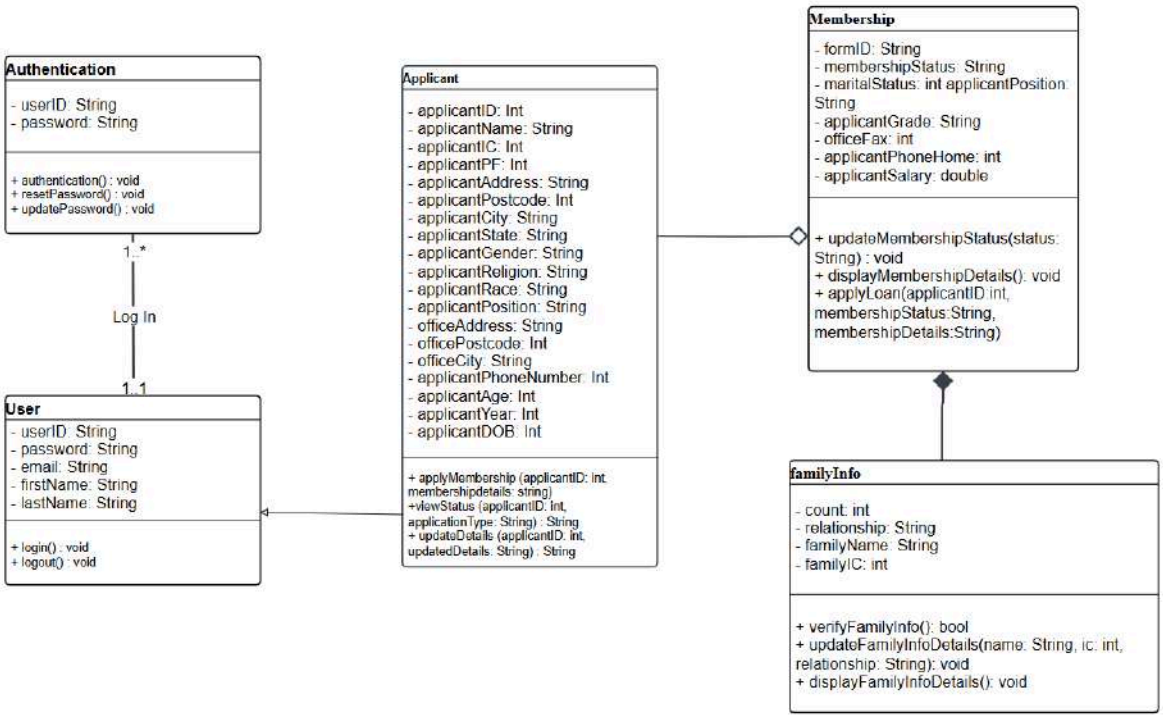


Figure 2.12: Class diagram for Login to Apply Membership Subsystem

Figure 2.13 class diagram shows the class diagram for Review Applicant's Form. The Administrator class has a one-to-many relationship with both Loan and Membership classes, meaning one administrator can review multiple loans and membership. The loan class has a composition relationship with the GuarantorInfo class, indicating that a guarantor is tightly bound to the loan and cannot exist independently. Similarly, the Membership class has a composition relationship with the familyInfo and saving class, meaning the family information is an essential part of the membership and cannot exist separately.

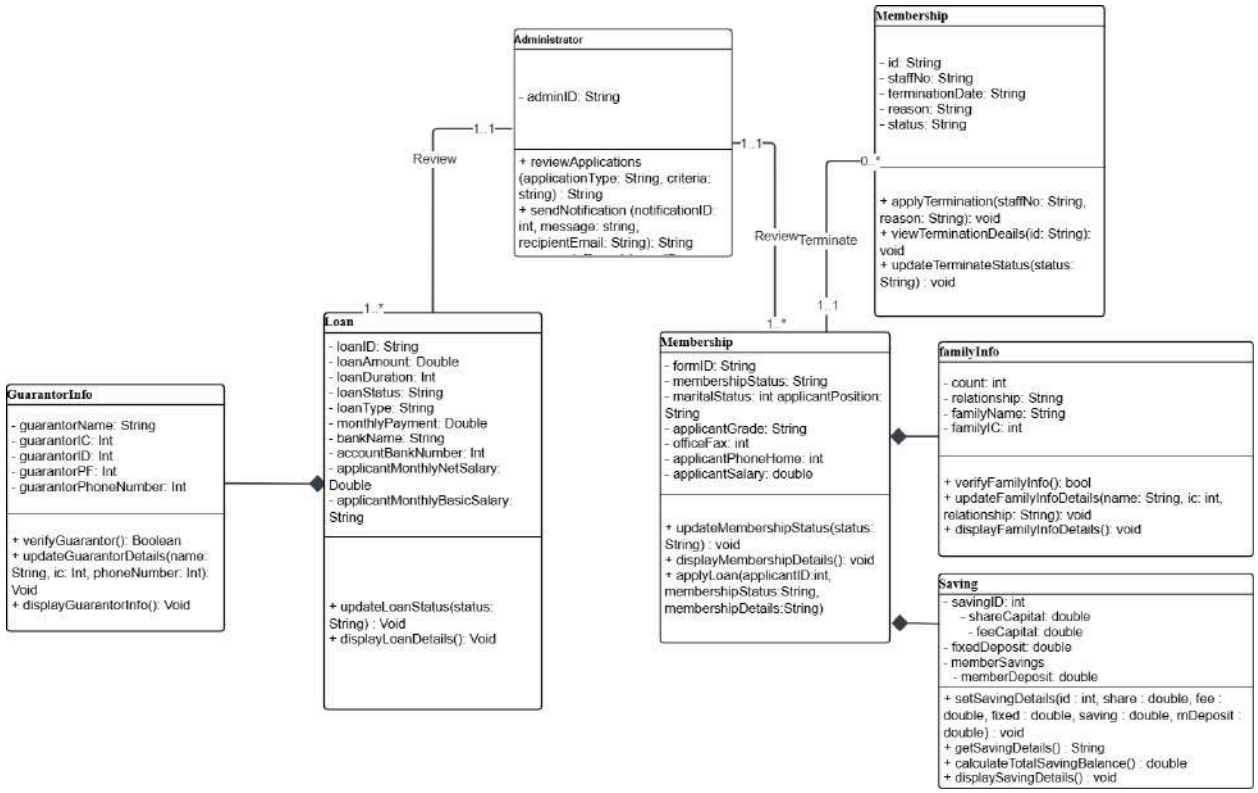


Figure 2.13: Class Diagram for Review Applicant's Form Subsystem

Figure 2.14 shows the class diagram for Approve Membership which includes the relationship between ALK class and Membership class. ALK class provides methods like approveMembership() to manage membership approvals and clarifications. The membership class stores the membership details through attributes. The relationship between the two classes is one-to-many, where an ALK can approve multiple membership applications.

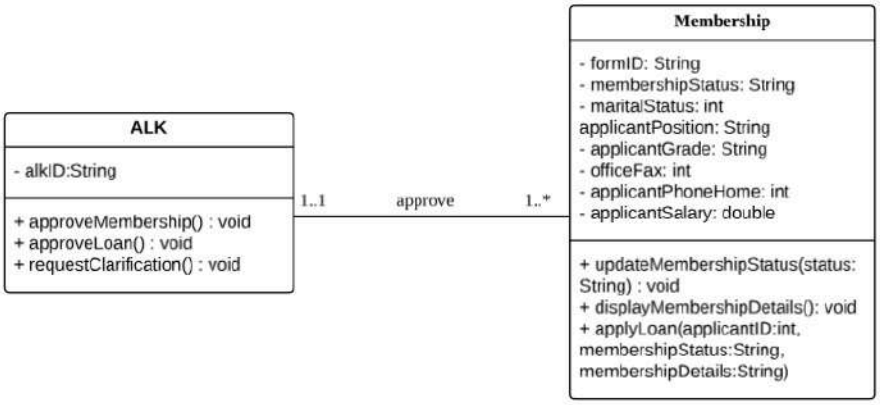


Figure 2.14: Class Diagram for Approve Membership Subsystem

Figure 2.15 class diagram shows the class diagram for Apply for Loan that shows two key relationships. First, the Membership and Loan classes have a one-to-many relationship indicating that a single membership can apply for several loans. Second, there is a composition relationship between the Loan and GuarantorInfo classes, which means that each loan application must include guarantor information because the guarantor is a critical component of the loan process.

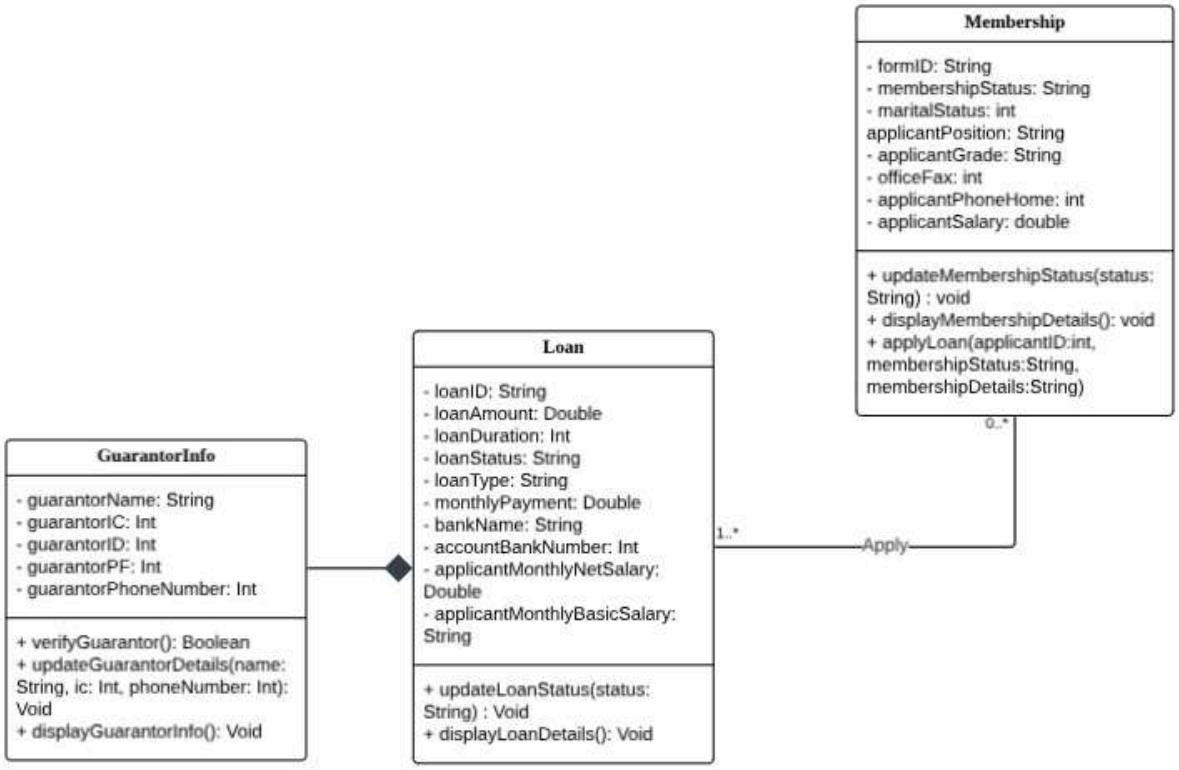


Figure 2.15: Class Diagram for Apply for Loan Subsystem

Figure 2.16 class diagram shows the class diagram for Calculate Loan that shows one key relationship. The user class has a one-to-one relationship with the loanCalculator class which allows users to calculate the estimation of monthly repayment.

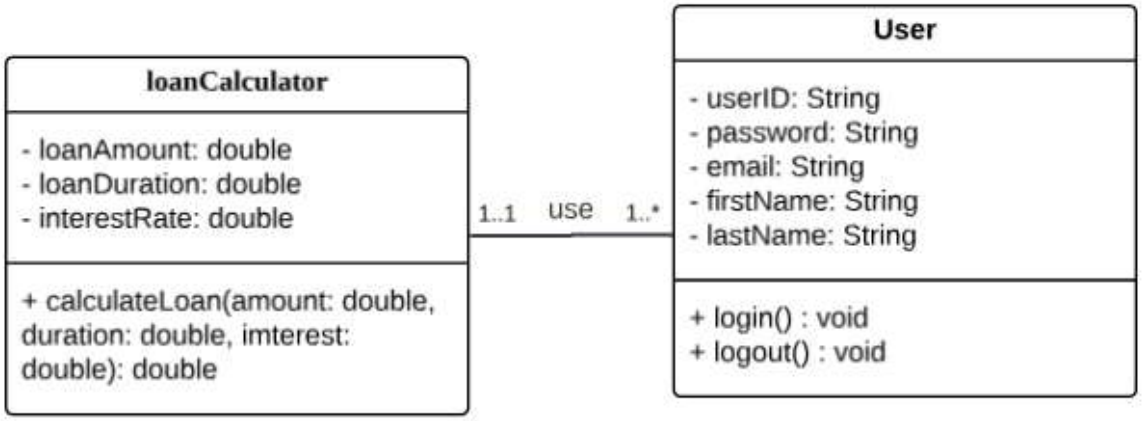


Figure 2.16: Class Diagram for Calculate Loan Subsystem

Figure 2.17 shows the class diagram for Loan Approval which includes the relationship between ALK class and Loan class. ALK class provides methods like approveLoan() to manage loan approvals. The loan class stores the membership details through attributes. The relationship between the two classes is one-to-many, where an ALK can approve multiple loan applications.

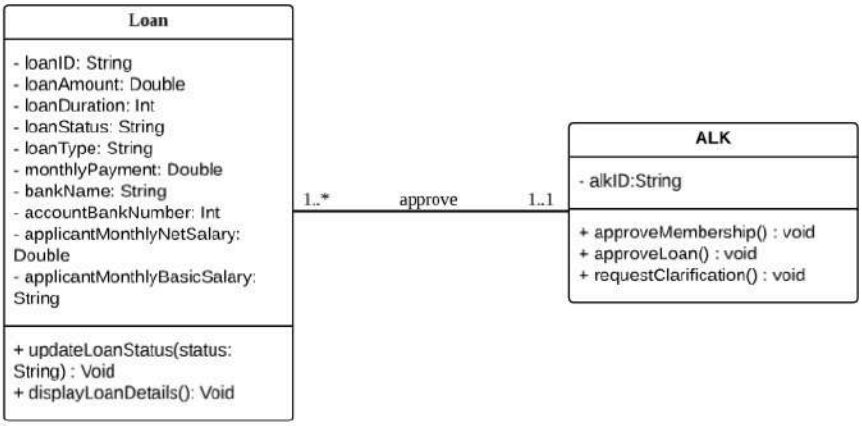


Figure 2.17: Class Diagram for Loan Approval Subsystem

Figure 2.18 illustrates the class diagram for preparing statements and reports, detailing the relationships between various entities involved in loan, savings, and membership management. The diagram includes classes such as LoanRepayment, Loan, Applicant, Membership, Saving, and SavingType, each with specific attributes and methods. The methods provided facilitate updates and retrieval of relevant data, supporting efficient statement and report generation.

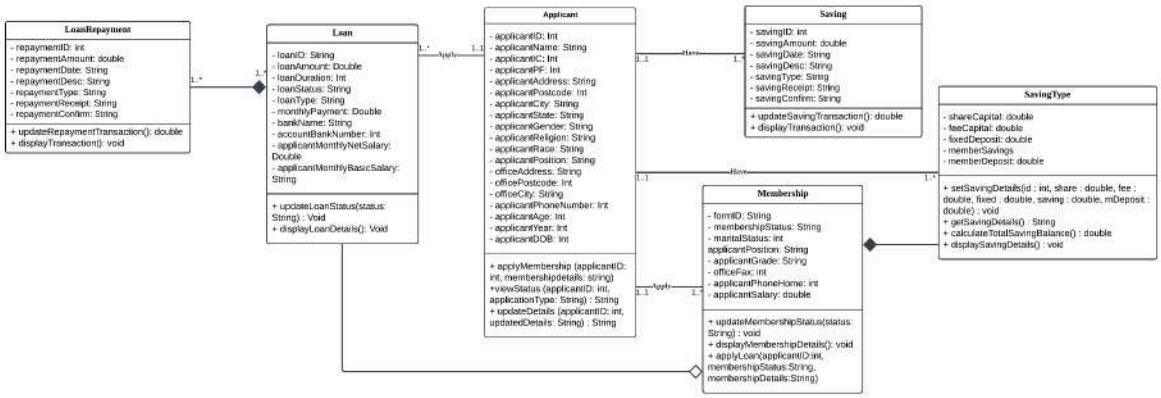


Figure 2.18: Class Diagram for Prepare Statement and Report Subsystem

Figure 2.19 presents the class diagram for the Notify via Email. In this diagram, there are two primary classes: Admin and Notification. The Admin class is responsible for sending notifications, establishing a one-to-many relationship with the Notification class. This means that one Admin can send multiple notifications to different recipients. The Notification class holds the details about each notification, such as the content and recipient information.

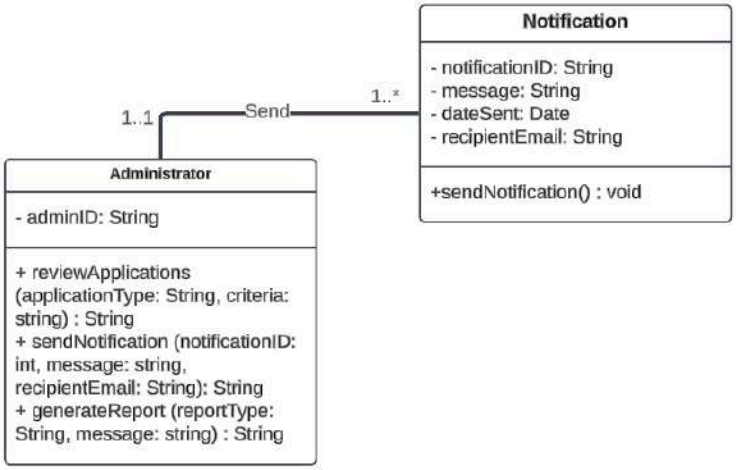


Figure 2.19: Class Diagram for Notify via Email Subsystem

Figure 2.20 is the Terminate entity representing the process of membership termination, linking it to the Membership Entity. Each membership can have one termination request while multiple termination requests can exist for different memberships. The entity includes attributes such as id, staffNo, terminationDate, reason and status ensuring a structured termination process.

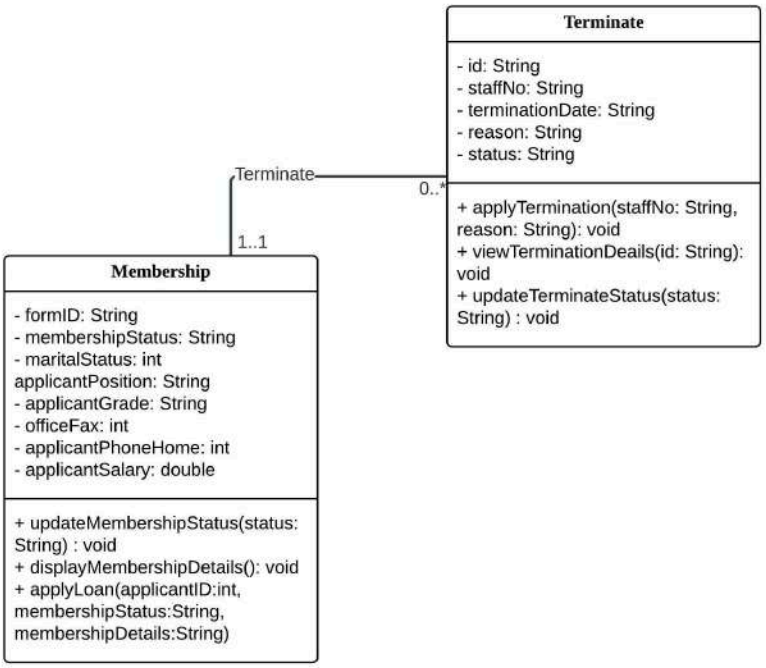


Figure 2.20: Class Diagram for Terminate Membership Subsystem

Figure 2.21 shows the class diagram for managing transactions. It includes classes like LoanRepayment, Loan, Membership, and Saving, each handling different financial activities. These classes store important details and have functions to update and display transaction records, making it easier to manage loan repayments, savings, and memberships.

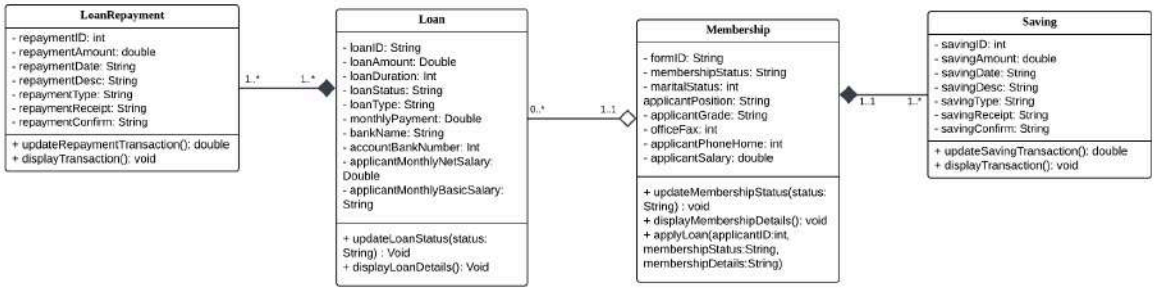


Figure 2.21: Class Diagram for Manage Transaction Subsystem

Figure 2.22 shows the class diagram for approve terminate membership application. It includes classes like ALK, Terminate and Membership. Each application will be undergoing

approval process by ALK. Each application will have updated status once ALK approved or rejected application.

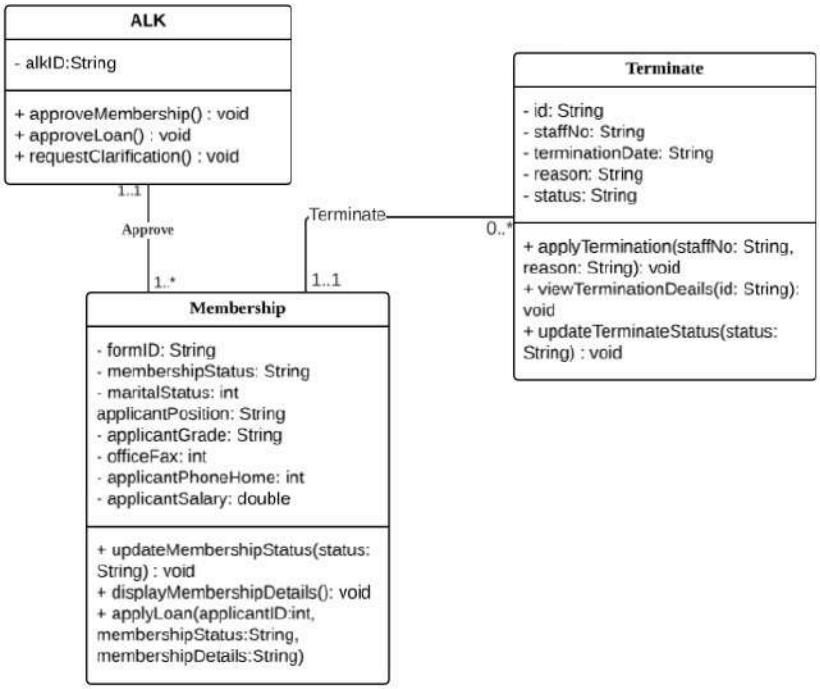


Figure 2.22: Class Diagram for Approve Terminate Application Subsystem

2.4 Process View - Sequence Diagram

This view illustrates how the system operates at runtime and focuses on how its components interact through communication and collaboration. It shows how different processes work together to achieve the system's goals, with each process responsible for specific tasks. To represent these interactions, sequence diagrams will be used for relevant use cases, showing the step-by-step flow of messages between components. Branching will capture alternate flows within the same diagram, and if the diagram becomes too cluttered, separate diagrams will be created for each scenario or alternate flow. This view provides a clear understanding of the system's dynamic behavior and the relationships between its components.

Figure 2.23 illustrates the applicant’s login process. The applicant accesses the login page, enters their details and the system will validate the details. If the details are valid, access to the membership application form is granted. Otherwise, an error message is displayed. The membership form is then accessed and shown to the applicant.

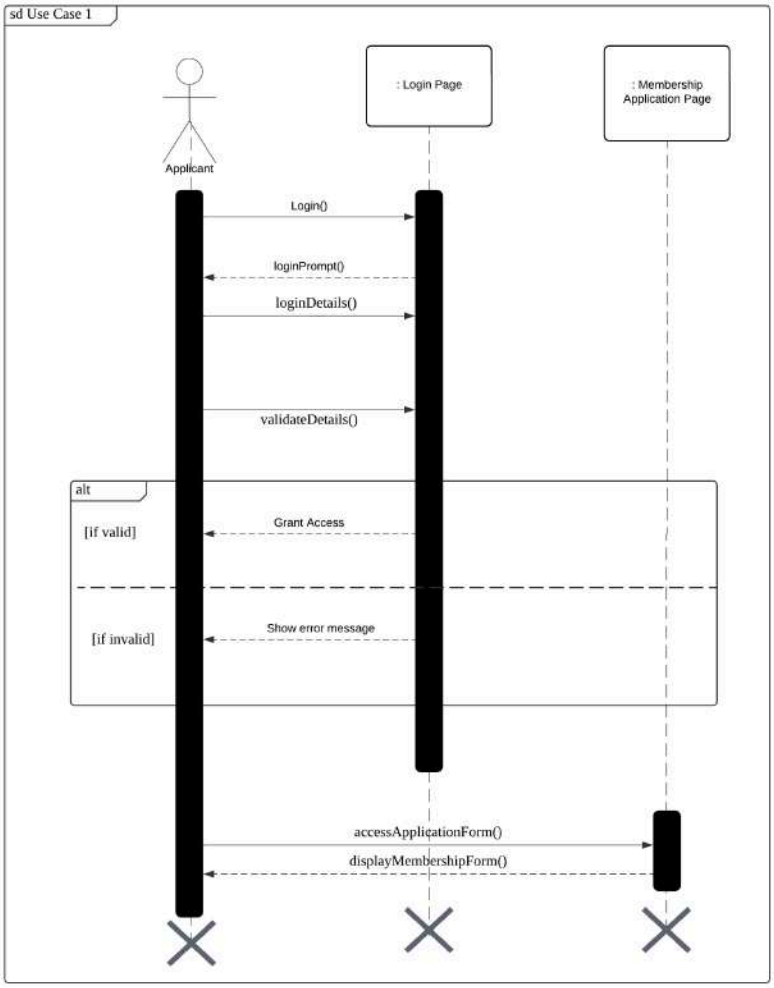


Figure 2.23: SD001 Sequence Diagram for Login for Apply Membership

Figure 2.24 shows the process of an admin reviewing membership, loan, or membership termination applications. The admin logs in, navigates to the membership list, and views batch application details. The system loops through each application in the batch, marking it as either “reviewed” or “incomplete”.

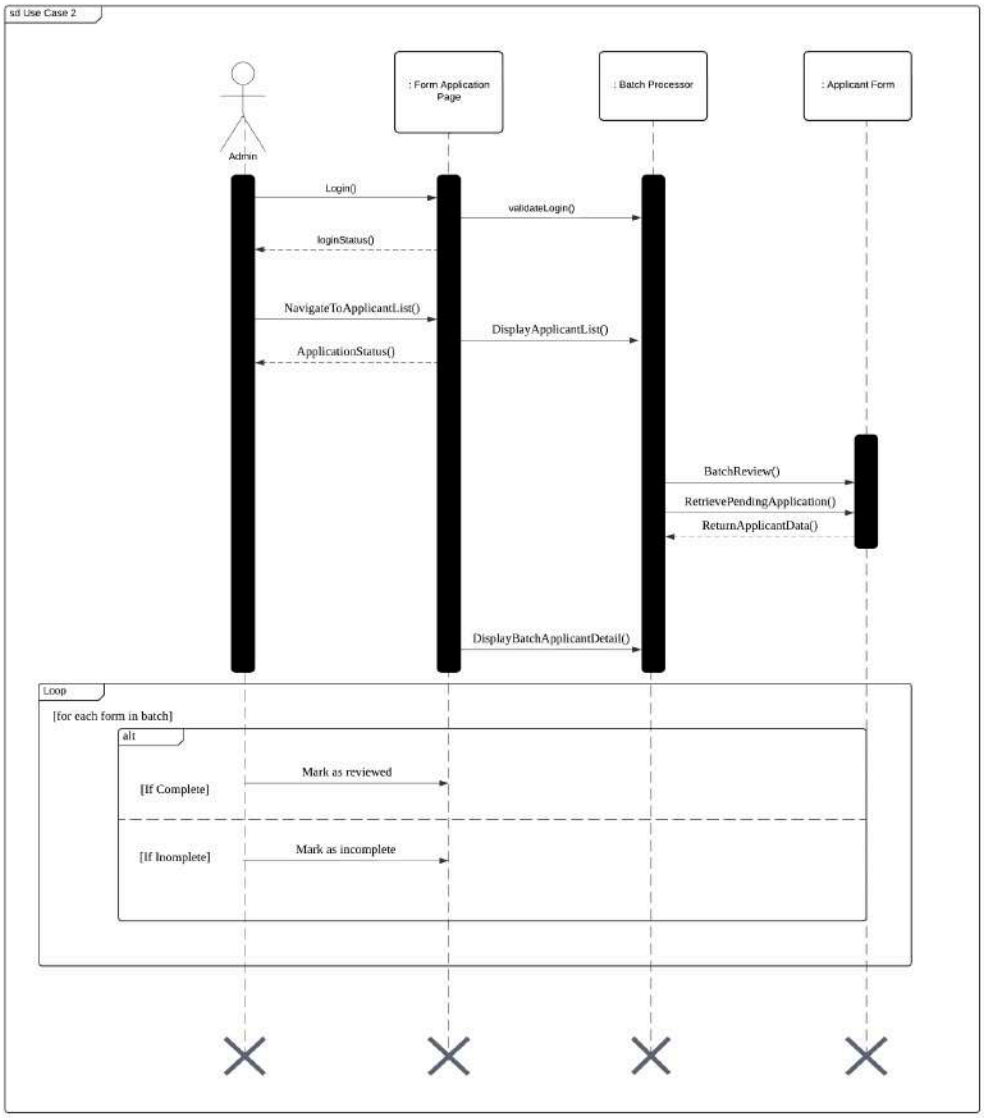


Figure 2.24: SD002 Sequence Diagram for Review Applicant's Form Subsystem

Figure 2.25 shows a sequence diagram for use case approved membership. ALK logs into the system, which either grants access or reuters an error. ALK then selects any membership application from the pending list, evaluates and decides whether to approve or not. Finally, the system updates the membership status

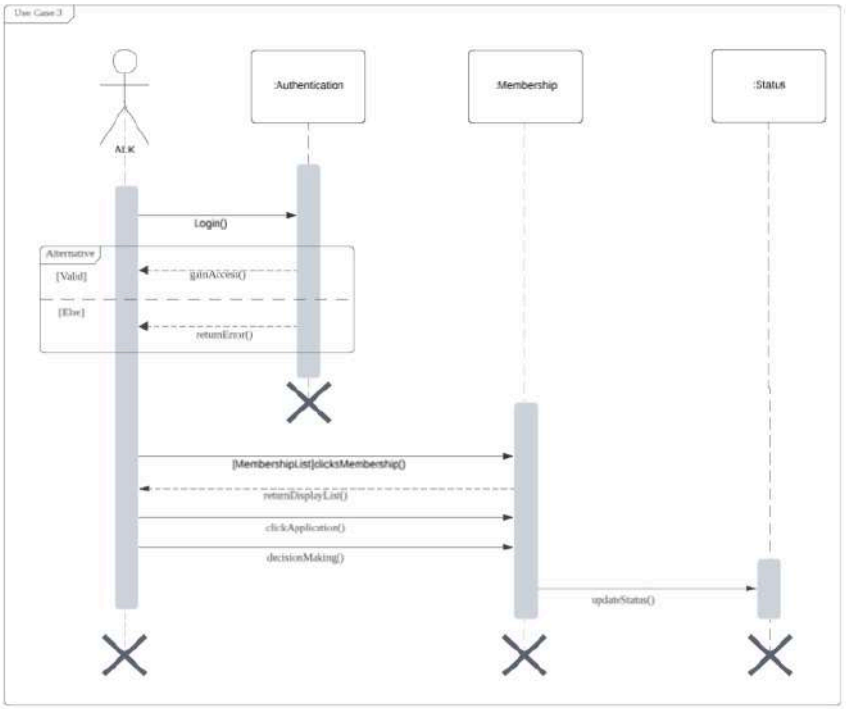


Figure 2.25: SD003 Sequence Diagram for Approve Membership Subsystem

Figure 2.26 is the sequence diagram for the application loan process, representing the interaction between the applicant and the system for applying for the loan. It starts when the user logs in and accesses the loan application page. The user fills out the loan application with compulsory details, then the system validates it. If the details are correct, the application is submitted and saved into the system's database. The system notifies the user after accepting the successful submission. This diagram illustrates the important steps involved, including user interactions, form validation and database operations.

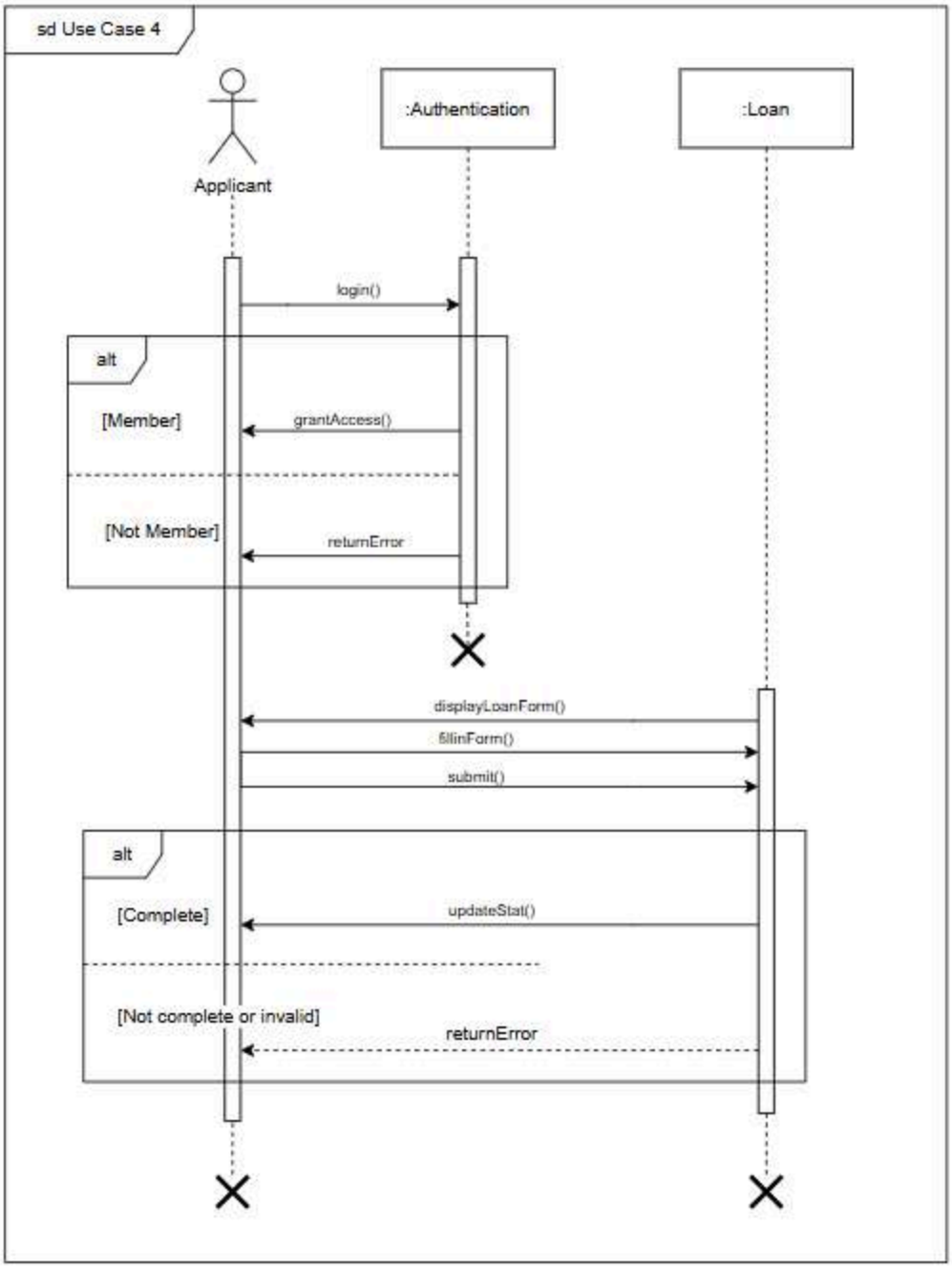


Figure 2.26: SD004 Sequence Diagram for Apply for Loan Subsystem

Figure 2.27 is a sequence diagram for calculation loan highlighting the interaction between members and the system. The process begins with the member logging into the system by entering their details. The system then verifies the details. If valid, the system grants access to the members. If invalid, the system returns an error message. The member selects

the ”Loan Calculator” button, and the system rate then submits the details. The Loan Calculator calculates and generates the loan repayment by processing these inputs. If the inputs are valid, the system displays the results. If invalid, it returns an error message that requires the member to reenter the inputs. The interaction makes sure the member has correct financial calculations based on their provided details.

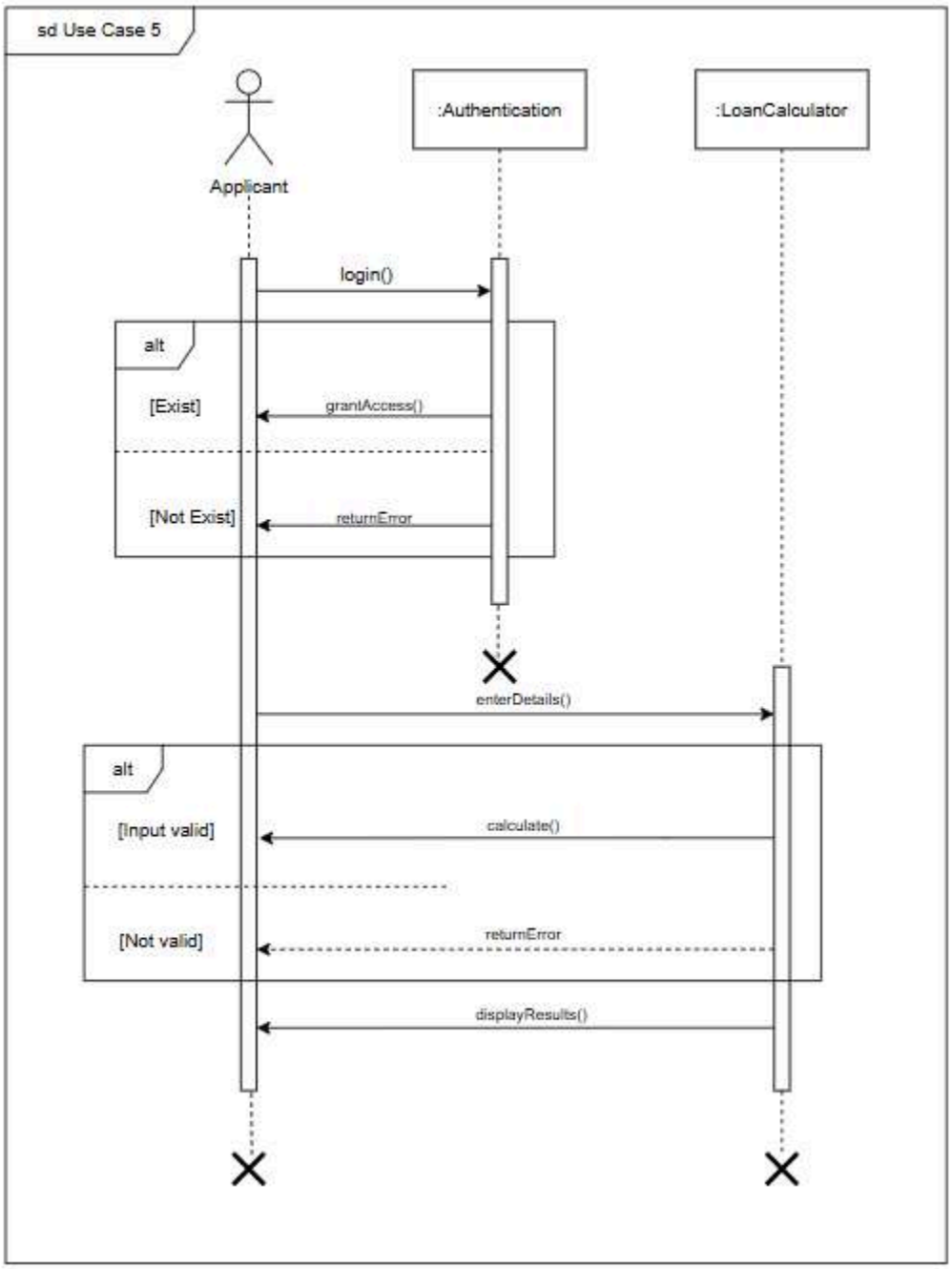


Figure 2.27: SD005 Sequence Diagram for Calculate Loan Subsystem

Figure 2.28 shows a sequence diagram for use case approved loan applications. ALK logs into the system, which either grants access or reuters an error. ALK then selects any loan application from the pending list, evaluates and decides whether to approve or not. If necessary, ALK uses the compare button to compare against loan terms before finalizing the decision. Finally, the system updates the loan status.

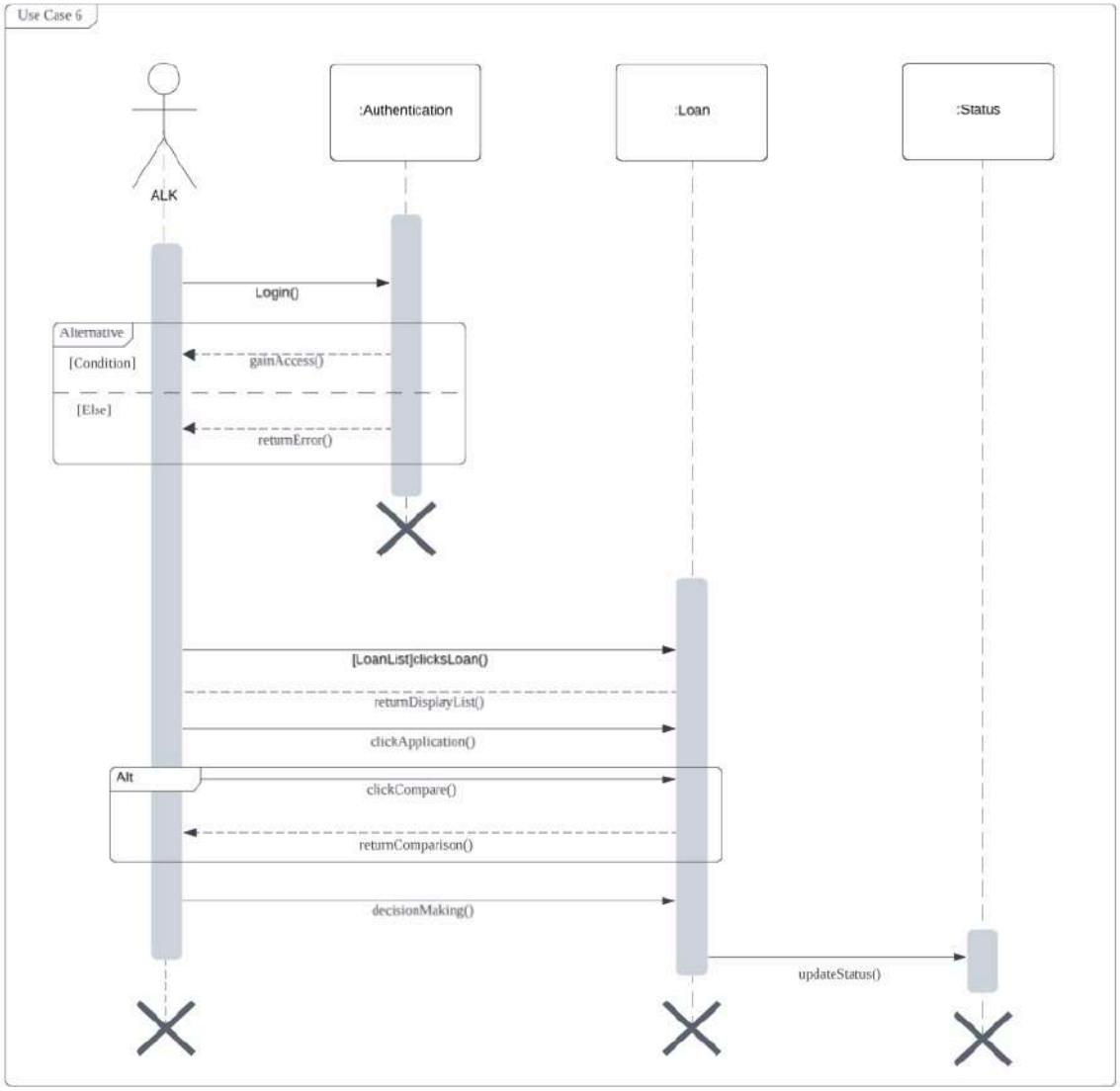


Figure 2.28: SD006 Sequence Diagram for Approve Loan Application Subsystem

Figure 2.29 shows the sequence diagram for the process of preparing financial statements and reports by an admin, interacting with the Authentication and Report systems. The admin begins by logging in, where successful authentication grants access to the admin

page, while failure triggers an error message. Once authenticated, the admin retrieves member data and generates financial statements for selected members. If statement generation is successful, a confirmation message is displayed; otherwise, an error message appears. Additionally, corporate reports containing membership, loan, and termination statistics are generated, providing monthly and yearly financial summaries.

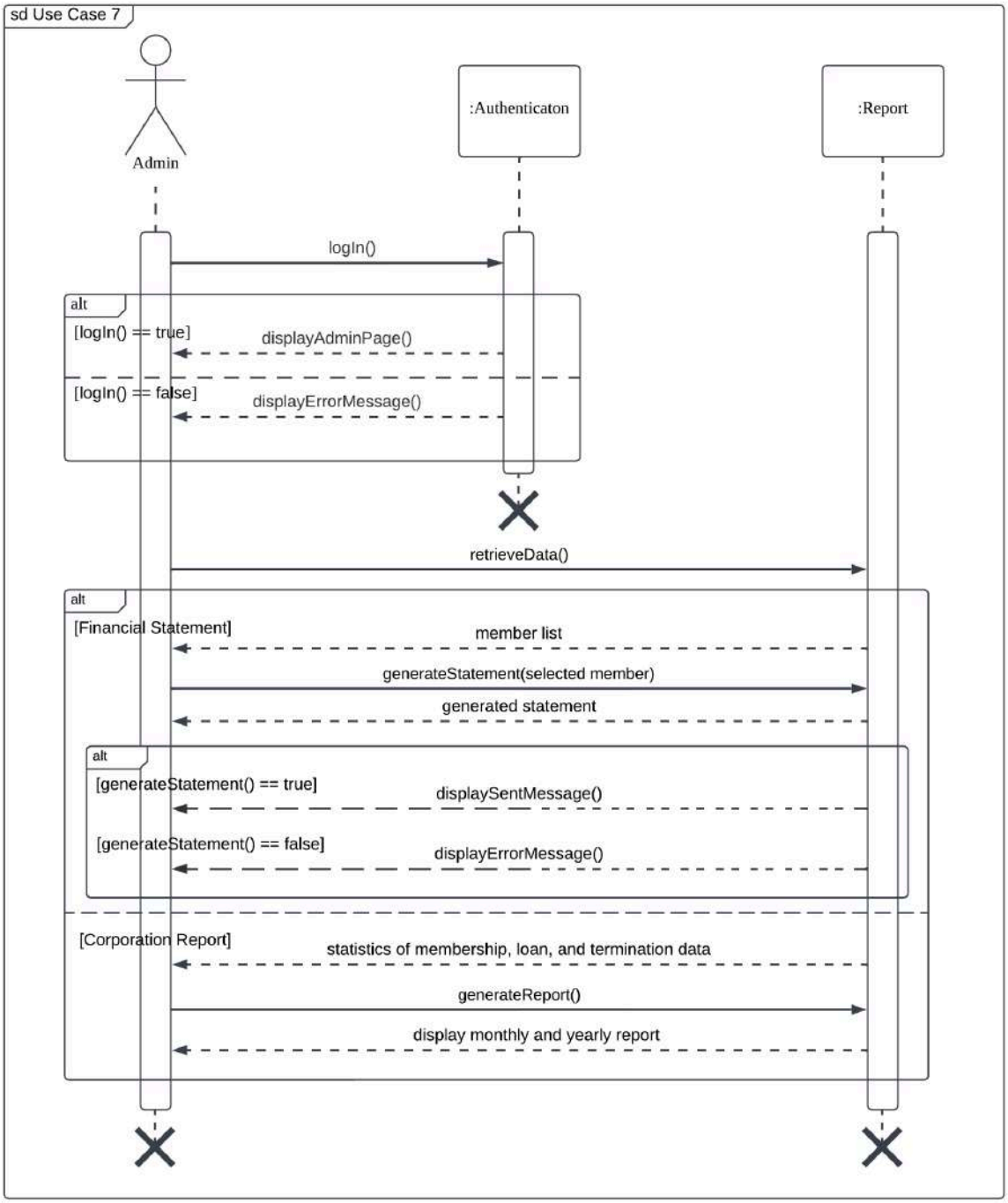


Figure 2.29: SD007 Sequence Diagram for Prepare Statement and Report Subsystem

Figure 2.30 illustrates the sequence diagram for the use case Notify via Email, regarding prepared statements and status updates. The Admin initiates the notification process by selecting a notification type, triggering interactions between the Notification and Report components. If a Generated Statement Notification is selected, the system retrieves the prepared statement for the selected member and sends it via email, followed by updating the notification log. Similarly, for Status Notification, an email is sent to the selected applicant, and the send status is updated. The process also supports batch processing, where multiple members or applicants are notified simultaneously. The final step verifies whether the email was successfully sent, displaying either a success or error message accordingly.

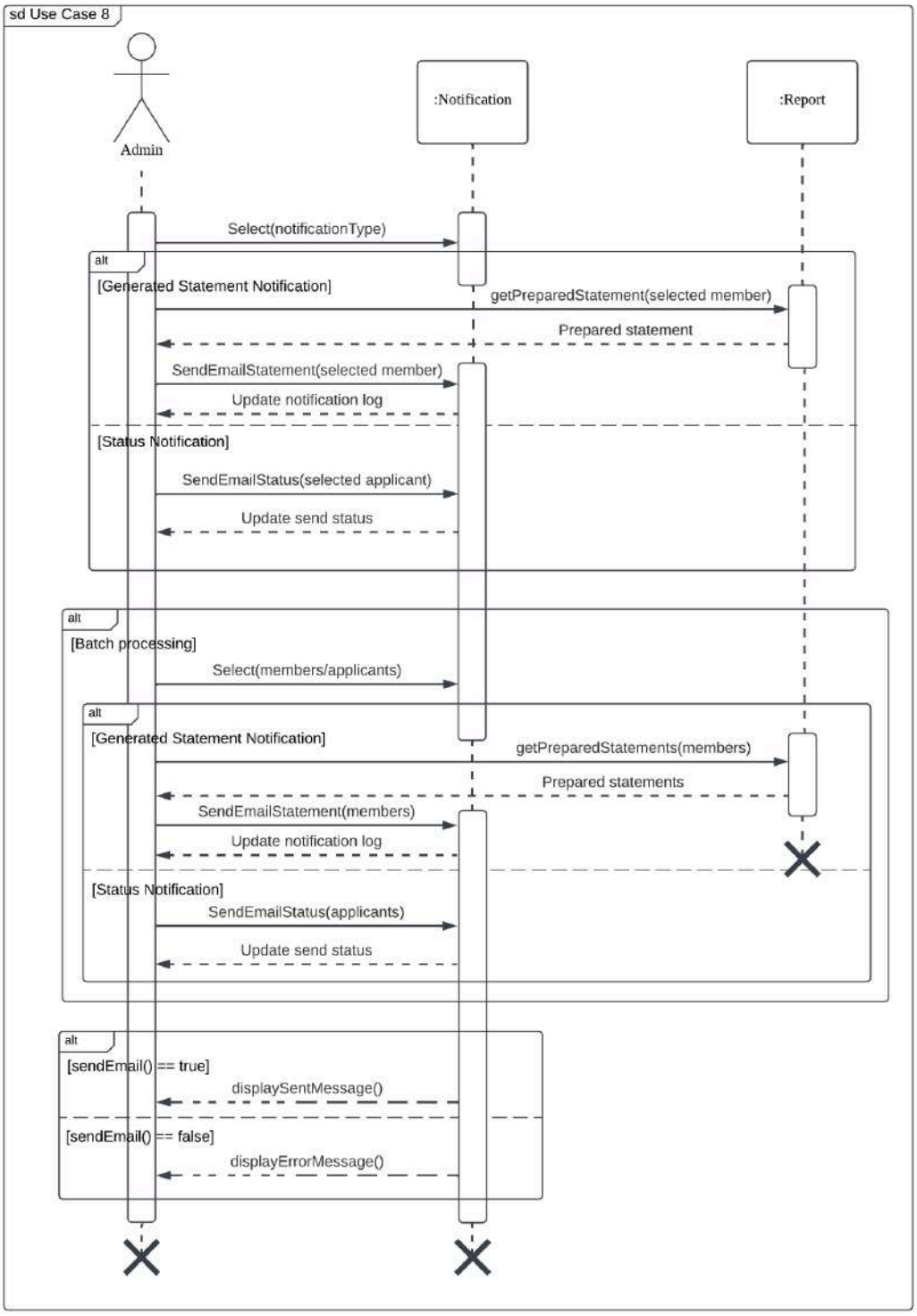


Figure 2.30 SD008 Sequence Diagram for Notify via Email Subsystem

Figure 2.31 illustrates the process of managing financial transactions by an admin, interacting with the Transaction and Financial Record systems. The admin selects a transaction

type, which can be a Monthly Transaction, an Irregular Transaction, or Batch Processing Monthly. For monthly transactions, the system deducts the salary of a selected member, while irregular transactions involve selecting a member and specifying savings or loan repayment. In batch processing, multiple members are selected for financial updates. Each transaction update is confirmed, and the system displays either a success or error message based on the update status.

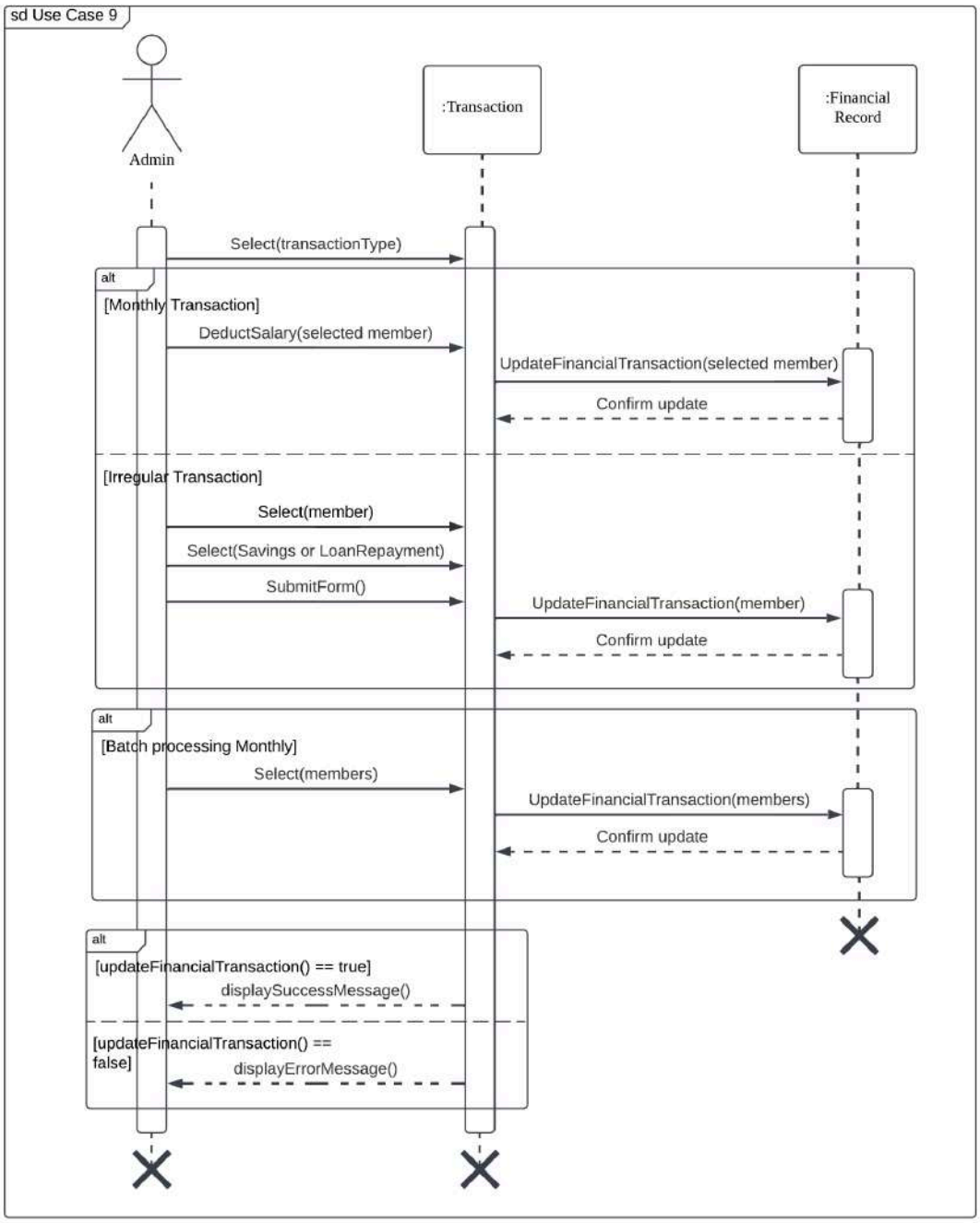


Figure 2.31: SD009 Sequence Diagram for Manage Transaction Subsystem

Figure 2.32 is a sequence diagram for Terminate Membership use case that illustrates the interaction between the Applicant, Authentication system and Terminate Membership module. The Applicant logs in and then the system verifies their membership status. If access is granted, the termination form is displayed. The Applicant fills in and submits the form. The system will validate the request, if complete, it updates the status of application. Otherwise, it returns an error to the Applicant.

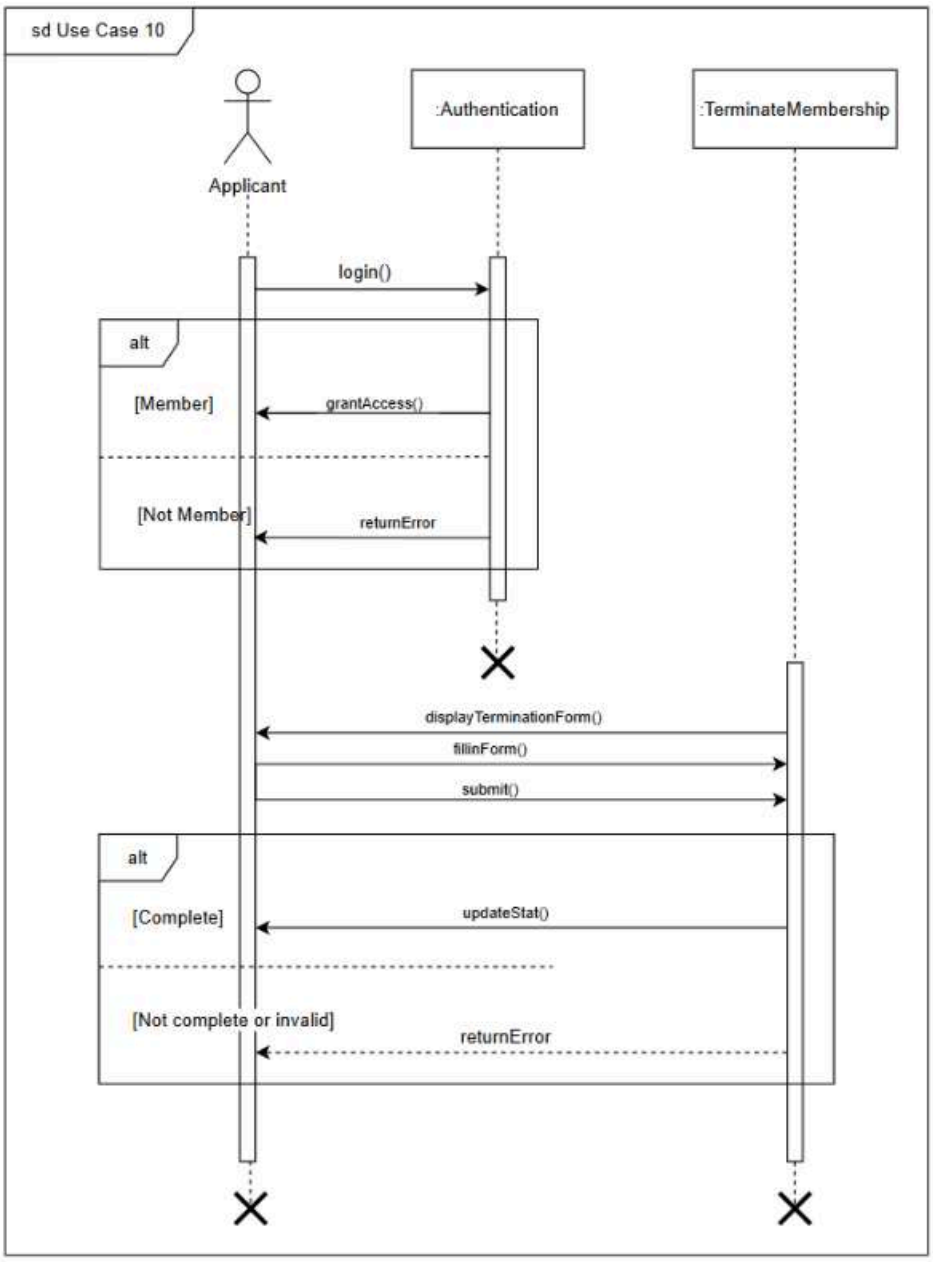


Figure 4.32: SD010 Sequence Diagram for Terminate Membership Subsystem

Figure 2.33 is a sequence diagram for the Approve Terminate Application use case that illustrates the interaction between the ALK and Terminate Membership module. The ALK will log in and then the system will display the termination application reviewed by the admin. The ALK then performs decision-making whether to approve or reject the application. If the ALK approved, then the applicant's membership will end otherwise the membership will continue as usual.

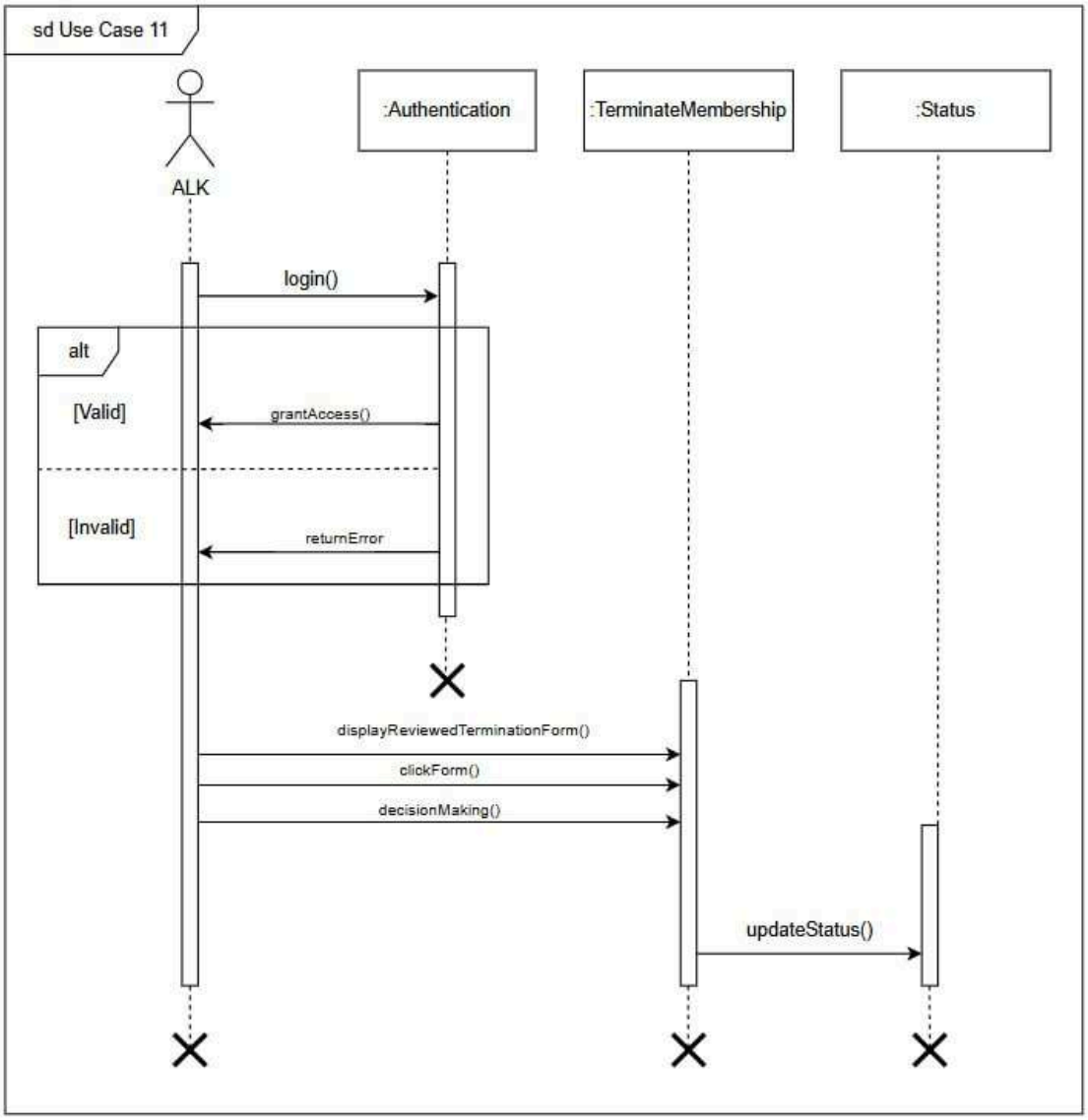


Figure 2.33: SD011 Sequence Diagram for Approve Terminate Application Subsystem

2.5 Deployment View - Deployment Diagram

This view focuses on how the system's components are distributed across hardware and software environments. It shows how the software is deployed on servers, devices, or systems, and how they interact with each other. Represented through a deployment diagram, this view illustrates the physical distribution of components across nodes and their communication paths. It helps to understand the system's layout, showing how components are allocated to machines or servers and how they interact over networks. The diagram also highlights the relationships between hardware, software, and external systems, ensuring the system runs efficiently in its environment.

Figure 2.34 illustrates the deployment diagram of the Kada Koperasi Automated System and highlights its key components along with their interactions. The system comprises four main nodes: the Client, Server, Database Server, and External Email Server. The Client node, represented by the artifact <<artifact>> Browser, communicates with the Server node, which includes <<artifact>> Internet Server and <<artifact>> Application Server to handle requests and execute business logic. The Application Server interacts with the Database Server, containing <<artifact>> MySQL DB, to manage and store cooperative data securely. Additionally, the External Email Server, represented by the artifact <<artifact>> Gmail SMTP Service, enables email notifications to send statement reports to members of koperasi KADA via the SMTP protocol.

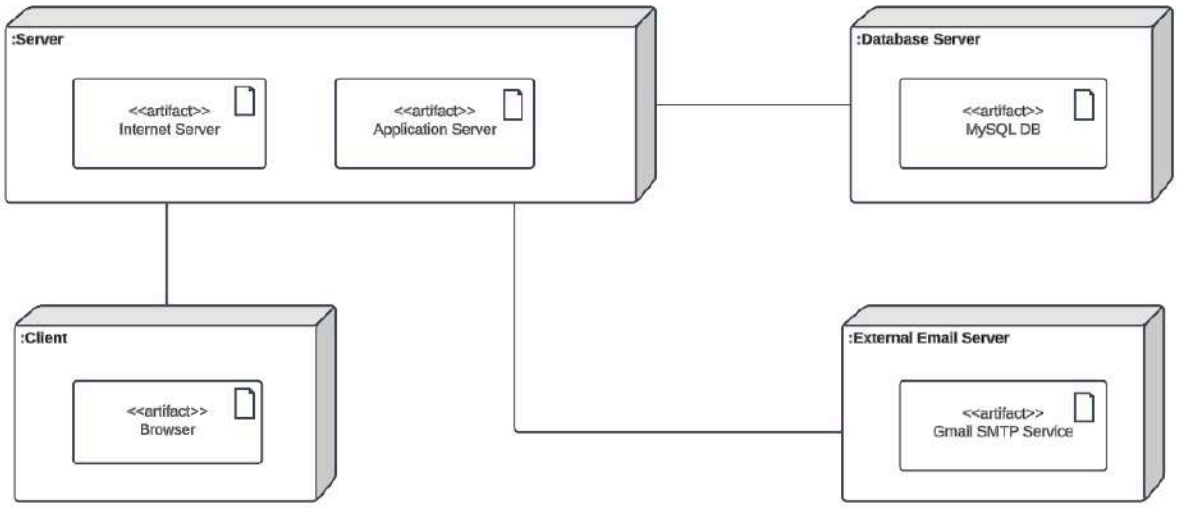


Figure 2.34: Deployment diagram for Koperasi KADA'S Automated System

3. Data Design

This section deals with the system by describing the information domain, using data entities and their relationships to represent the structured data. Various key entities have been listed, including Admin, ALK, Applicant, Loan, Membership and Notification. All of these entities are attributed according to their respective properties and behaviours. The major data or systems entities are stored into a relational database. Table 3.1 shows the description of entities in the database.

Table 3.1: Description of Entities in the Database

No.	Entity Name	Description
1	Administrator	General term describing the staff responsible for managing and reviewing membership and loan application
2	AdminApplicationFeedback	General term describing the reason of each application rejection by Admin
3	ALK	General term describing the authorized personnel responsible for approving membership and loan application
4	AlkApplicationFeedback	General term describing the reason of each application rejection by ALK
5	Applicant	General term describing all users logged in as applicants to apply membership.
6	BankInfo	General term describing the bank account information provided by applicants for processing membership applications.
7	FamilyInfo	General term for the details of the family members associated with an applicant.
8	Guarantor	General term describing a person who agrees to take responsibility for repaying a loan if the applicant has not paid for it.
9	Historymodimem	General term describing the past membership applications and the modification allowed

10	historymodiloan	General term describing the past loan applications and the modification allowed
11	Loan	General term describing the financial agreement where an applicant borrows a specific amount of money to be repaid over a period with interest.
12	LoanRepayment	General term describing the financial activities related to loans, such as disbursements, repayments, and transaction history.
13	LoanType	General term describing the different categories or plans of loans offered
14	Managing	General term describing a notification update related to an approval process
15	Membership	General term describing to the record of applicants who are members of the Koperasi and
16	MembershipEnd	General term describing details of membership termination including reason, date of application, approval and review.
17	Saving	General term describing the funds saved by members, managed through deposits and balance tracking used as collateral for loans.
18	SavingType	General term describing type of savings the company had
19	Status	General term represents transaction records of saving.
20	User	General term describing all the people who used the system.
21	UserType	General term describing type of user who accesses the system

3.1 **Data Dictionary**

This section covers a data dictionary for the Entity Relation Diagram (ERD) in figure 3.1 which is a structured document that provides detailed information about the data used in a system. It describes the system's entities, attributes and type, and serves as a reference for

understanding the data structure. By ordering attribute names alphabetically, developers, designers and stakeholders can communicate with greater consistency, clarity and ease. The data dictionary describes classes and attribute names in the classes to offer a clear overview of the system's data components.

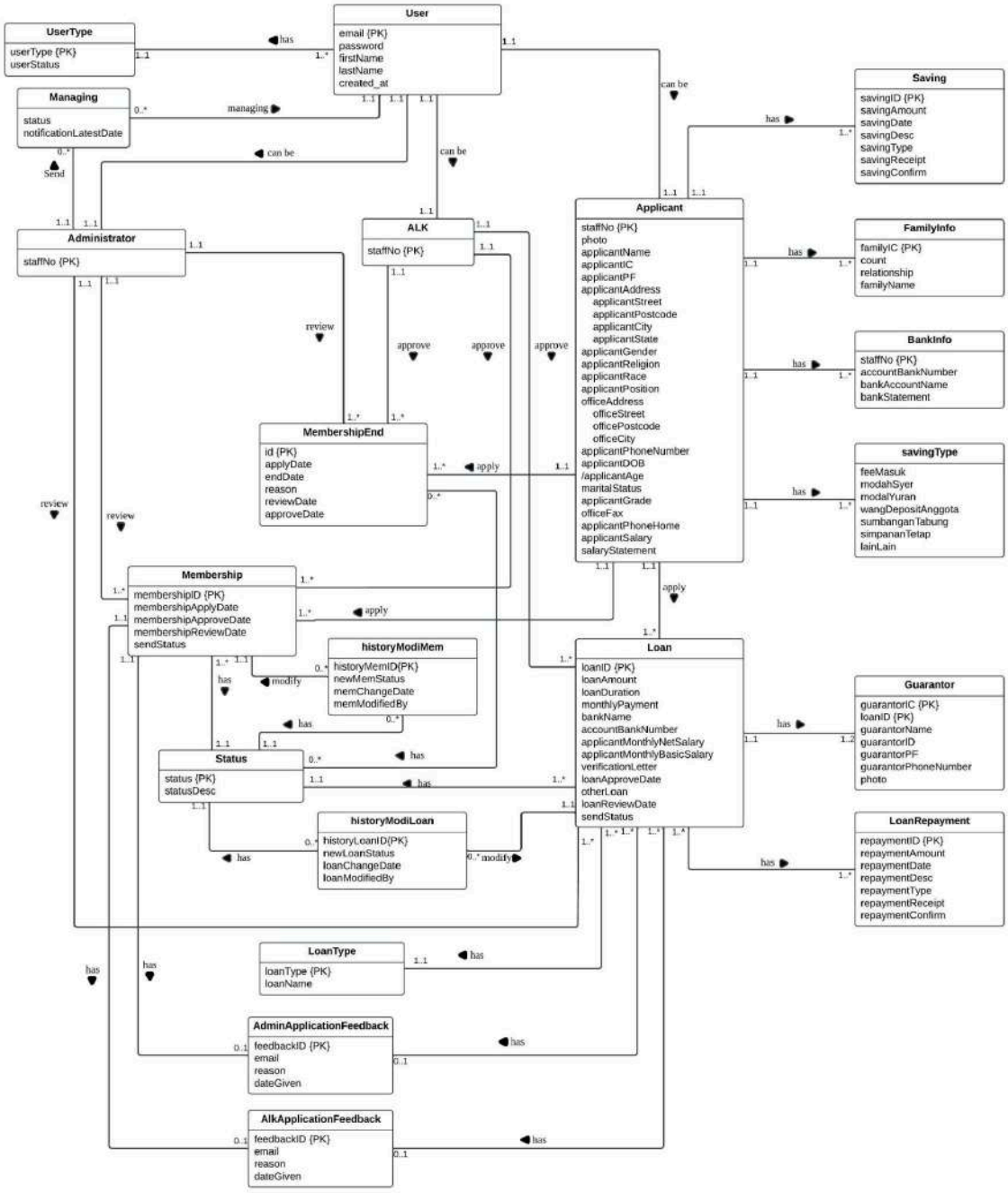


Figure 3.1: Entity Relationship Diagram (ERD)

3.1.1 Entity: Admin

Table 3.2: Data Dictionary for Admin

Attribute Name	Type	Description
email	VARCHAR(50)	Admin's email address
firstName	VARCHAR(25)	Admin's first name
lastName	VARCHAR(25)	Admin's last name
password	VARCHAR(50)	Password for admin's system login
staffNo	INT(10)	Admin's staff number
userID	INT(10)	Unique identifier for admin

3.1.2 Entity: AdminApplicationFeedback

Table 3.3: Data Dictionary for AdminApplicationFeedback

Attribute Name	Type	Description
feedbackID	INT (20)	Uniquely identifies feedback comment
email	VARCHAR(30)	Email
reason	LONGTEXT	Feedback messages to send
dateGiven	DATE	Date email was sent

3.1.3 Entity: ALK

Table 3.4: Data Dictionary for ALK

Attribute Name	Type	Description
email	VARCHAR(50)	ALK's email address
firstName	VARCHAR(25)	ALK's first name
lastName	VARCHAR(25)	ALK's last name
password	VARCHAR(50)	Password for ALK's system login

staffNo	INT(10)	ALK's staff number
userID	INT(10)	Unique identifier for ALK

3.1.4 Entity: AlkApplicationFeedback

Table 3.5: Data Dictionary for AlkApplicationFeedback

Attribute Name	Type	Description
feedbackID	INT (20)	Uniquely identifies feedback comment
email	VARCHAR(30)	Email
reason	LONGTEXT	Feedback messages to send
dateGiven	DATE	Date email was sent

3.1.5 Entity: Applicant

Table 3.6: Data Dictionary for Applicant

Attribute Name	Type	Description
applicantAddress	VARCHAR(50)	Home address of the applicant
applicantAge	INT(3)	Current Age of the applicant. This is the derivation of date of birth.
applicantCity	VARCHAR(30)	City of the applicant's address
applicantDOB	DATE	Date of birth of the applicant.
applicantGender	CHAR(1)	Gender of the applicant
applicantGrade	VARCHAR (10)	Professional grade of the applicant's job position
applicantIC	INT(12)	Identification Card number of the applicant
applicantName	VARCHAR(30)	Full name of applicant

applicantPF	VARCHAR(30)	Personal File number of the applicant
applicantPhoneHome	VARCHAR (15)	Home phone number of the applicant
applicantPhoneNumber	VARCHAR(15)	Primary contact number of the applicant
applicantPosition	VARCHAR(30)	Job position of the applicant
applicantPostcode	INT(5)	Postcode of the applicant's address
applicantRace	VARCHAR(25)	Race or ethnicity of the applicant
applicantReligion	VARCHAR(25)	Religion of the applicant
applicantSalary	DECIMAL (10,2)	Monthly Income of the applicant
applicantState	VARCHAR(30)	State of the applicant's address
maritalStatus	VARCHAR (10)	Marital status of the applicant
officeAddress	VARCHAR(50)	Address of the applicant's workplace
officeCity	VARCHAR(30)	City of the applicant's workplace
officeFax	VARCHAR (15)	Fax number of the applicant's workplace
officePostcode	INT(5)	Postcode of the applicant's workplace

3.1.6 **Entity: BankInfo**

Table 3.7: Data Dictionary for BankInfo

Attribute Name	**Type**	**Description**
accountID	INT(20)	Account Number for transactions
bankAccountType	VARCHAR(20)	Name of the bank

3.1.7 **Entity: FamilyInfo**

Table 3.8: Data Dictionary forFamilyInfo

Attribute Name	**Type**	**Description**
count	INT(20)	Number of family member
familyIC	VARCHAR(15)	Family member Identification Number
familyName	VARCHAR(25)	Name of the family member
relationship	VARCHAR(10)	Relationship of applicant with family member

3.1.8 **Entity: Guarantor**

Table 3.9: Data Dictionary for Guarantor

Attribute Name	**Type**	**Description**
guarantorIC	VARCHAR(20)	Guarantor identification card
guarantorID	INT(20)	Unique identifier for guarantor
guarantorName	VARCHAR(30)	Guarantor full name
guarantorPF	VARCHAR(20)	Guarantor personal file number
guarantorPhoneNumber	VARCHAR(20)	Guarantor phone number
loanID	INT (5)	Loan identification number
photo	LONGBLOB	Guarantor photo

3.1.9 **Entity : HistoryModiMem**

Table 3.10: Data Dictionary for historymodimem

Attribute Name	**Type**	**Description**
historyMemID	INT(5)	Unique identifier for each past applications

newMemStatus	INT(3)	New status after modification
memChangeDate	DATE	Date status modified
memModifiedBy	INT(5)	Modifier's staff number

3.1.10 **Entity : historymodiloan**

Table 3.11: Data Dictionary for historymodiloan

Attribute Name	**Type**	**Description**
historyLoanID	INT(5)	Unique identifier for each past applications
newLoanStatus	INT(3)	New status after modification
loanChangeDate	DATE	Date status modified
loanModifiedBy	INT(5)	Modifier's staff number

3.1.11 **Entity: Loan**

Table 3.12: Data Dictionary for Loan

Attribute Name	**Type**	**Description**
accountBankNumber	BIGINT(25)	Account bank number
applicantBasicSalary	DECIMAL(10,2)	Applicant basic salary
applicantNetSalary	DECIMAL(10,2)	Applicant net salary
bankName	VARCHAR(30)	Bank name
loanAmount	DECIMAL(10,2)	Loan Amount
loanApplyDate	DATE	Loan apply date
loanApproveDate	DATE	Loan approve date
loanDuration	INT(2)	Loan Duration
loanReviewDate	DATE	Loan review date
loanID	INT(5)	Loan identification number
monthlyPayment	DECIMAL(10,2)	Monthly payment of loan
otherLoan	INT(5)	Other loan type

sendStatus	INT(3)	Status for send email
verificationLetter	LONGBLOB	Applicant verified employer

3.1.12 **Entity: LoanRepayment**

Table 3.13: Data Dictionary for LoanRepayment

Attribute Name	**Type**	**Description**
RepaymentID	INT(5)	Uniquely identifies saving transaction
RepaymentAmount	DECIMAL(10, 2)	Repayment amount
RepaymentDate	DATE	Repayment date
RepaymentDesc	VARCHAR(200)	Repayment description
RepaymentType	VARCHAR(30)	Repayment type
RepaymentReceipt	VARCHAR(15)	Repayment receipt
RepayementConfirm	VARCHAR(15)	Repayment confirmation

3.1.13 **Entity: LoanType**

Table 3.14: Data Dictionary for LoanType

Attribute Name	**Type**	**Description**
loanName	VARCHAR(25)	Loan name
loanType	INT(7)	Unique identifier for the loan type

3.1.14 **Entity: Managing**

Table 3.15: Data Dictionary for Managing

Attribute Name	**Type**	**Description**
notificationLatestDate	VARCHAR(25)	Date of notification being sent
status	VARCHAR(25)	Status of notification

3.1.15 **Entity: Membership**

Table 3.16: Data Dictionary for Membership

Attribute Name	**Type**	**Description**
membershipStatus	VARCHAR(20)	Current status of the membership request
membershipApplyDate	DATE	Date when the membership starts
membershipEndDate	DATE	Date when the membership ends
membershipReviewDate	DATE	Date when the membership form is review
membershipApproveDate	DATE	Date when the membership is approved

3.1.16 **Entity: MembershipEnd**

Table 3.17: Data Dictionary for MembershipEnd

Attribute Name	**Type**	**Description**
applyDate	DATE	Terminate membership apply date
approveDate	DATE	Approve date
endDate	DATE	Membership end date
id	INT (5)	Uniquely identifies membership end application form
reason	VARCHAR(255)	Reasons message to end membership
reviewDate	DATE	reviewDate

3.1.17 **Entity: Saving**

Table 3.18: Data Dictionary for Saving

Attribute Name	Type	Description
savingID	INT(5)	Uniquely identifies saving transaction
savingAmount	DECIMAL(10, 2)	Saving amount
savingDate	DATE	Saving date
savingDesc	VARCHAR(200)	Saving description
savingType	VARCHAR(30)	Saving payment type
savingReceipt	VARCHAR(15)	Saving receipt
savingConfirm	VARCHAR(15)	Saving confirmation

3.1.18 **Entity: SavingType**

Table 3.19: Data Dictionary for SavingType

Attribute Name	Type	Description
feeMasuk	DECIMAL(10, 2)	Fee capital amount
modahSyer	DECIMAL(10, 2)	Fixed deposit amount
modalYuran	DECIMAL(10, 2)	Member deposit amount
wangDepositAnggota	DECIMAL(10, 2)	Member savings amount
sumbanganTabung	DECIMAL(10, 2)	Share capital amount
simpananTetap	DECIMAL(10, 2)	Fixed saving amount

3.1.19 **Entity: Status**

Table 3.20: Data Dictionary for Status

Attribute Name	Type	Description
status	INT (3)	Status
statusDesc	VARCHAR (25)	Status description

3.1.20 **Entity: User**

Table 3.21: Data Dictionary for User

Attribute Name	Type	Description
email	VARCHAR(25)	Email for communication and receiving any update information.
firstName	VARCHAR(25)	Given name of the user
lastName	VARCHAR(20)	Family Name of the user
password	VARCHAR(15)	Password for user authentication
staffNo	INT(10)	Staff Number of the user
userID	INT AUTO_INCREMENT	Unique identifier assigned for each user

3.1.21 **Entity: UserType**

Table 3.22: Data Dictionary for UserType

Attribute Name	Type	Description
UserType	INT(5)	Type of user
UserStatus	VARCHAR(10)	Type of user description

4. User Interface Design

The interface includes sections designed for different types of users which are applicants, administrators, and ALK.

For applicants, the interface starts with a login page where users enter their credentials to access the system. Once logged in, they see a menu with options like applying for membership, applying for a loan, applying for a termination of membership and using a loan calculator. The membership application involves filling out a form in two steps where users provide personal details and submit their application. For loan applications, users can calculate repayment amounts, fill out forms with personal and guarantor information, sign agreements. Furthermore, the termination of membership application follows a similar process where users fill out a form with their details and state their reason. All applications can check the status of their application after submitting the application. These features ensure users can easily manage their membership and loan applications.

For administrators, the interface focuses on managing the application, including membership and loan applications, financial transactions, notifications, and reporting. Admins log in through an admin page to access tools for reviewing applications, updating membership statuses, and verifying applicants' financial and guarantor information before proceeding with approvals. They also manage transactions by recording payments, tracking outstanding balances, and updating payment statuses. Email notifications are sent to inform users about application statuses and financial statements. Additionally, administrators can generate detailed reports on transactions, loan approvals, and membership activities, which are then emailed to relevant parties for record-keeping and decision-making.

The ALK interface is designed for approving or rejecting applications. After logging in, ALK users can choose to review either membership or loan application. For membership forms, ALK users can approve or reject applications directly. For loan applications, they can check the loan details, compare them with the terms, and review supporting documents like payslips. The system clearly shows whether the application is successful or not, allowing ALK users to make quick decisions.

In summary, the interfaces are designed to help users manage applications, administrators handle reviews, and ALK staff approve or reject the applications efficiently.

4.1 Screen Images

The Koperasi KADA's Automated System's first-time user should see the login page when he/she opens the website like in Figure 4.1. If the user has not registered, he/she should be able to do that on the login page. If the user is already registered, the user can login and will be redirected to which type of user they are either user, admin or Ahli Lembaga Koperasi (ALK). After Logging in, Figure 4.2 allows users to choose from various menu options, including applying for membership, applying for a loan, and calculating loan details as well as applying for a membership termination.

Figure 4.1: Login Page

Figure 4.2: Applicant's Menu

When users choose to apply for membership, they are directed to Figure 4.3, where they must fill out the membership form. After clicking the "Next" button, they proceed to Figure 4.4 to complete their family information, bank information, and saving information. Once all required information is filled in and they have to agree with an agreement like in Figure 4.7 before submitting the form. Lastly, the application status will be shown as shown in Figure 4.8, indicating whether the membership application is pending approval, is being reviewed, or needs further information.

Figure 4.3: Membership Form

Figure 4.4: Family Information

Figure 4.5: Bank Information

Figure 4.6: Saving Information

Figure 4.7: Applicant's Agreement

Figure 4.8: Membership Status

The prerequisite to apply for a loan is, the user must have a Koperasi KADA membership in order to apply for a loan using the website. Before filling out the loan application form as in Figure 4.10 with the loan type, amount, and duration, they can utilize the loan calculator to estimate monthly repayments like in Figure 4.9. After filling out loan information, the system will display member's information as in Figure 4.11. In Figure 4.12, the user must provide the details of a guarantor and have an agreement to the terms like in Figure 4.13 before submitting the form. Lastly, the application status will be shown as shown in Figure 4.14, indicating whether the loan is pending approval, is being reviewed, or needs further information.

Figure 4.9: Loan Calculator

Figure 4.10: Loan Application Form

Figure 4.11: Applicant's Information

Figure 4.12: Guarantor Information

Figure 4.13: Agreement

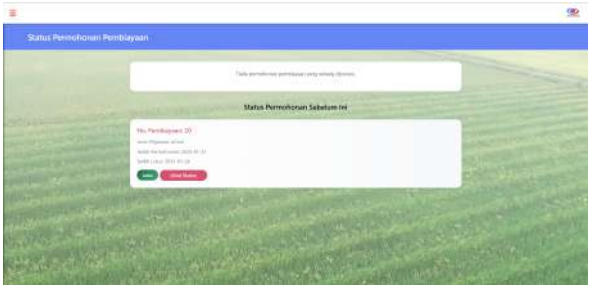


Figure 4.14: Loan Status

Furthermore, if a member wishes to terminate their membership, they can do so by filling out the application in Figure 4.15 and providing reason as in Figure 4.16. After completing the application, the applicant needs to have an agreement by clicking in the checkbox in Figure 4.17. Then, the application status will also be shown like the other two applications, as in Figure 4.18, indicating whether the loan is pending approval, is being reviewed, or needs further information.

Figure 4.15: Terminate Application

Figure 4.16: Reason of Termination

Figure 4.17: Agreement

Figure 4.18: Terminate Application Status

Admin is prompted to login first by entering Admin ID and Password as shown in Figure 4.1. After successfully logging in as an admin, it will show options to manage various aspects of the organization. These options include reviewing the membership list, reviewing the loan application list, reviewing membership termination application list, and reviewing members approved by ALK as shown in Figure 4.19.

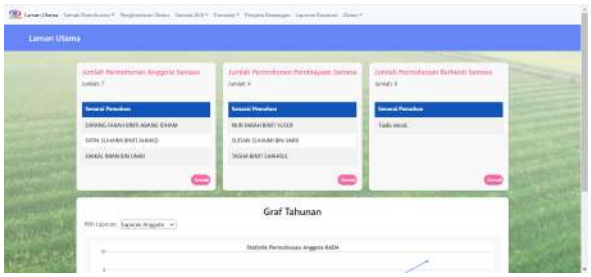


Figure 4.19: Admin Menu

After logging in as an admin and selecting the "Membership list", “Loan Application List” or “Membership Termination Application List” option as shown in Figure 4.20, 4.21, and 4.22 admin will see a list of members and their review status (either "Completed" or "Not Completed"). To review a member's form, click the applicant’s name. After completing the review of all details, return to the main review form to update the review status to "Completed" if the application is completed. Otherwise, Admin will update the review status to “Not Completed”.

Last but not least, Admin can perform batch processing by clicking any intended names using the checkbox to approve or reject simultaneously.

Figure 4.20: Membership Applicant's Name List **Figure 4.21: Loan Application Name List**

Figure 4.22: Membership Termination Application Name List

Next, Figure 4.23 illustrates the ALK's menu which shows the activity handled by ALK. To approve membership, ALK can click the Senarai Permohonan Anggota to view the list of applications reviewed by Admin. Figure 4.24 shows the list of names that are waiting to be approved or rejected. ALK can simply click the button 'Diterima' or 'Ditolak' to perform decision making for each application.

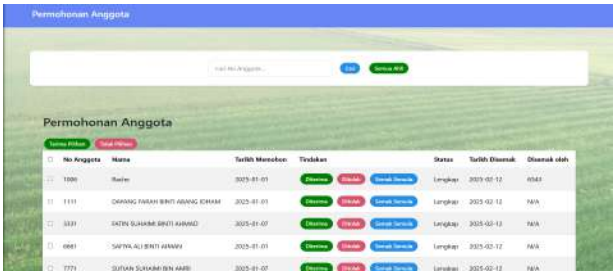


Figure 4.23: ALK's Menu **Figure 4.24: Membership List**

If the decision-maker is unsure and wants to recheck the application, ALK can click the 'Semak Semula' button to view the applicant's application. Figure 4.25 shows the applicant's information after clicking 'Semak Semula'. Last but not least for approving membership, ALK

can perform batch processing by clicking any intended names using the checkbox to approve or reject simultaneously. Figure 4.26 shows the batch processing function.

Figure 4.25: Application's Information **Figure 4.26 Batch Processing**

Moving to approving loans, ALK can choose 'Senarai Permohonan Pembiayaan in ALK' menu to access the loan applications reviewed by Admin. Same goes for loans, the ALK can directly perform decision-making by clicking the 'Diterima' or 'Ditolak' button directly from the list. ALK also can view the applications by clicking the 'Semak Semula' button. Figure 4.27 and Figure 4.28 show the list of names with buttons and applicant's loan application respectively. After viewing applicant's information, ALK can use the 'Banding' button to perform comparison of that particular application with the company's loan policies. Figure 4.29 shows the location of the 'Banding' button and Figure 4.30 illustrates the comparison result.

Figure 4.27: Loan List **Figure 4.28: Applicant's Info**

Figure 4.29: Banding Button **Figure 4.30: Comparison Result**

ALK is also responsible for approving membership termination applications. Figure 4.31 shows the list of applications reviewed by Admin. ALK can perform decision-making simply by clicking the ‘Diterima’ or ‘Ditolak’ button. ALK also can view the applicant’s information as shown in Figure 4.32. Lastly, all membership, loan and terminate membership approval supports batch processing using checkbox to perform decision-making to more than one applications simultaneously.

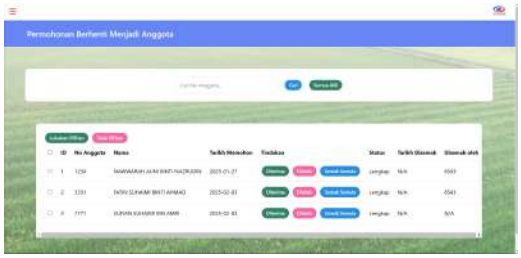


Figure 4.31: Terminate List

Figure 4.32: Application’s Info

Figure 4.33 and Figure 4.34 display the interface for administrators to manage monthly salary deductions and irregular transactions. Administrators can review and update transactions by clicking the “Keluarkan” button or filling the form. Meanwhile, Figure 4.35 and Figure 4.36 show the applicant's interface for viewing savings and loan repayment records. Applicants can check their transaction details to ensure the displayed financial information is clear and easily accessible.

Figure 4.33: Monthly Salary Deduction

Figure 4.34: Irregular Transaction

Figure 4.35: Savings Record

Figure 4.36: Loan Repayment Record

Next, Figure 4.37 displays a list of membership, loan, and termination applications that have been reviewed by ALK and require applicants to be notified via email. Figure 4.38 provides an option to generate a statement for the approved loan applicant list. When the administrator clicks the "Hasilkan" button, the system automatically generates the report statement, as shown in Figure 4.39. The administrator can then send the report statement via email for applicant reference. Finally, Figure 4.40 presents the corporation report, which includes both monthly and yearly summaries. This report helps in monitoring the overall financial performance and membership activities within the cooperative.

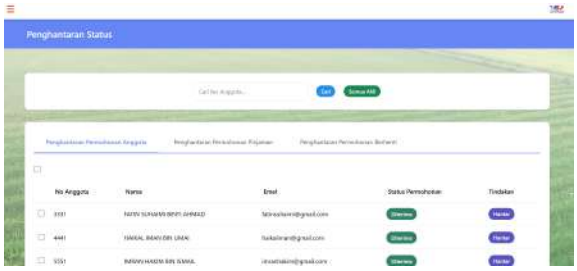


Figure 4.37: List of Applicant's Status

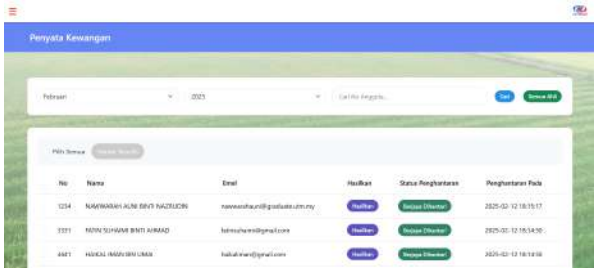
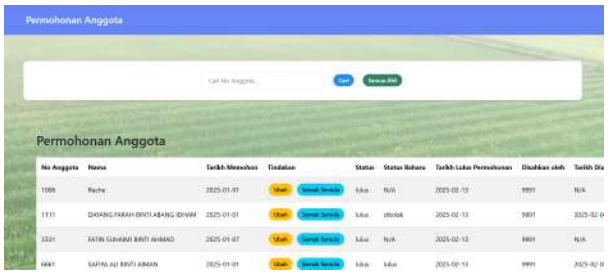


Figure 4.38: List of Member of Koperasi

Figure 4.39: Generated Statement

Figure 4.40: Corporation report

Lastly, ALK can modify the status of each application. This function was used when ALK unintentionally performed a wrong decision-making process. Figure 4.41 and Figure 4.42 illustrate the modification interface of membership and loan application respectively.



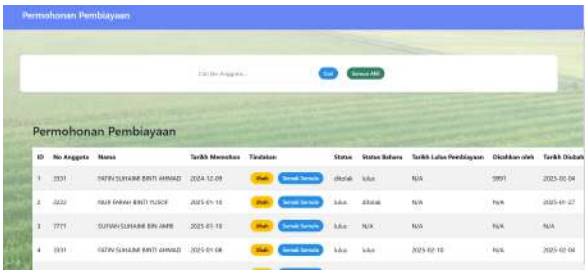


Figure 4.41: Membership Modification

Figure 4.42: Loan Modification

5. Traceability

This section generally ensures that every requirement, feature and functionality of a system is connected to its corresponding design, implementation and testing processes. Traceability is often maintained through tools like the Traceability Matrix, which maps the relationships between requirements and their corresponding artifacts.

Table 7.1 and Table 7.2 shows the requirement traceability matrix, identifies functional requirements, which are mapped to specific system components, such as the use cases, sequence diagram and related entities. For example, the P001 is associated with the entities User, Applicant, Membership, and FamilyInfo since a membership application should address the authentication of a user, details of an applicant and information about the family. P002 will review the membership form. Similarly, P004 includes Admin, ALK, Membership, Loan and Approval to handle the approval process of memberships and loans. Further, P003 includes entities such as Loan, LoanType, LoanCalculator and GuarantorInfo for loan applications, tracking of loan types and transactions and verification guarantors. Lastly, P005 is associated with Admin, Applicant, Saving, LoanTransaction, Notification for preparing statement reports and sending email notifications.

Table 7.1: Requirement Matrix

Package, UC, Sequence Diagram /Entity Class	User	User Type	Applicant	Administrator	ALK	Membership	FamilyInfo	BankInfo	SavingType	MembershipEnd	Managing
P001, UC001, SD001	X	X	X			X	X	X	X		
P002, UC002, SD002	X	X		X		X	X	X	X		
P004, UC003,	X	X			X	X	X	X	X		

SD003											
P003, UC004, SD004	X	X	X			X					
P003, UC005, SD005	X	X	X								
P004, UC006, SD006	X	X			X						
P005, UC007, SD007	X	X		X		X			X		
P005, UC008, SD008	X	X	X	X		X					X
P007, UC009, SD009	X	X	X	X		X			X		
P006, UC010, SD010	X	X	X	X						X	
P004, UC011, SD011	X	X			X	X				X	

Table 7.2: Requirement Matrix (Continuation)

Package, UC, Sequence Diagram /Entity Class	Saving	Loan	Guarantor	Loan Type	LoanRepayment	historyModiMem	historyModiLoan	AdminApplicationFeedback	AlkApplicationFeedback	status
P001, UC001,										X

SD001										
P002, UC002, SD002										X
P004, UC003, SD003						X			X	X
P003, UC004, SD004		X	X	X						X
P003, UC005, SD005		X		X		X				
P004, UC006, SD006		X	X	X			X		X	X
P005, UC007, SD007	X	X		X	X					
P005, UC008, SD008		X							X	X
P007, UC009, SD009	X	X		X	X					
P006, UC010, SD010						X	X	X	X	X
P004, UC011, SD011									X	X